Selection Plots: Config Comparisons

April 15, 2026

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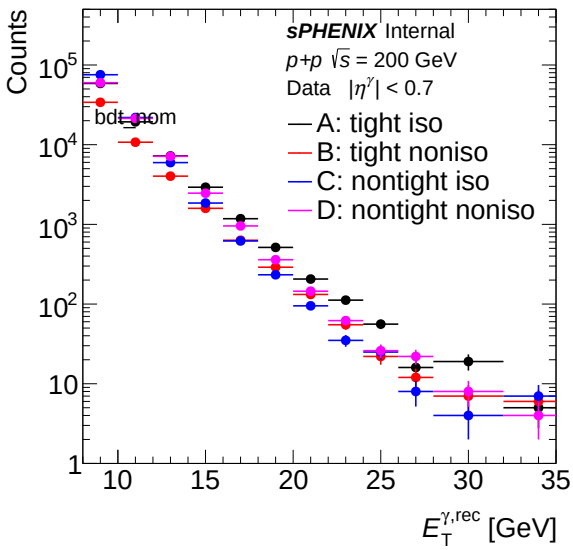
1 bdt_nom vs bdt_tightbdt50

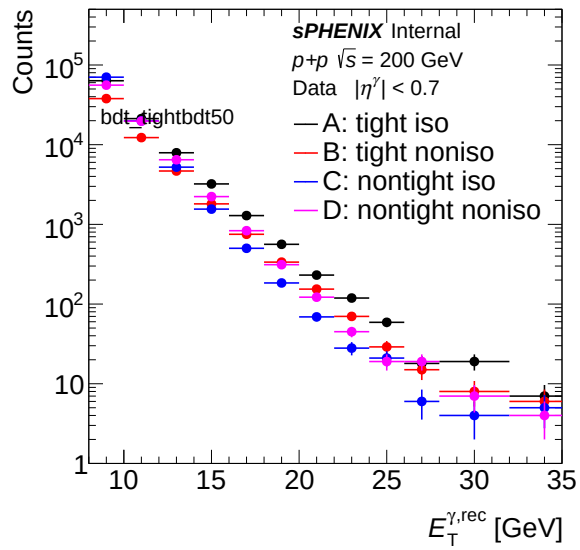
Config diff (A → B)

```diff
--- config_bdt_nom.yaml
+++ config_bdt_tightbdt50.yaml
@@ -20,5 +20,5 @@
   response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
   data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
+  var_type: "bdt_tightbdt50"
   final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"

@@ -77,11 +77,4 @@
   # Leave empty ("") to use the pre-made vertex_reweight_file instead.
   vertex_scan_data_file: ""
-  # Data-driven iterative truth-vertex reweight (standalone tool in
-  # efficiencytool/truth_vertex_reweight/). When on, forces vertex_reweight_on
-  # off and applies w(z_h) (single MC) or w(z_h)*w(z_mb) (double MC) per event.
-  # Flip truth_vertex_reweight_on to 1 to switch the nominal 1.5 mrad
-  # analysis from the f(z_r) reco-level reweight to the truth-level one.
-  truth_vertex_reweight_on: 0
-  truth_vertex_reweight_file: "/sphenix/user/shuhangli/ppg12/efficiencytool/truth_vertex_reweight/output
   fit_option: 1
   cluster_escale: 1.0
@@ -154,5 +147,5 @@
     bdt_min: 0.7
     bdt_min_slope: -0.015
-    bdt_min_intercept: 0.80
+    bdt_min_intercept: 0.7


@@ -210,5 +203,5 @@
     bdt_max: 0.5
     bdt_max_slope: -0.015
-    bdt_max_intercept: 0.80
+    bdt_max_intercept: 0.7
     bdt_min: 0.02
```

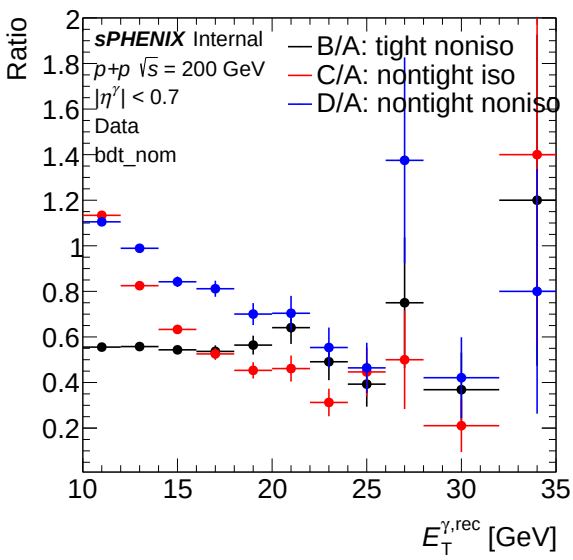
1.1 Sideband (Data)



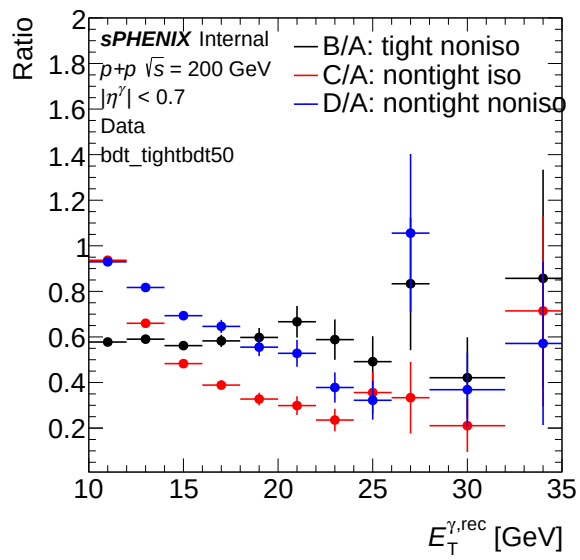
(a) Sideband ET spectrum (data) — bdt_nom



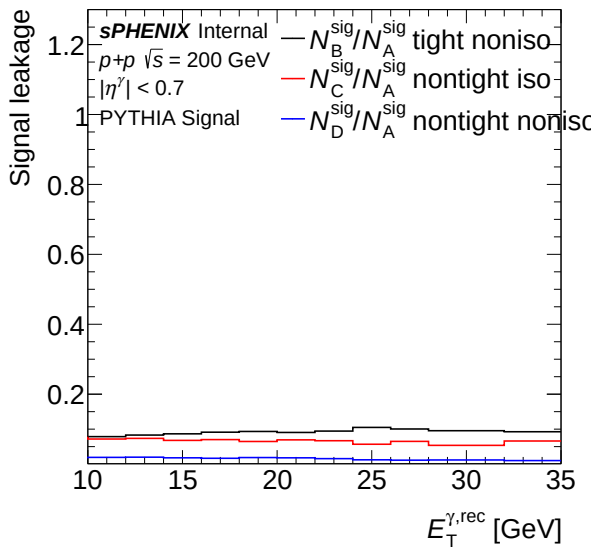
(b) Sideband ET spectrum (data) — bdt_tightbdt50



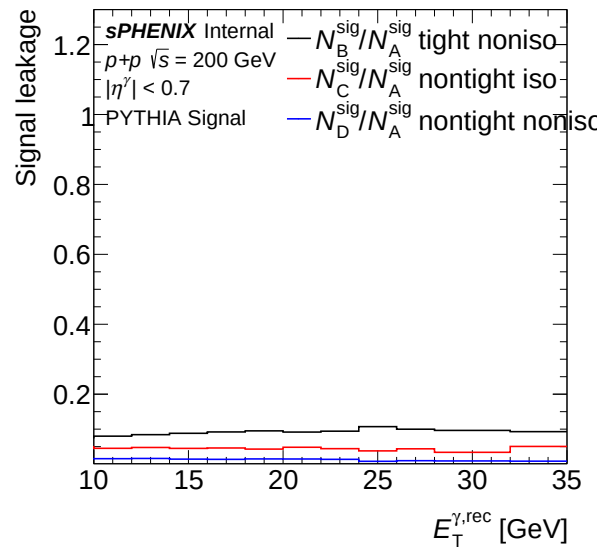
(a) Sideband ET ratio (data) — bdt_nom



(b) Sideband ET ratio (data) — bdt_tightbdt50

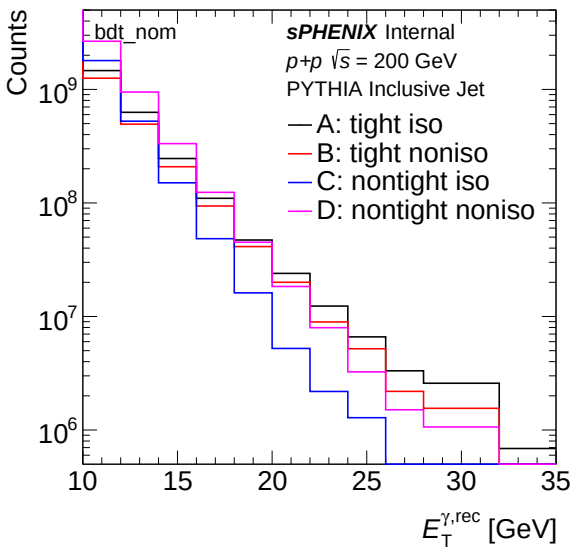


(a) Leakage fraction vs ET (data) — bdt_nom

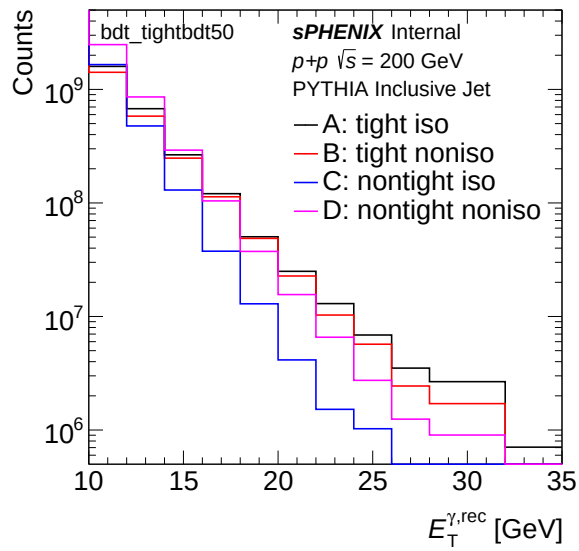


(b) Leakage fraction vs ET (data) — bdt_tightbdt50

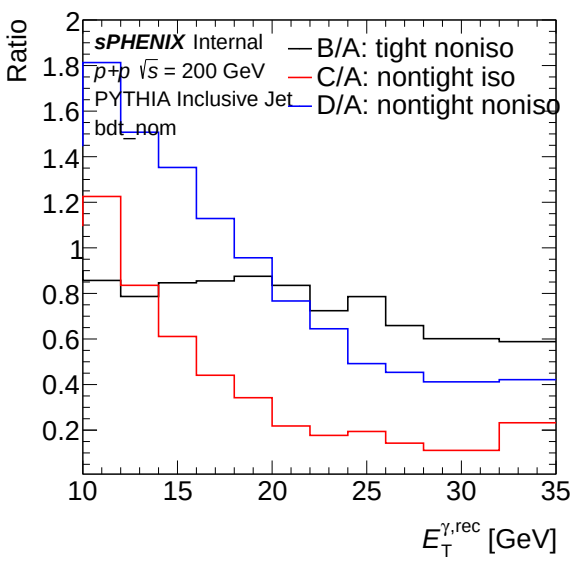
1.2 Sideband (MC)



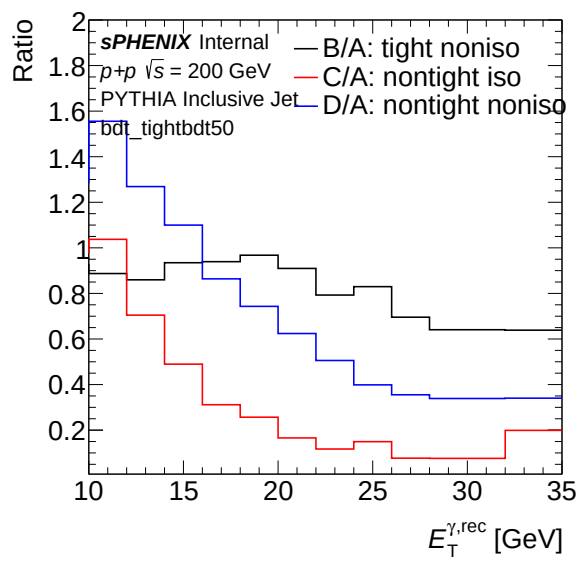
(a) Sideband ET spectrum (MC) — bdt_nom



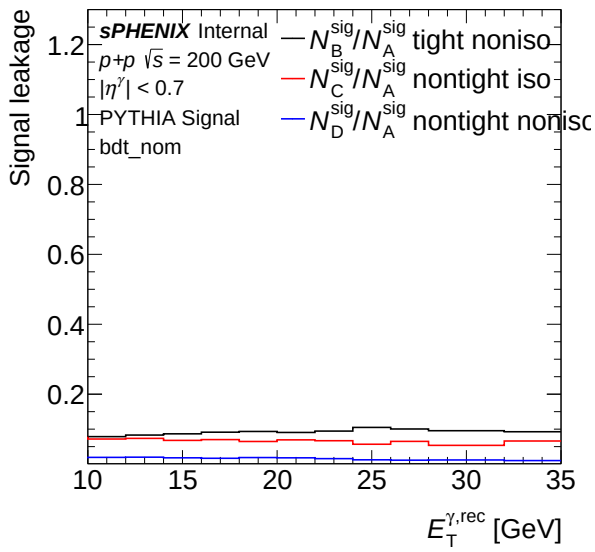
(b) Sideband ET spectrum (MC) — bdt_tightbdt50



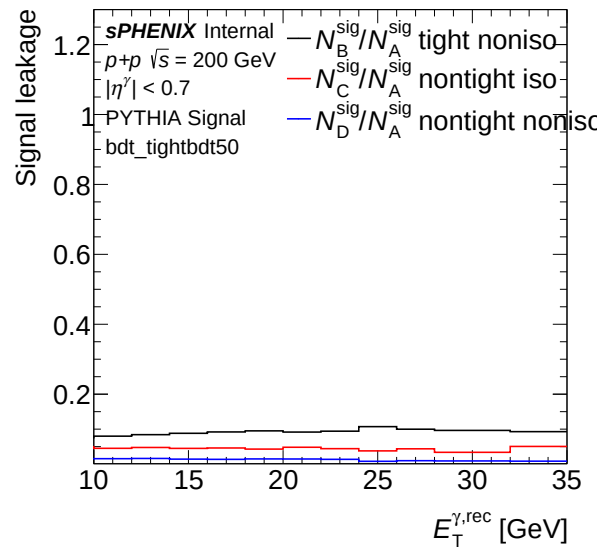
(a) Sideband ET ratio (MC) — bdt_nom



(b) Sideband ET ratio (MC) — bdt_tightbdt50

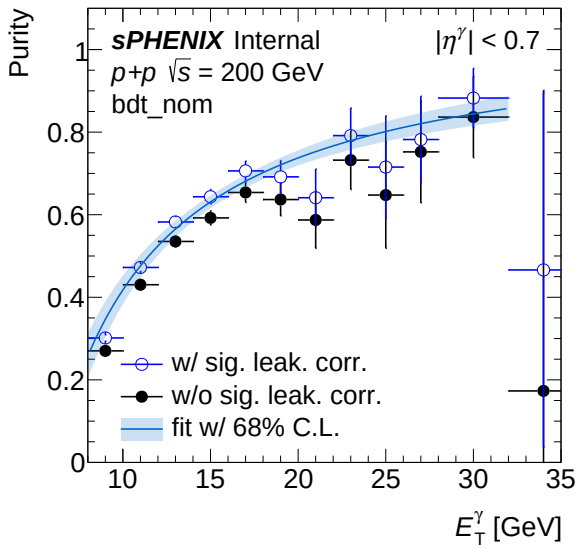


(a) Leakage fraction vs ET (MC) — bdt_nom

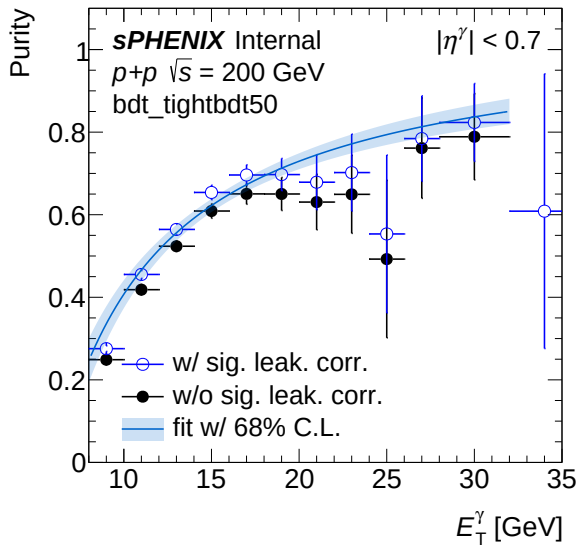


(b) Leakage fraction vs ET (MC) — bdt_tightbdt50

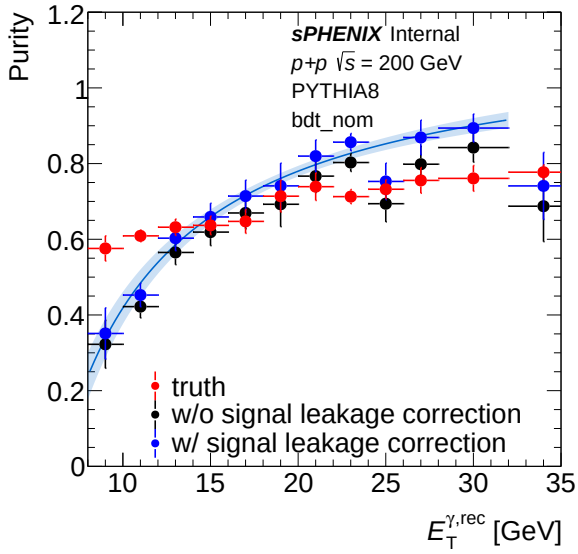
1.3 Purity



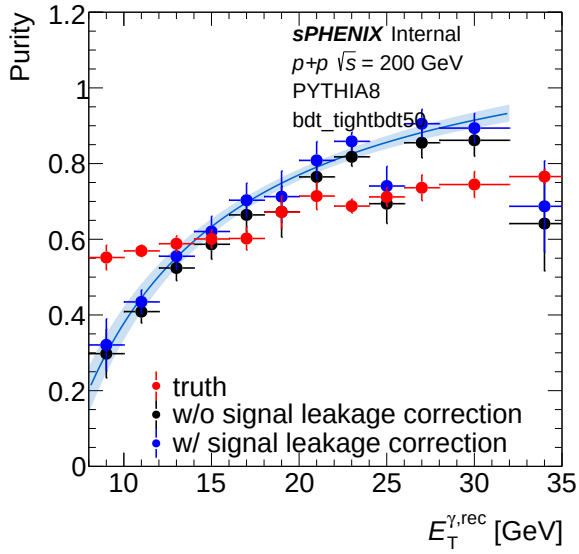
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_tightbdt50

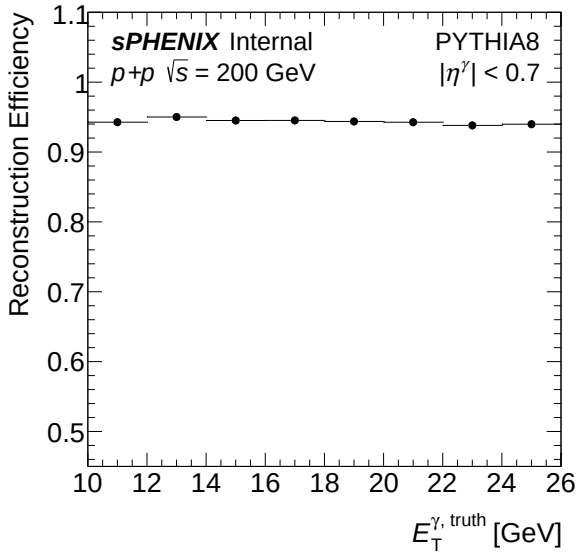


(a) Purity (MC) — bdt_nom

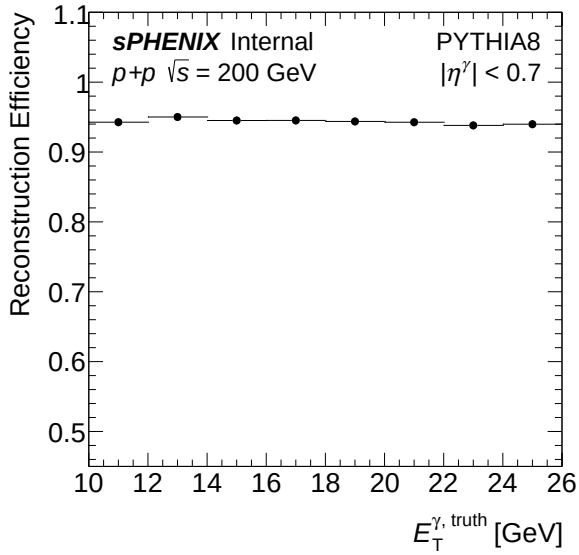


(b) Purity (MC) — bdt_tightbdt50

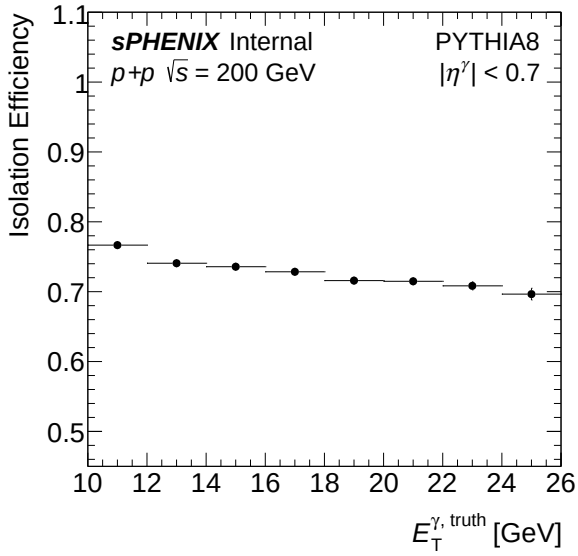
1.4 Efficiency



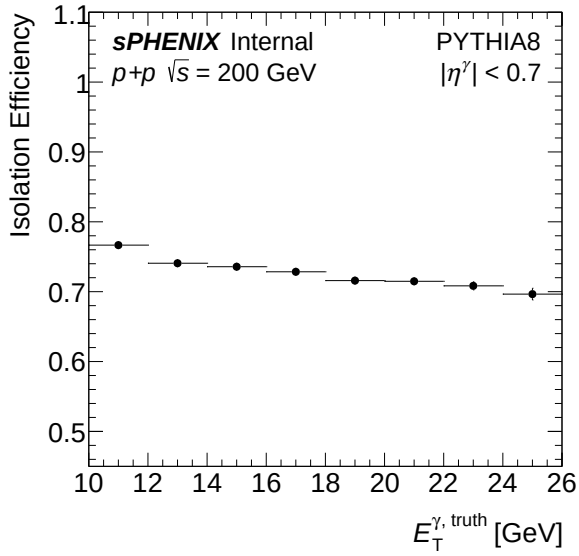
(a) Reconstruction efficiency — bdt_nom



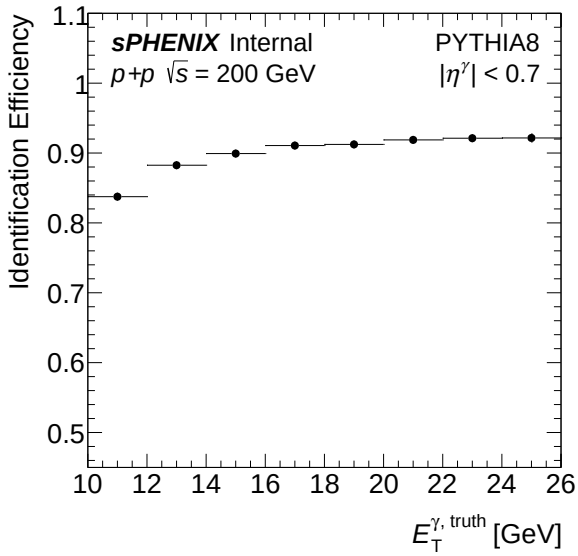
(b) Reconstruction efficiency — bdt_tightbdt50



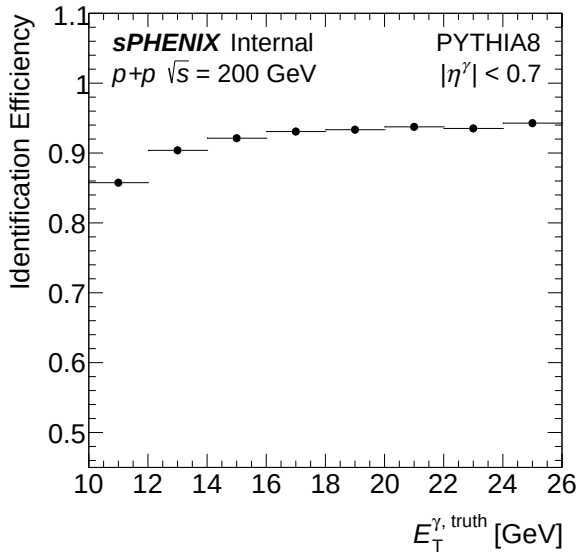
(a) Isolation efficiency — bdt_nom



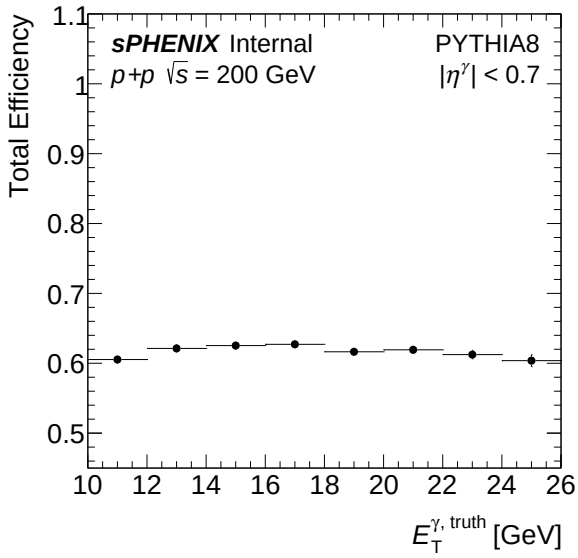
(b) Isolation efficiency — bdt_tightbdt50



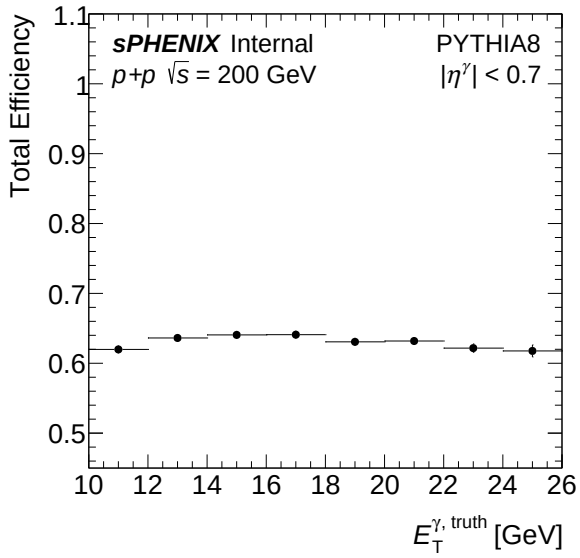
(a) Identification efficiency — bdt_nom



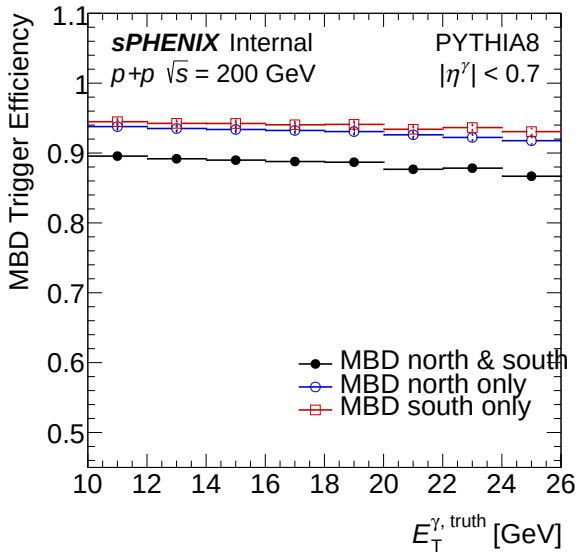
(b) Identification efficiency — bdt_tightbdt50



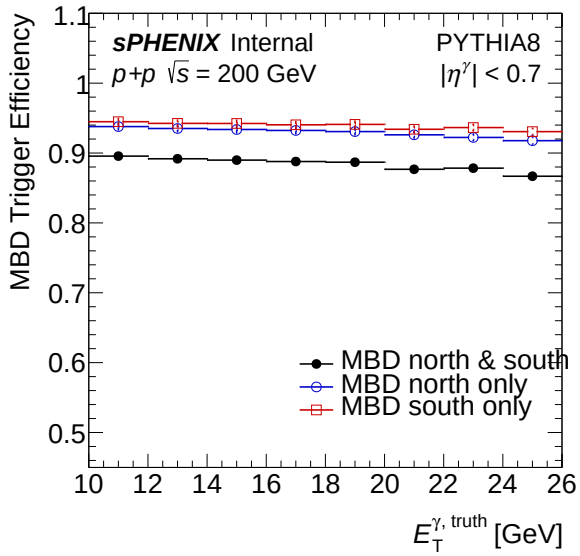
(a) Total efficiency — bdt_nom



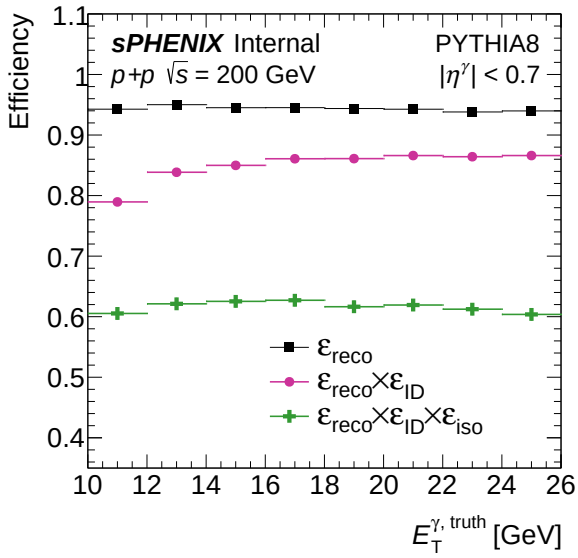
(b) Total efficiency — bdt_tightbdt50



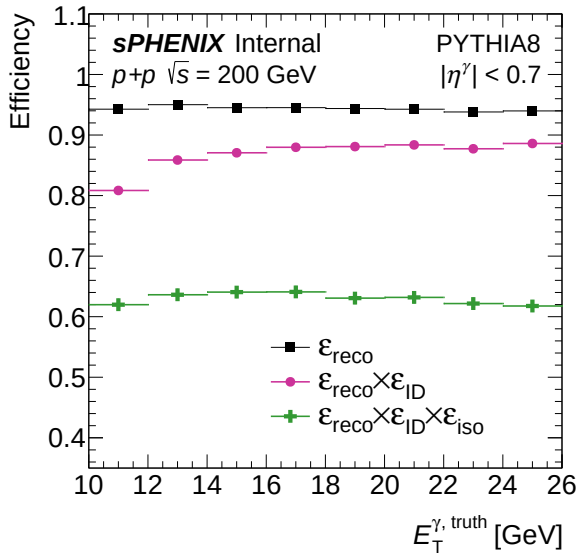
(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_tightbdt50

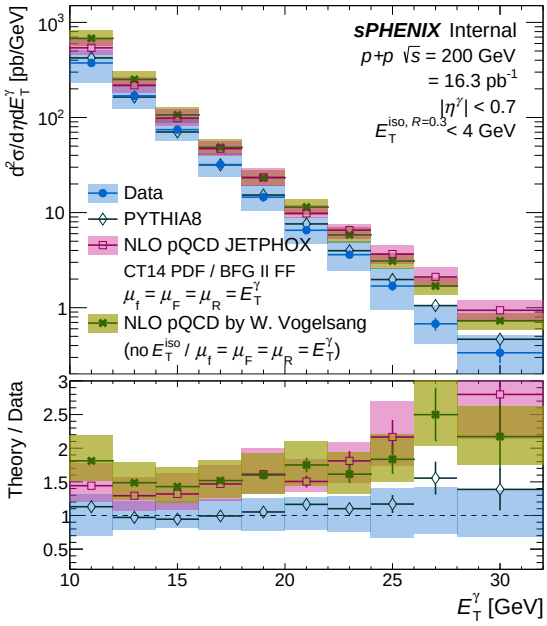


(a) Combined photon efficiency — bdt_nom

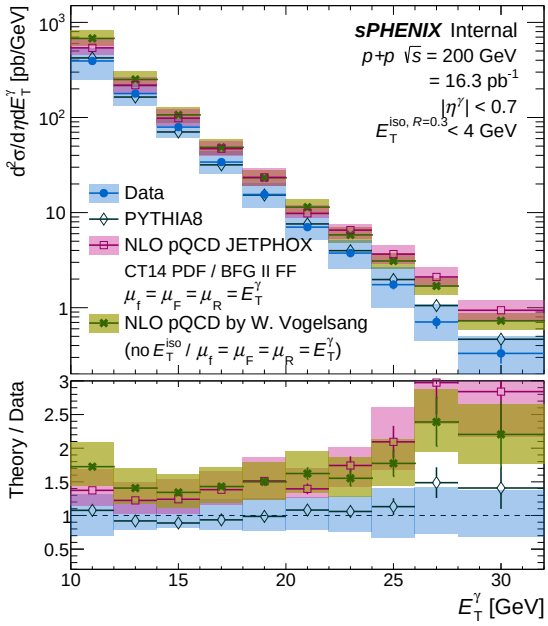


(b) Combined photon efficiency — bdt_tightbdt50

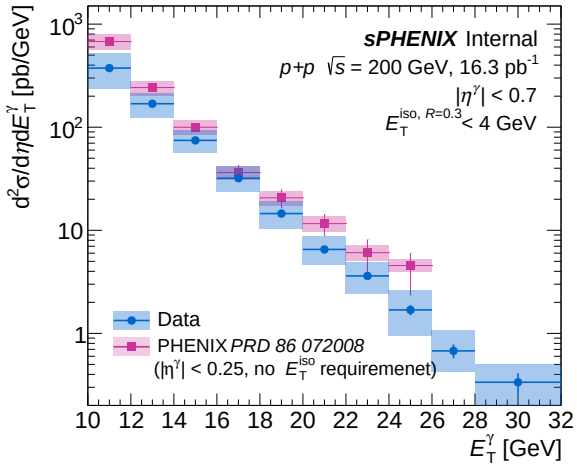
1.5 Final result



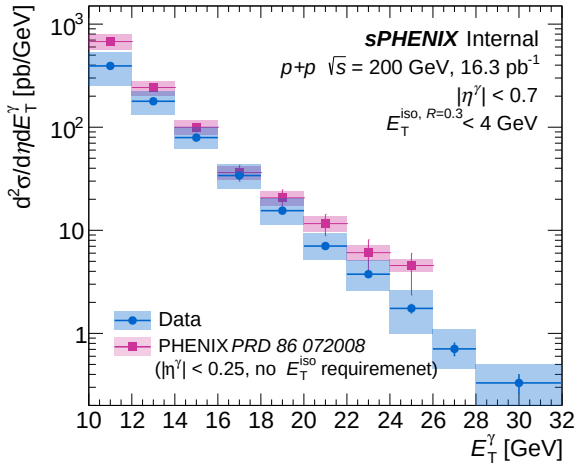
(a) Final cross section result — bdt_nom



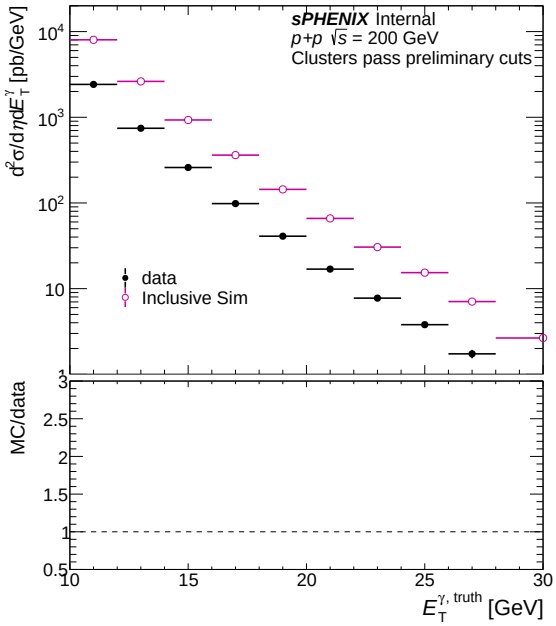
(b) Final cross section result — bdt_tightbdt50



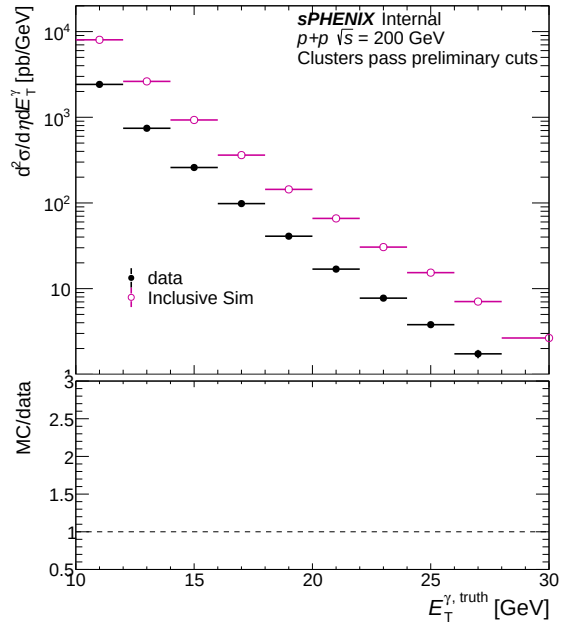
(a) Comparison with PHENIX — bdt_nom



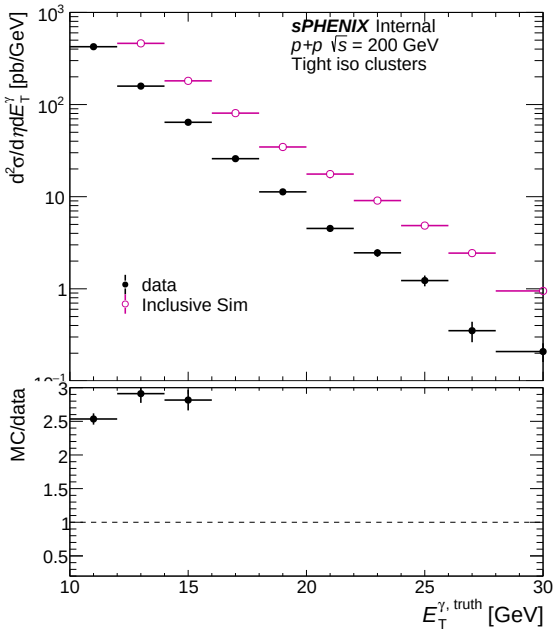
(b) Comparison with PHENIX — bdt_tightbdt50



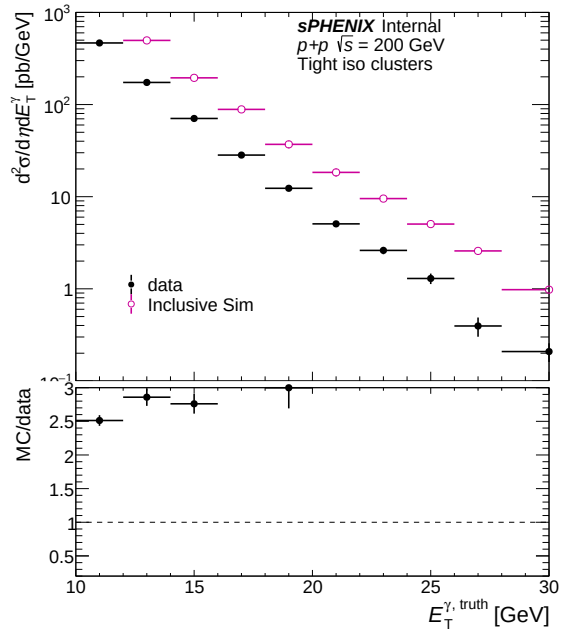
(a) Common cluster yield (data vs MC) — bdt_nom



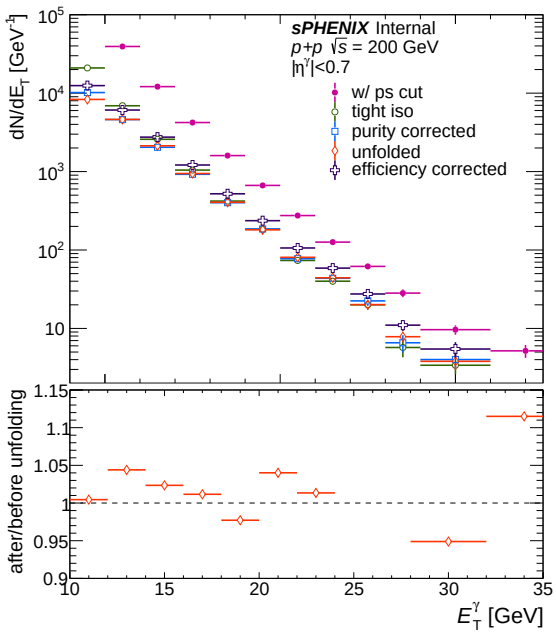
(b) Common cluster yield (data vs MC) — bdt_tightbdt50



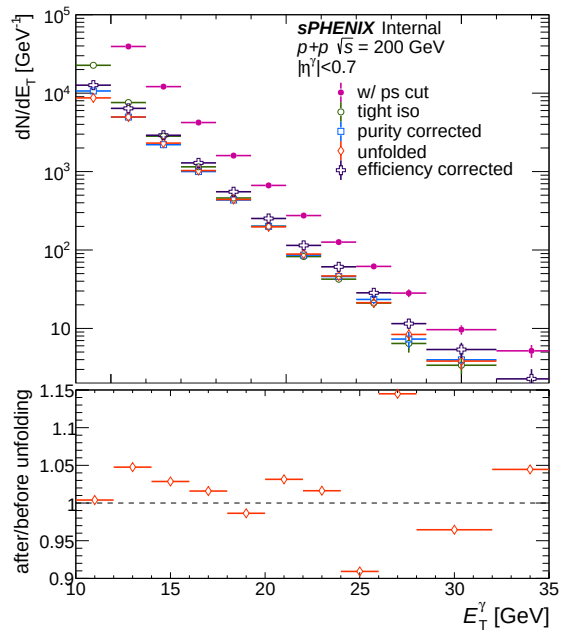
(a) Tight iso cluster yield (data vs MC) — bdt_nom



(b) Tight iso cluster yield (data vs MC) — bdt_tightbdt50

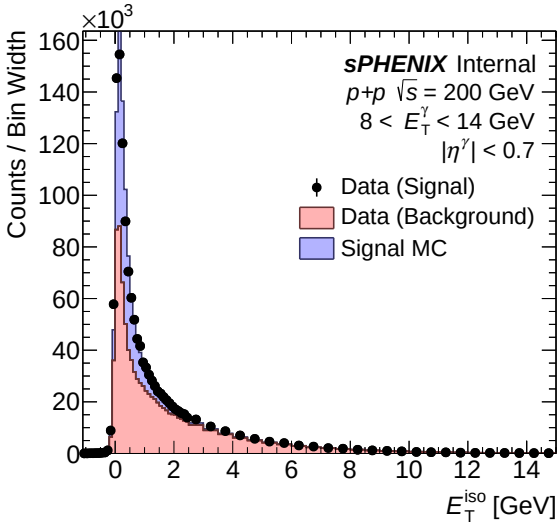


(a) All correction steps — bdt_nom

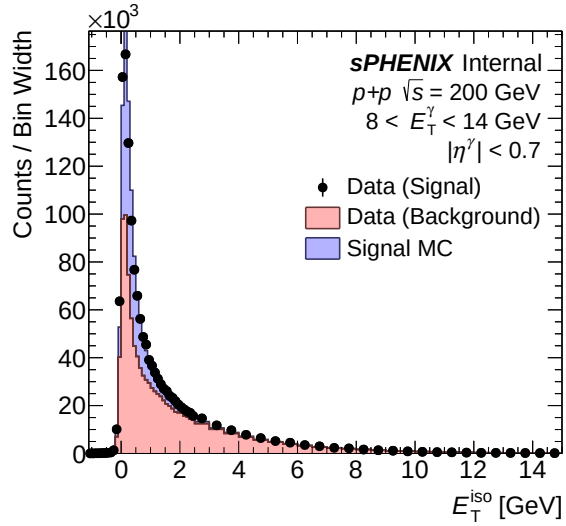


(b) All correction steps — bdt_tightbdt50

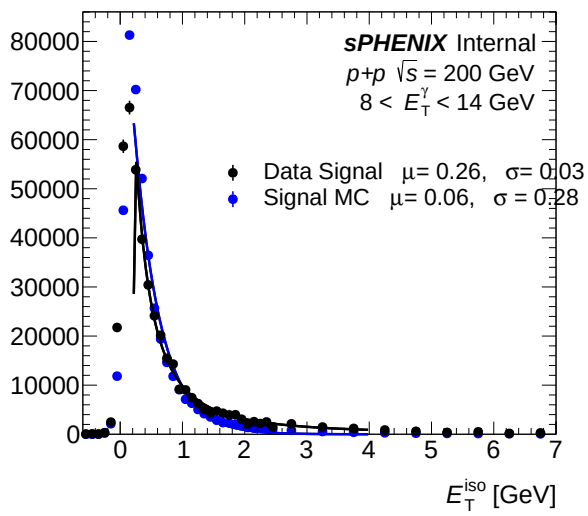
1.6 Isolation fit (CONF)



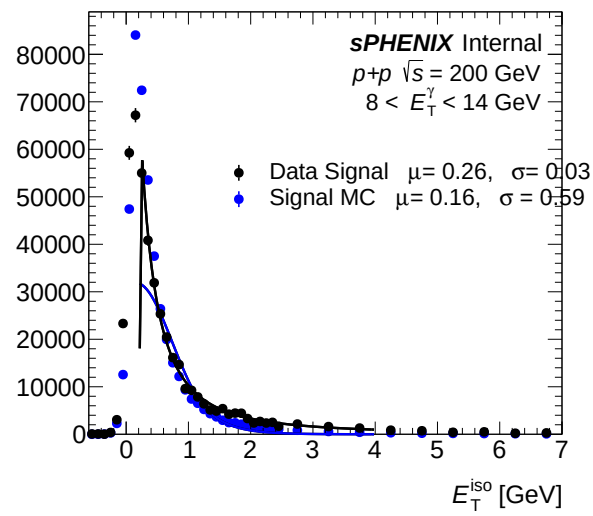
(a) IsoET template (pT bins 0–2) — bdt_nom



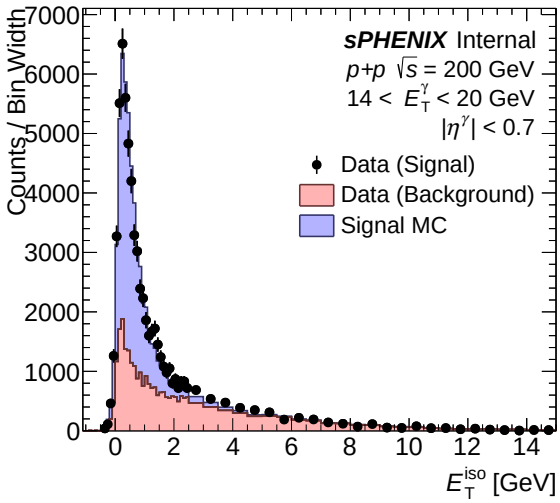
(b) IsoET template (pT bins 0–2) — bdt_tightbdt50



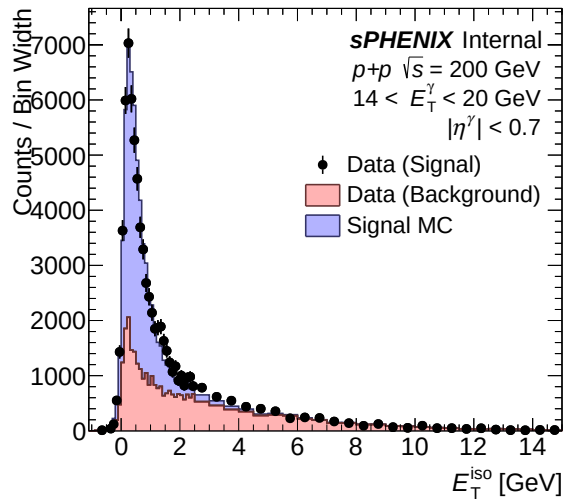
(a) IsoET fit template (pT bins 0–2) — bdt_nom



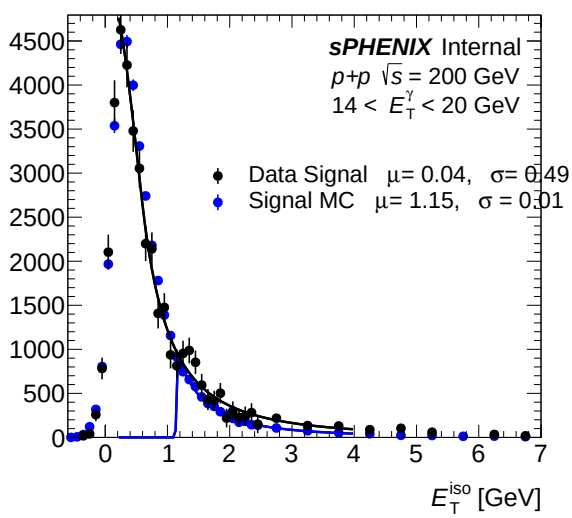
(b) IsoET fit template (pT bins 0–2) — bdt_tightbdt50



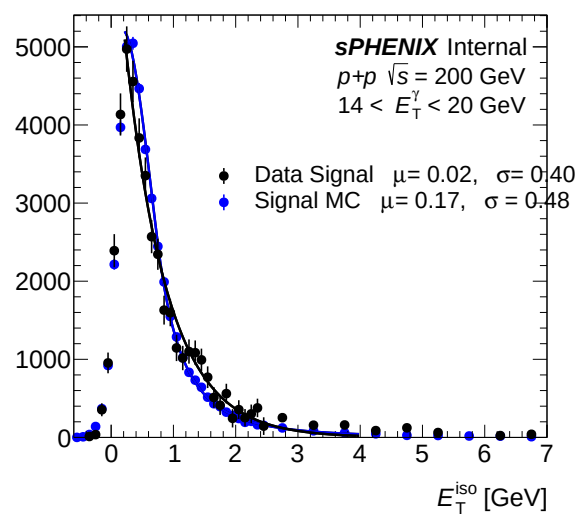
(a) IsoET template (pT bins 3–5) — bdt_nom



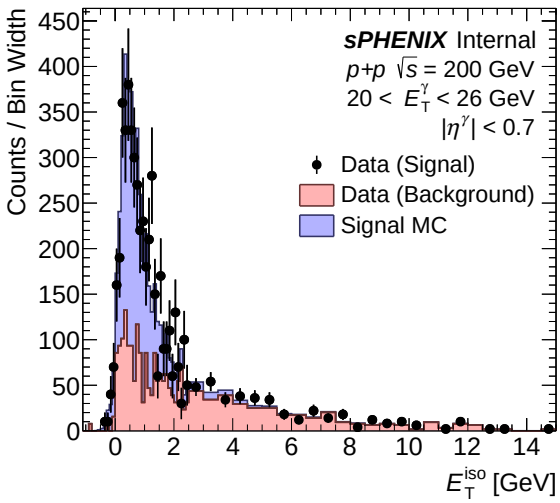
(b) IsoET template (pT bins 3–5) — bdt_tightbdt50



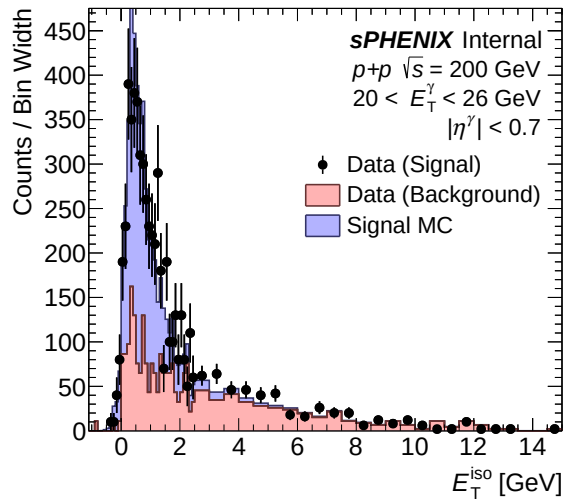
(a) IsoET fit template (pT bins 3–5) — bdt_nom



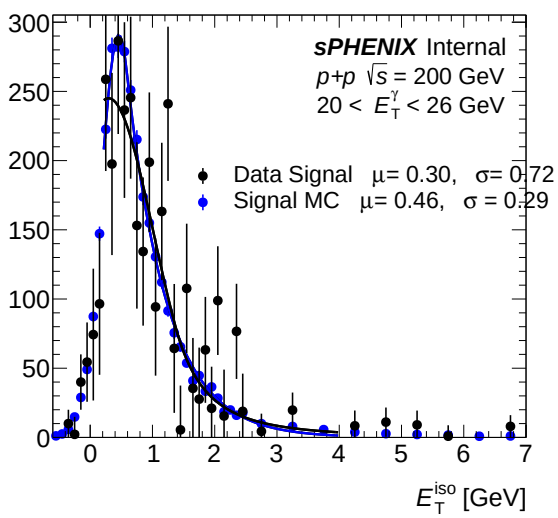
(b) IsoET fit template (pT bins 3–5) — bdt_tightbdt50



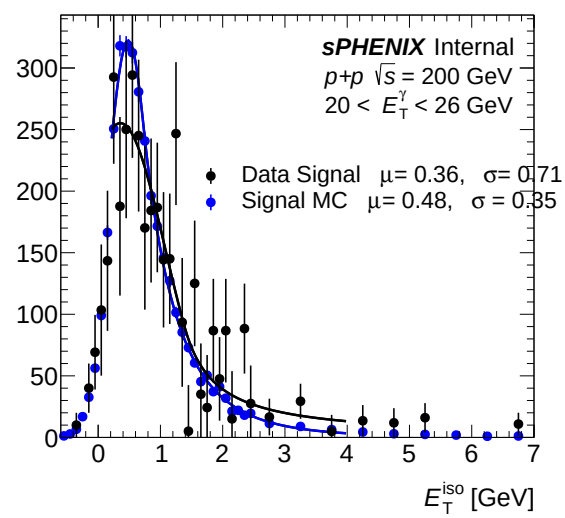
(a) IsoET template (pT bins 6–8) — bdt_nom



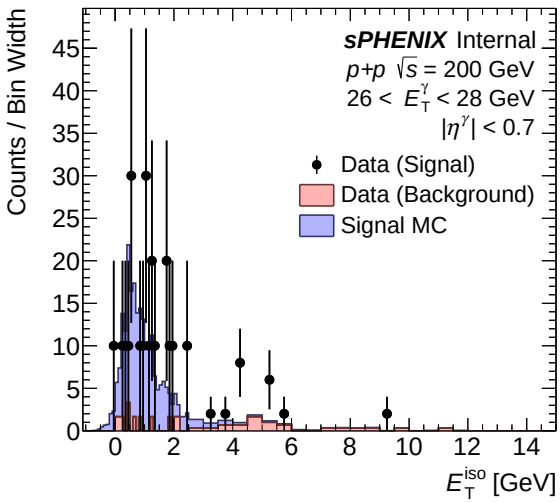
(b) IsoET template (pT bins 6–8) — bdt_tightbdt50



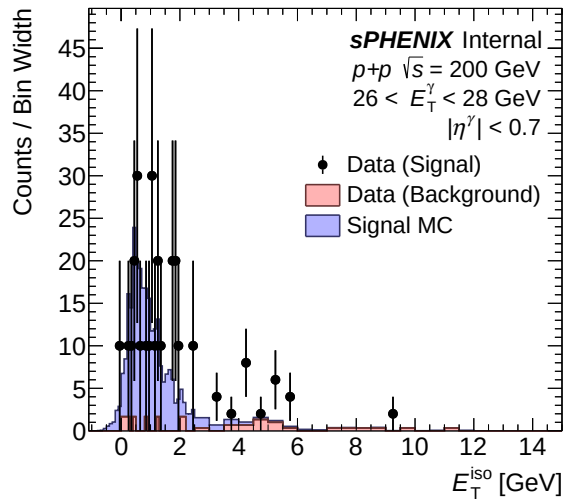
(a) IsoET fit template (pT bins 6–8) — bdt_nom



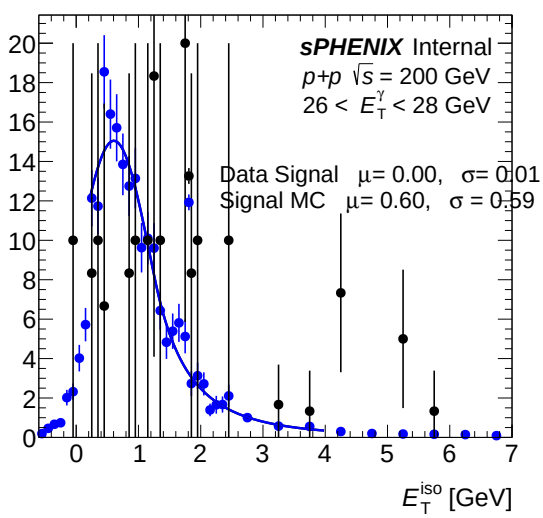
(b) IsoET fit template (pT bins 6–8) — bdt_tightbdt50



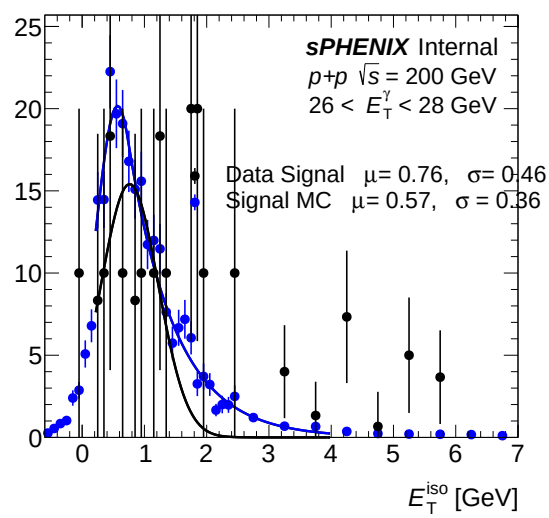
(a) IsoET template (pT bins 9–9) — bdt_nom



(b) IsoET template (pT bins 9–9) — bdt_tightbdt50

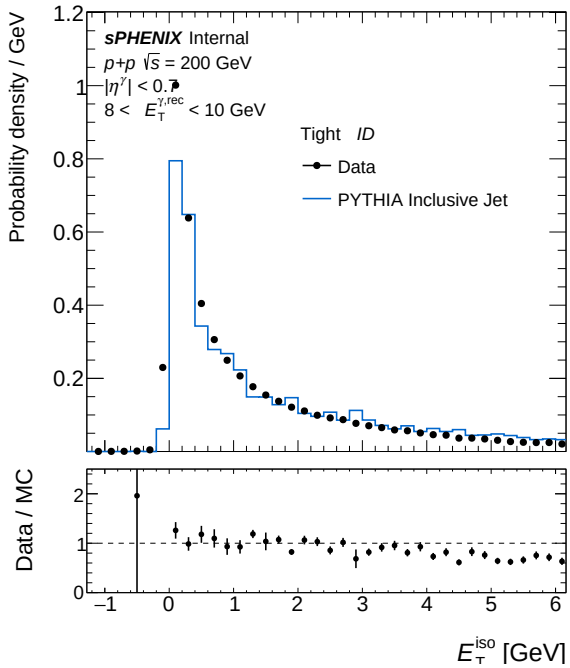


(a) IsoET fit template (pT bins 9–9) — bdt_nom

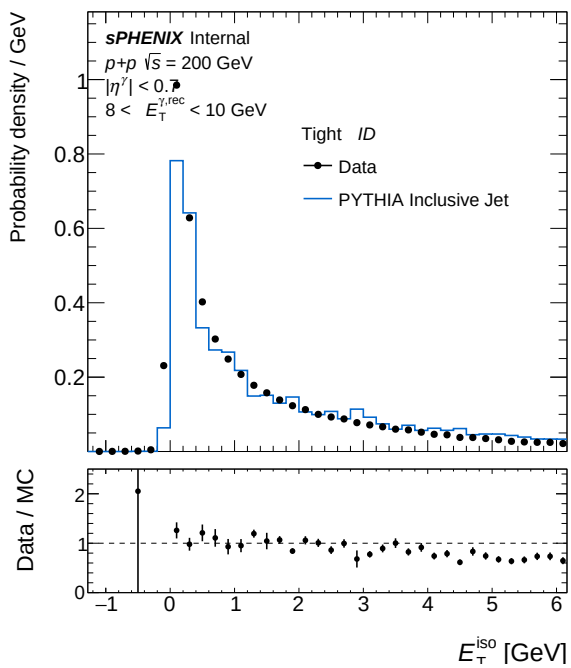


(b) IsoET fit template (pT bins 9–9) — bdt_tightbdt50

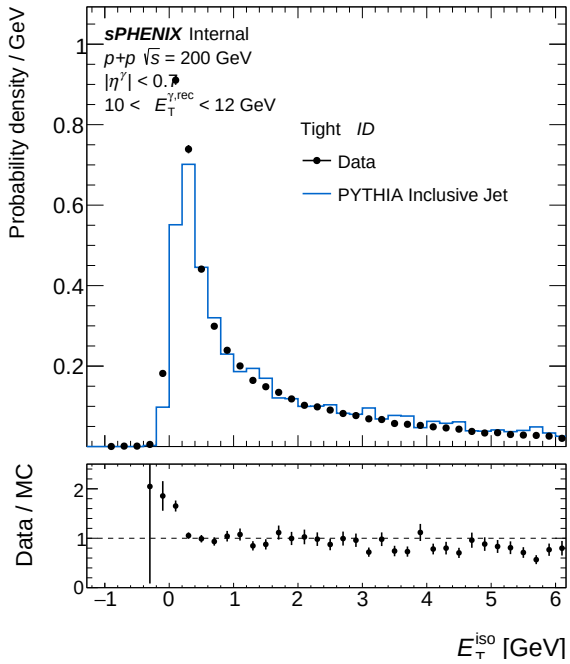
1.7 Isolation ET



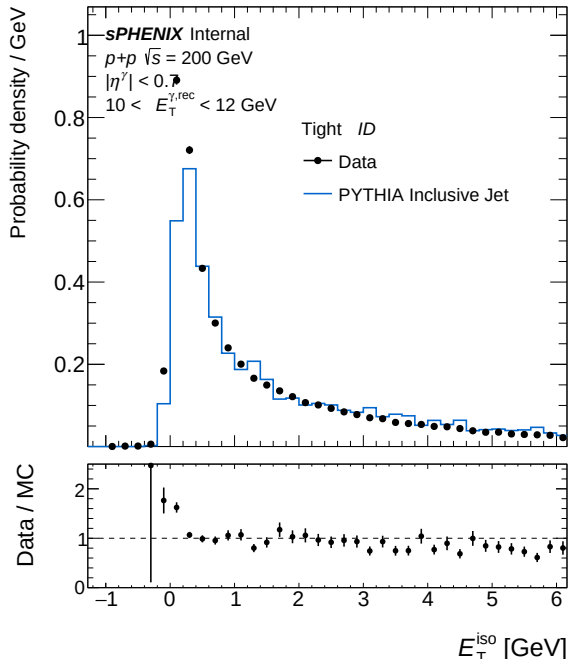
(a) IsoET Tight, pT bin 0 — bdt_nom



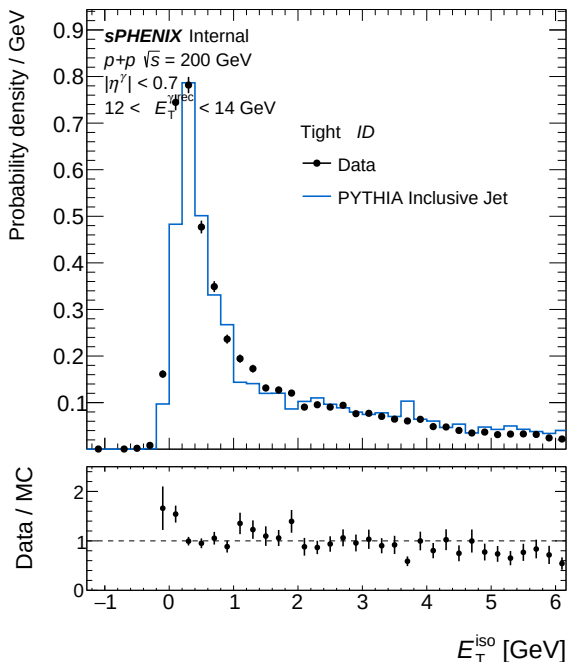
(b) IsoET Tight, pT bin 0 — bdt_tightbdt50



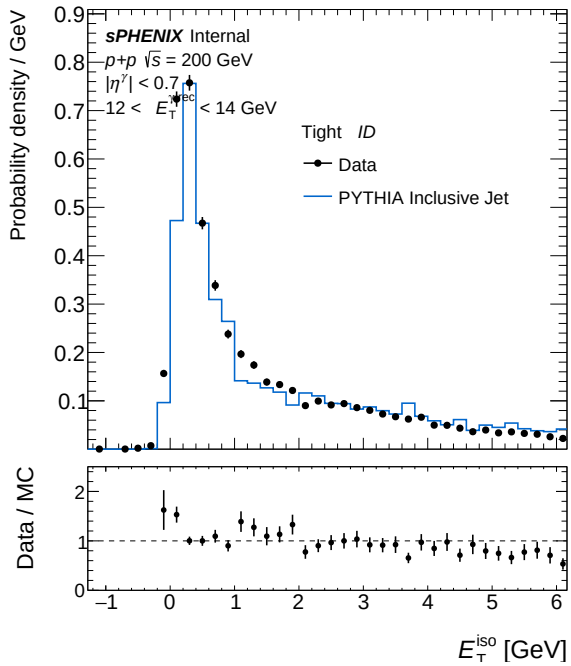
(a) IsoET Tight, pT bin 1 — bdt_nom



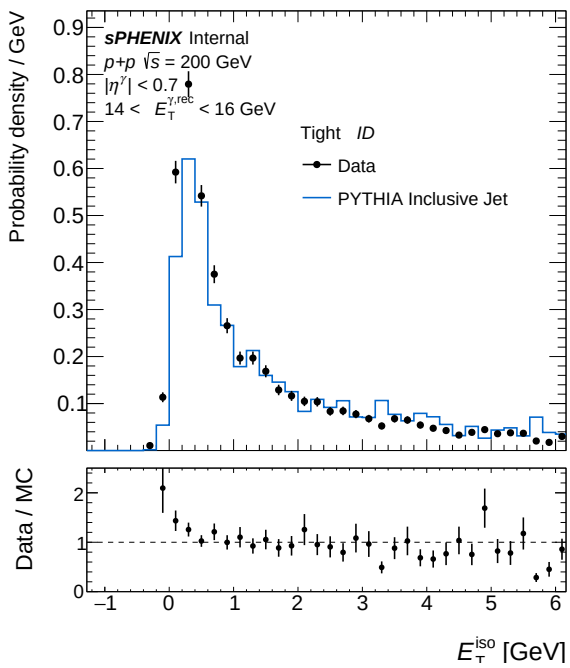
(b) IsoET Tight, pT bin 1 — bdt_tightbdt50



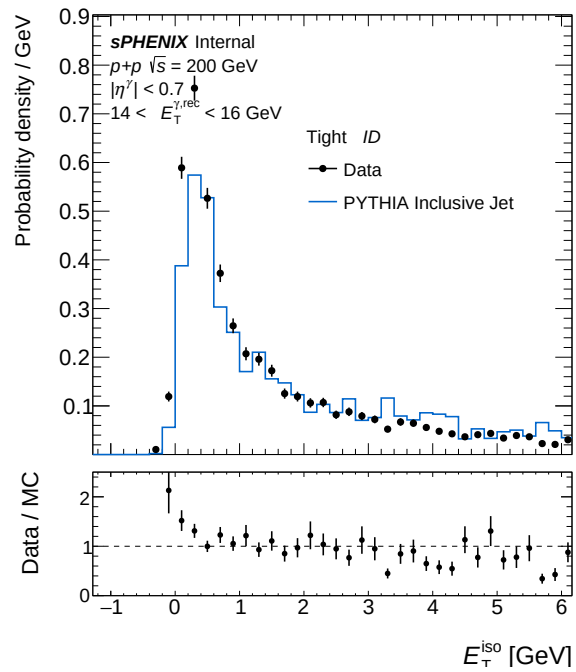
(a) IsoET Tight, pT bin 2 — bdt_nom



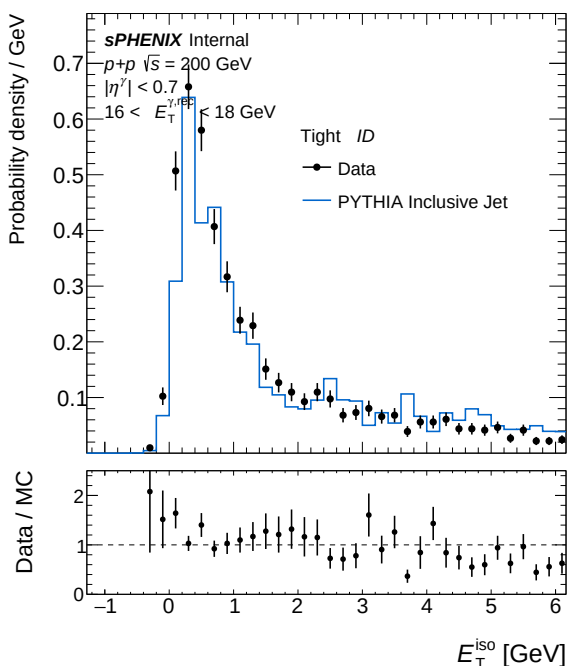
(b) IsoET Tight, pT bin 2 — bdt_tightbdt50



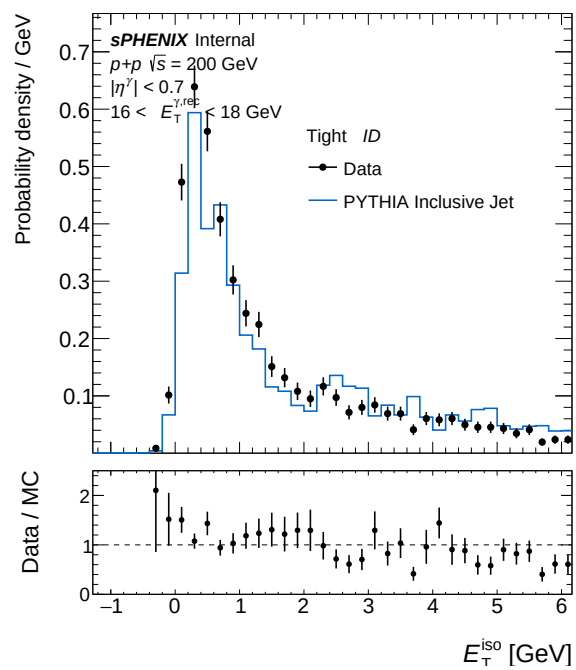
(a) IsoET Tight, pT bin 3 — bdt_nom



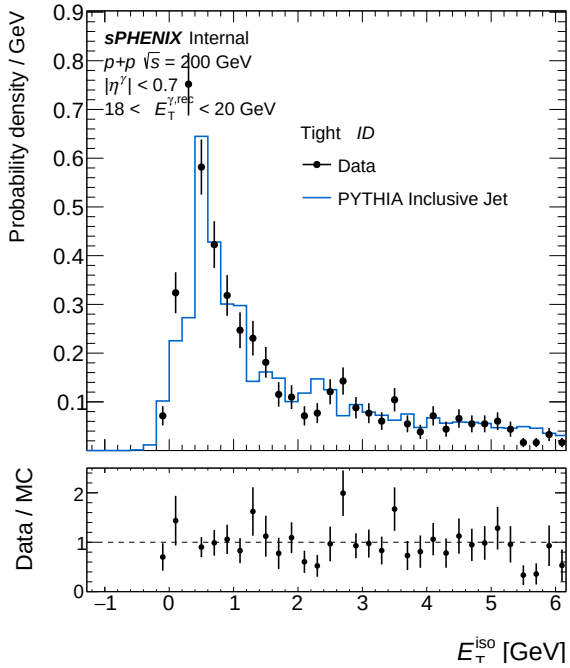
(b) IsoET Tight, pT bin 3 — bdt_tightbdt50



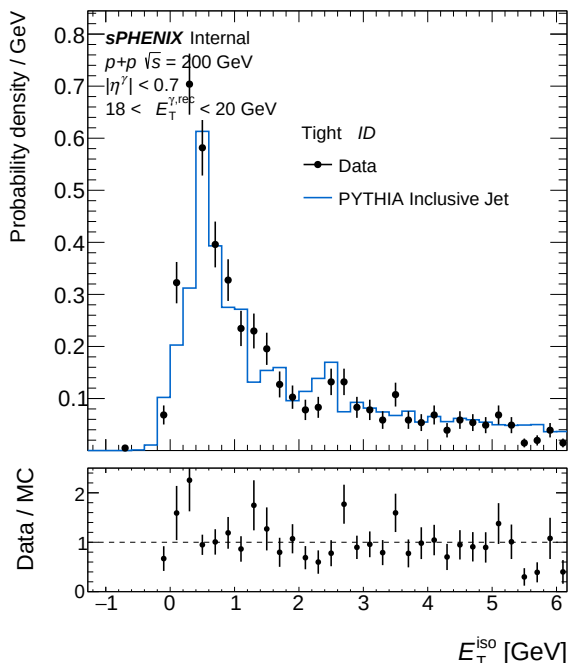
(a) IsoET Tight, pT bin 4 — bdt_nom



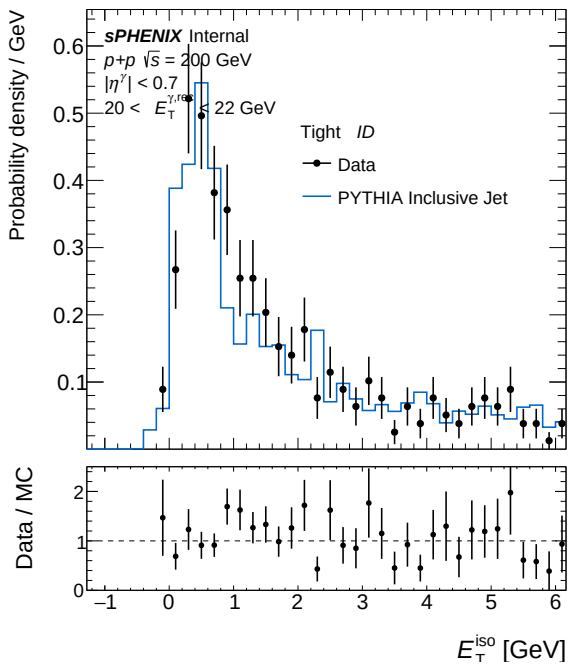
(b) IsoET Tight, pT bin 4 — bdt_tightbdt50



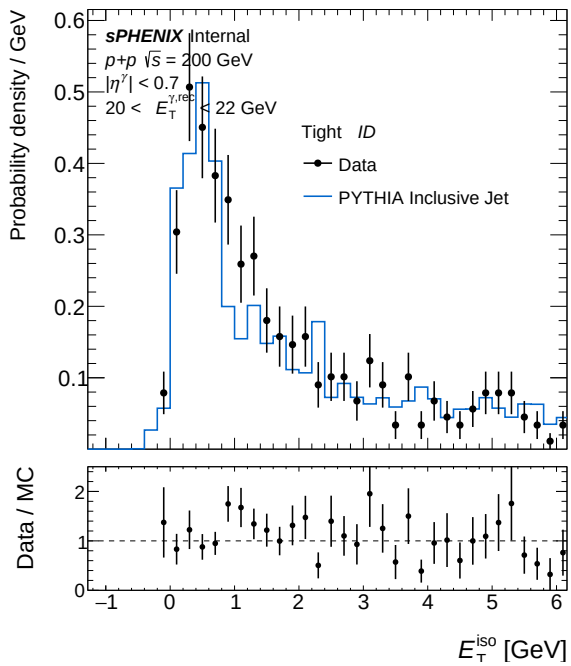
(a) IsoET Tight, pT bin 5 — bdt_nom



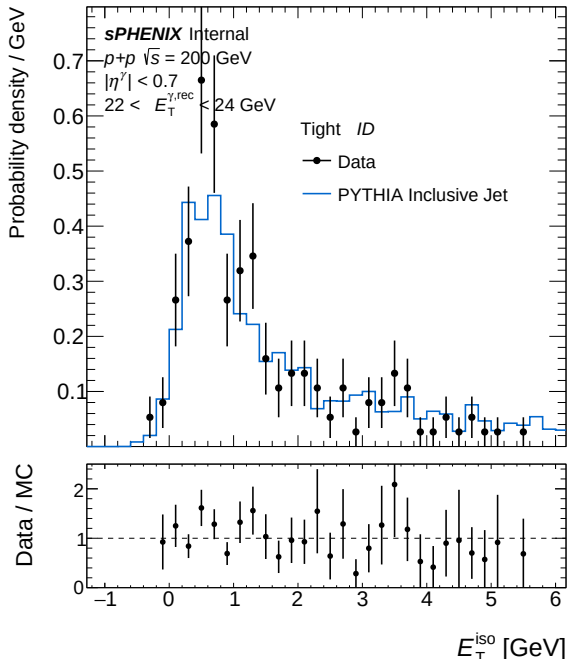
(b) IsoET Tight, pT bin 5 — bdt_tightbdt50



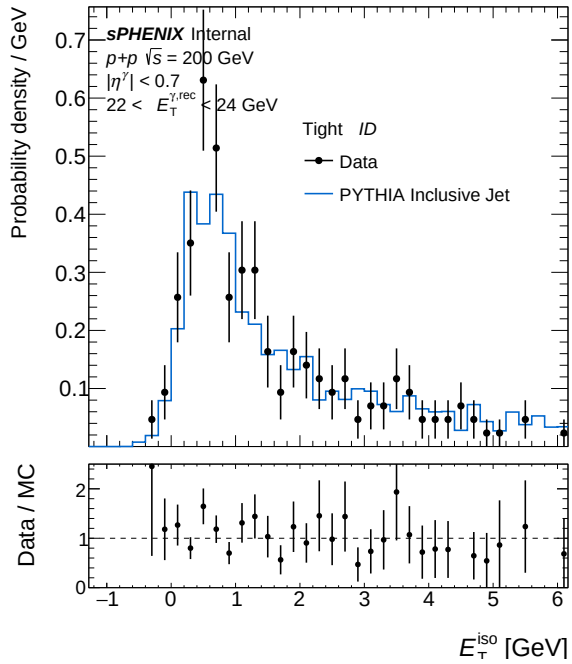
(a) IsoET Tight, pT bin 6 — bdt_nom



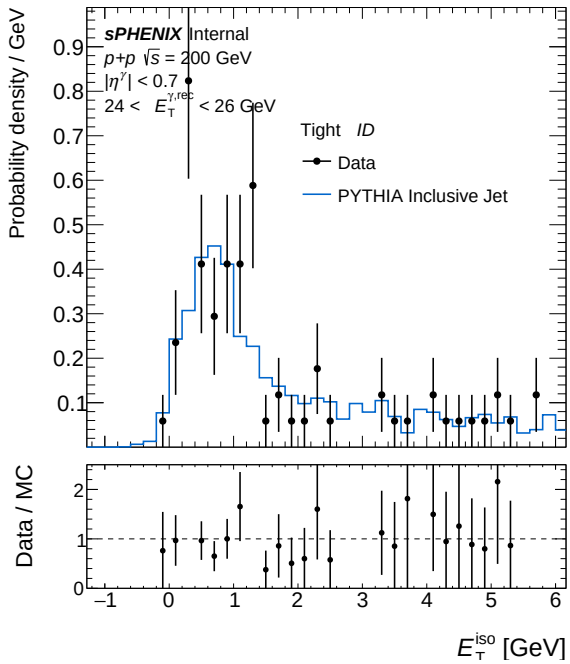
(b) IsoET Tight, pT bin 6 — bdt_tightbdt50



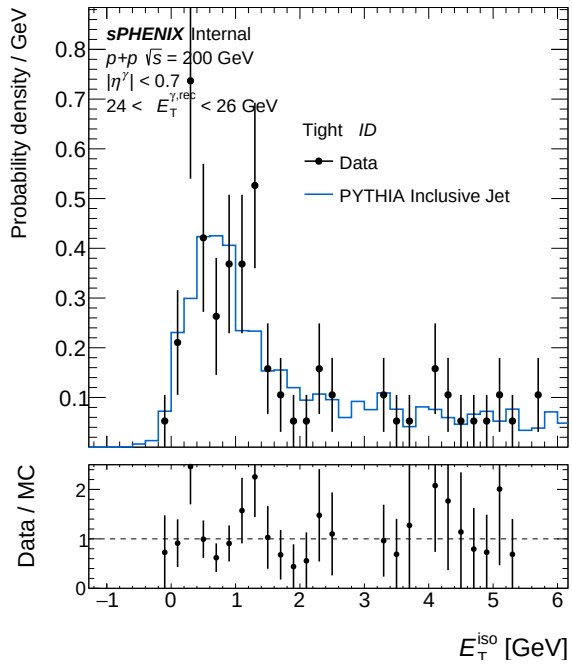
(a) IsoET Tight, pT bin 7 — bdt_nom



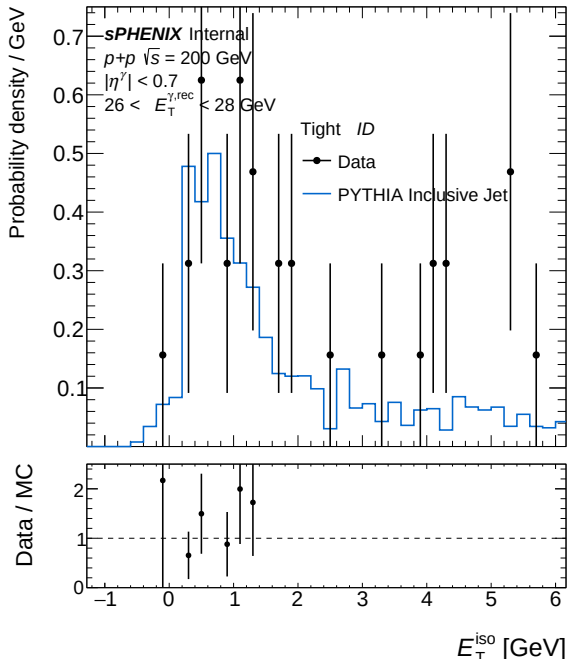
(b) IsoET Tight, pT bin 7 — bdt_tightbdt50



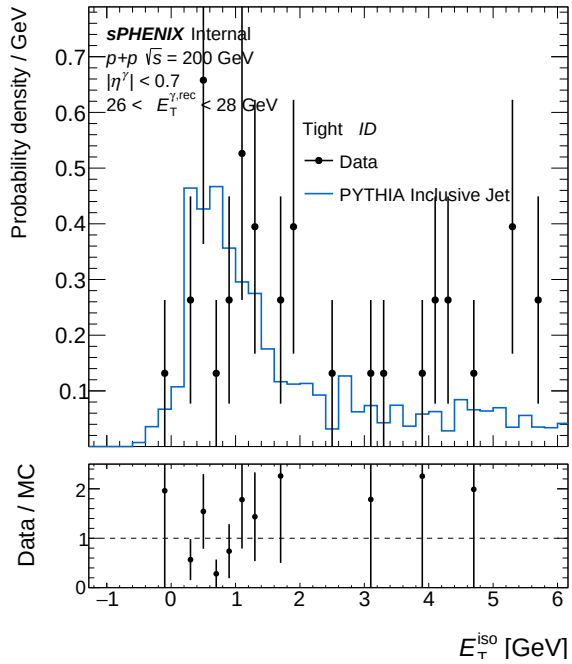
(a) IsoET Tight, pT bin 8 — bdt_nom



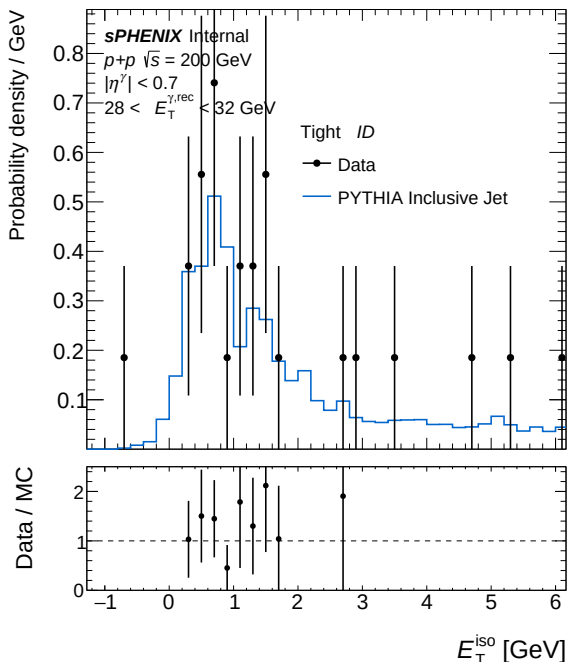
(b) IsoET Tight, pT bin 8 — bdt_tightbdt50



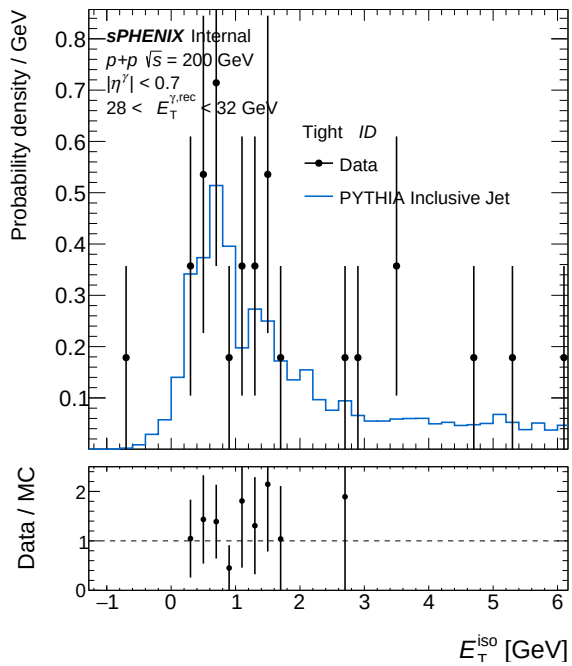
(a) IsoET Tight, pT bin 9 — bdt_nom



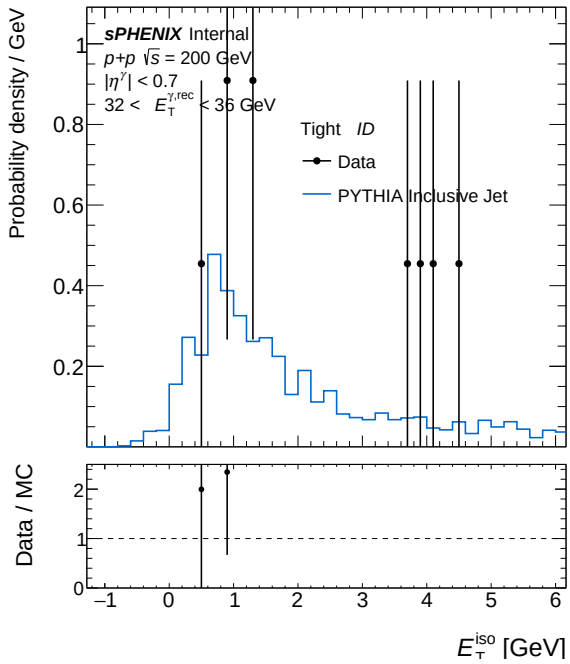
(b) IsoET Tight, pT bin 9 — bdt_tightbdt50



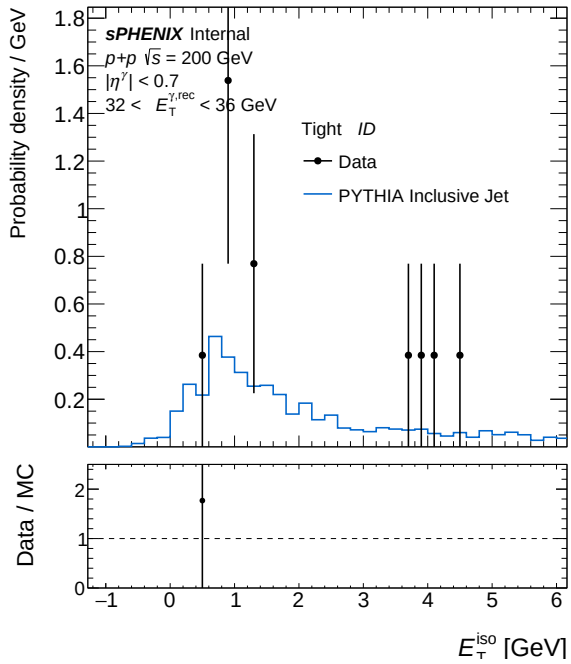
(a) IsoET Tight, pT bin 10 — bdt_nom



(b) IsoET Tight, pT bin 10 — bdt_tightbdt50



(a) IsoET Tight, pT bin 11 — bdt_nom



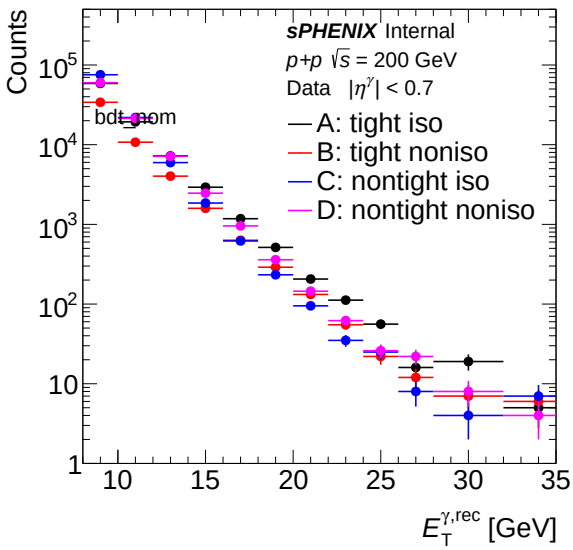
(b) IsoET Tight, pT bin 11 — bdt_tightbdt50

2 bdt_nom vs bdt_npb03

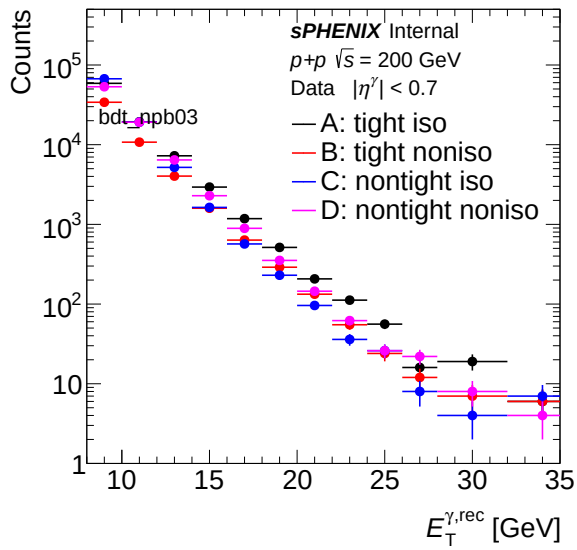
Config diff (A → B)

```diff
--- config_bdt_nom.yaml
+++ config_bdt_npb03.yaml
@@ -20,5 +20,5 @@
   response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
   data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
+  var_type: "bdt_npb03"
   final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"

@@ -77,11 +77,4 @@
   # Leave empty ("") to use the pre-made vertex_reweight_file instead.
   vertex_scan_data_file: ""
-  # Data-driven iterative truth-vertex reweight (standalone tool in
-  # efficiencytool/truth_vertex_reweight/). When on, forces vertex_reweight_on
-  # off and applies w(z_h) (single MC) or w(z_h)*w(z_mb) (double MC) per event.
-  # Flip truth_vertex_reweight_on to 1 to switch the nominal 1.5 mrad
-  # analysis from the f(z_r) reco-level reweight to the truth-level one.
-  truth_vertex_reweight_on: 0
-  truth_vertex_reweight_file: "/sphenix/user/shuhangli/ppg12/efficiencytool/truth_vertex_reweight/output
   fit_option: 1
   cluster_escale: 1.0
@@ -218,5 +211,5 @@

     npb_cut_on: 1
-    npb_score_cut: 0.5
+    npb_score_cut: 0.3

     prob_max: 1.0
```

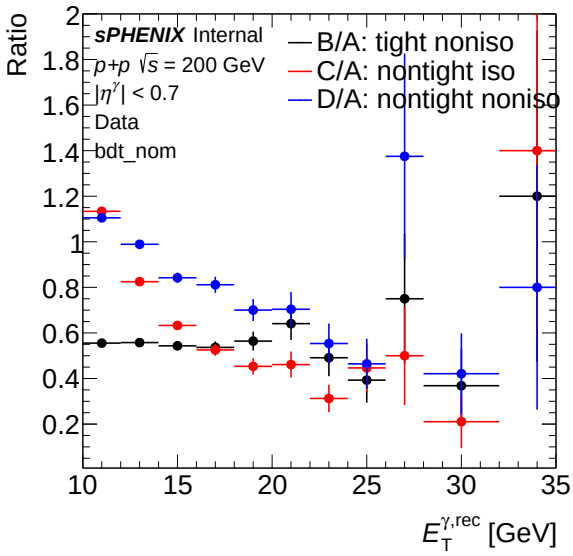
2.1 Sideband (Data)



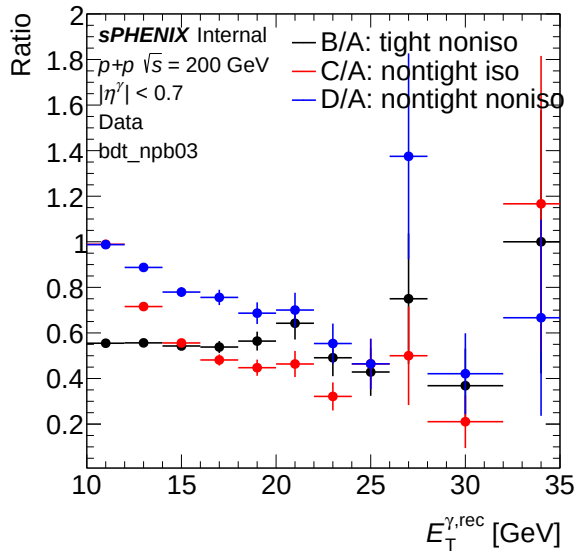
(a) Sideband ET spectrum (data) — bdt_nom



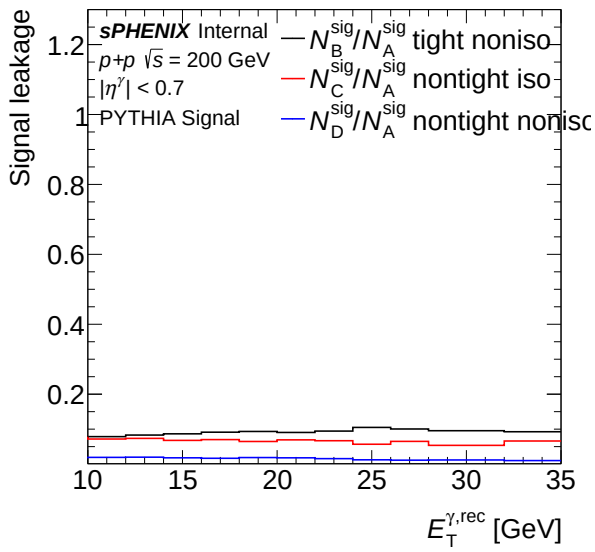
(b) Sideband ET spectrum (data) — bdt_npb03



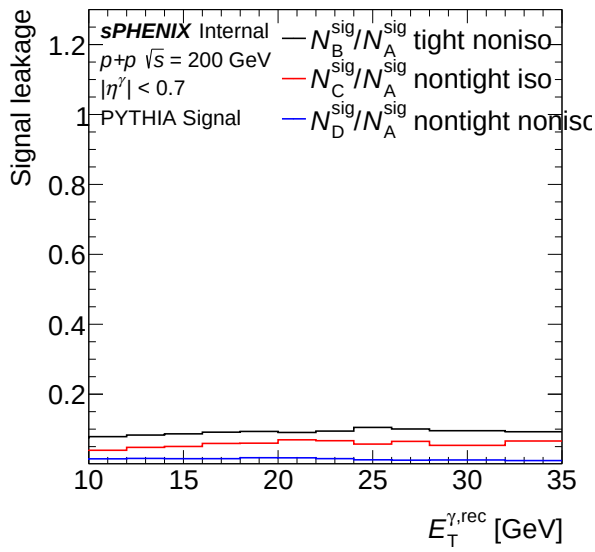
(a) Sideband ET ratio (data) — bdt_nom



(b) Sideband ET ratio (data) — bdt_npb03

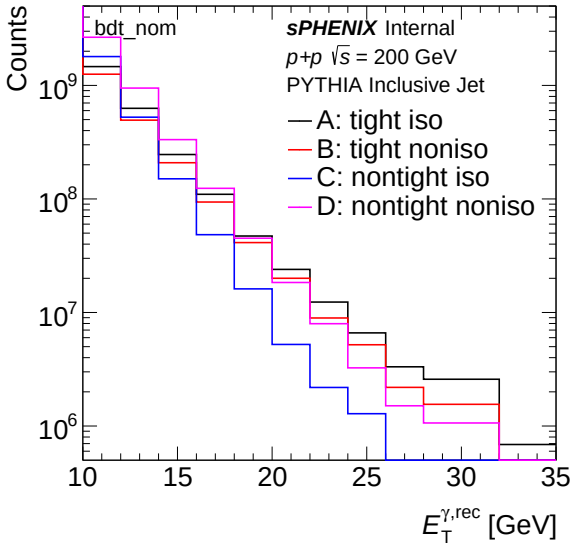


(a) Leakage fraction vs ET (data) — bdt_nom

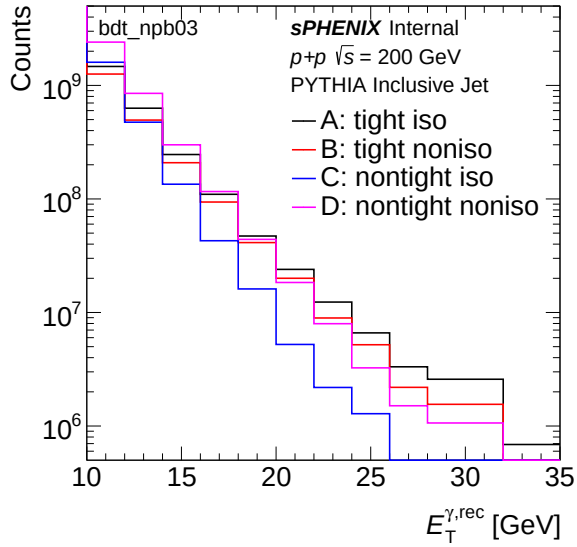


(b) Leakage fraction vs ET (data) — bdt_npb03

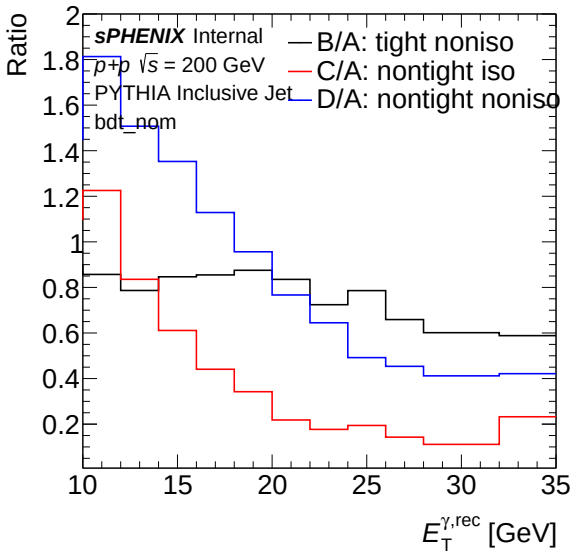
2.2 Sideband (MC)



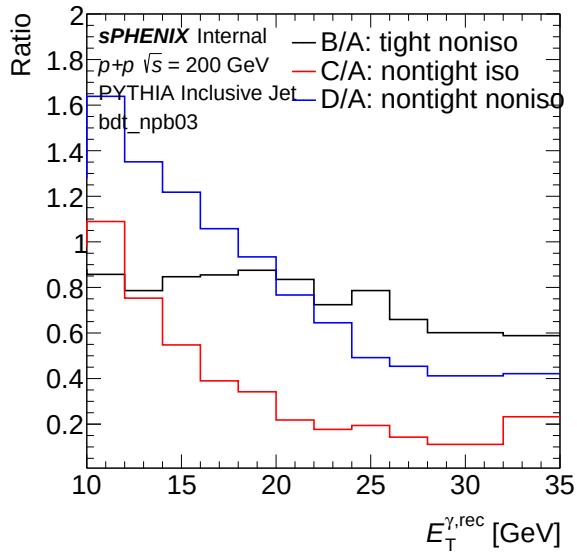
(a) Sideband ET spectrum (MC) — bdt_nom



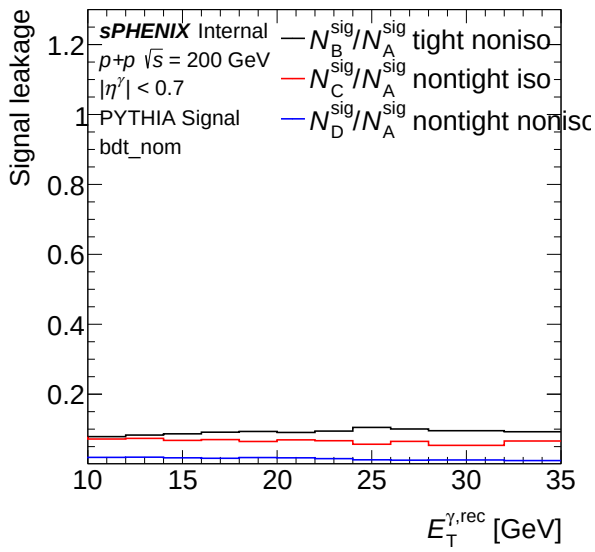
(b) Sideband ET spectrum (MC) — bdt_npb03



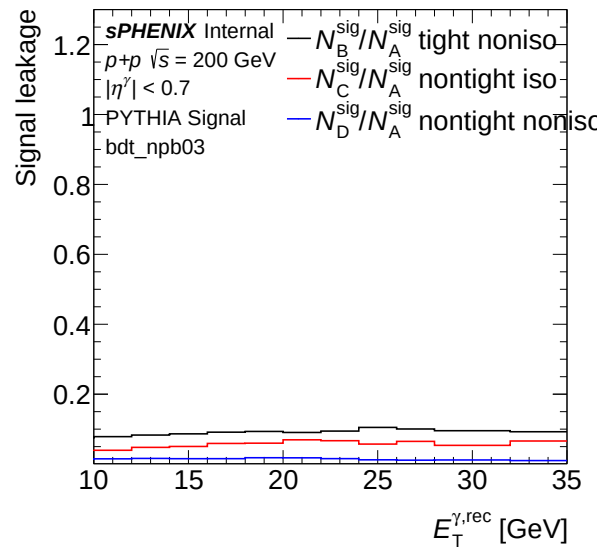
(a) Sideband ET ratio (MC) — bdt_nom



(b) Sideband ET ratio (MC) — bdt_npb03

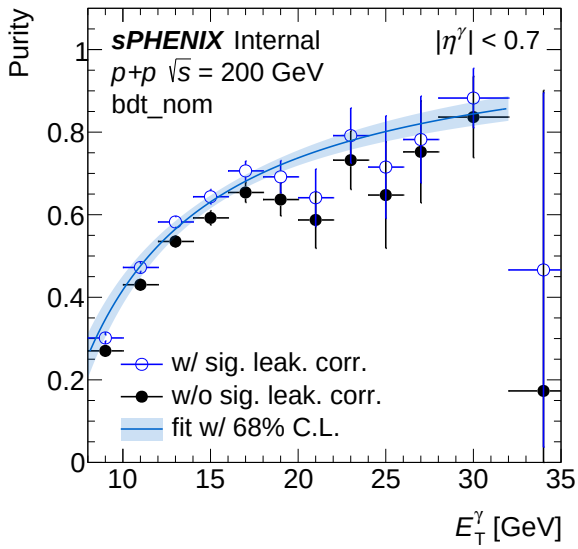


(a) Leakage fraction vs ET (MC) — bdt_nom

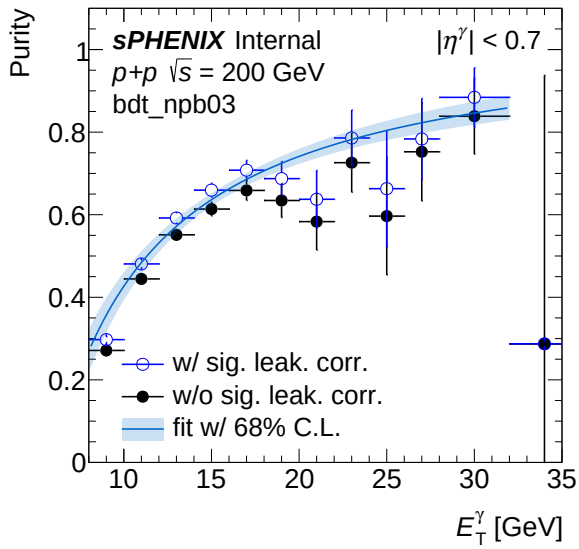


(b) Leakage fraction vs ET (MC) — bdt_npb03

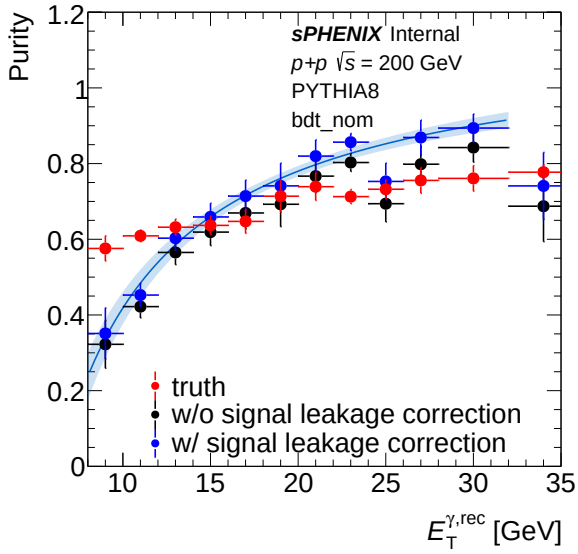
2.3 Purity



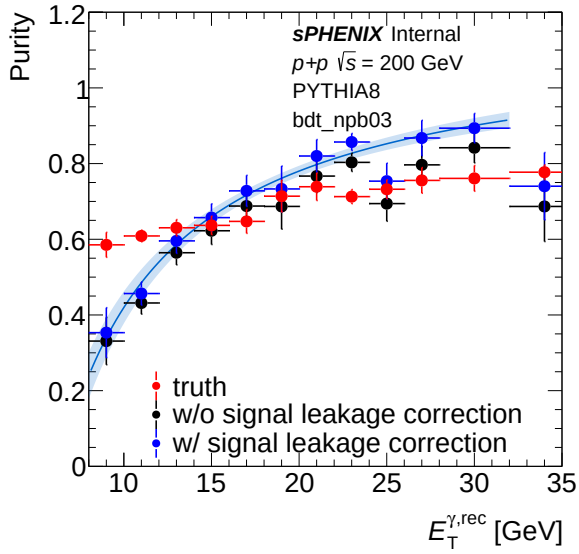
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_npb03

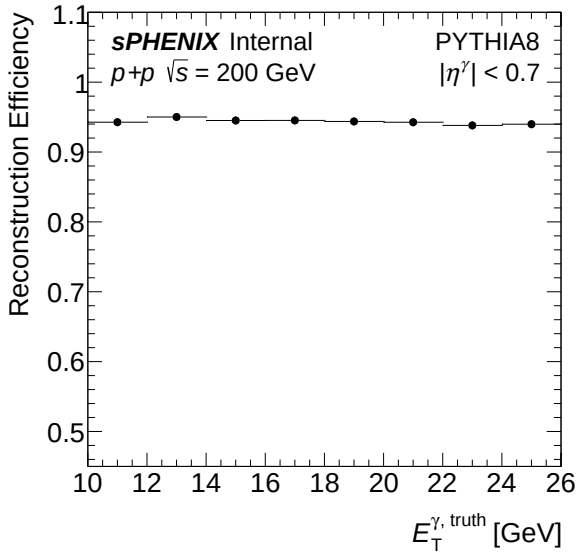


(a) Purity (MC) — bdt_nom

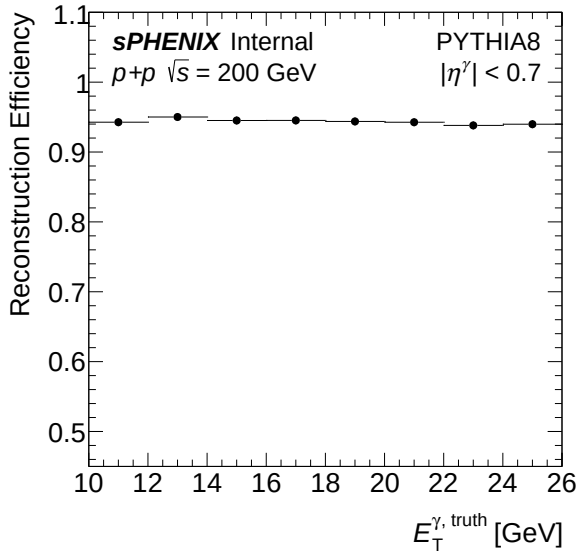


(b) Purity (MC) — bdt_npb03

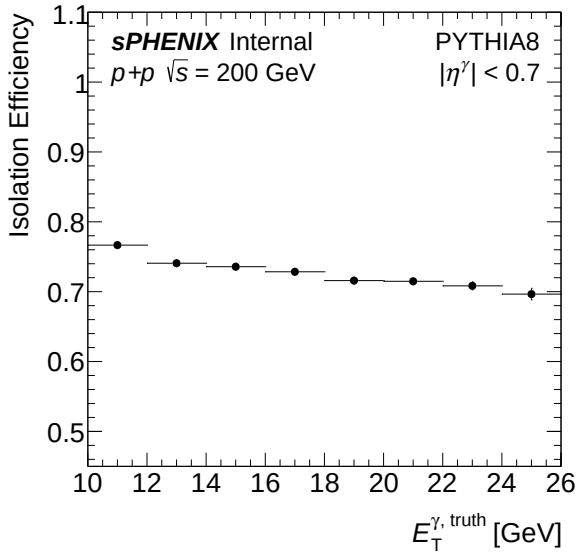
2.4 Efficiency



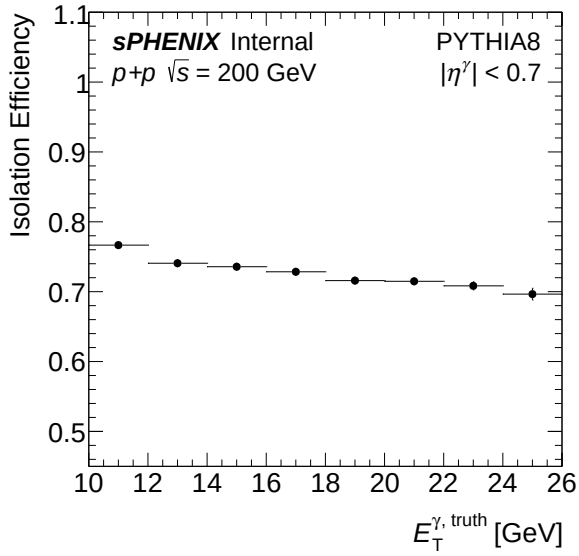
(a) Reconstruction efficiency — bdt_nom



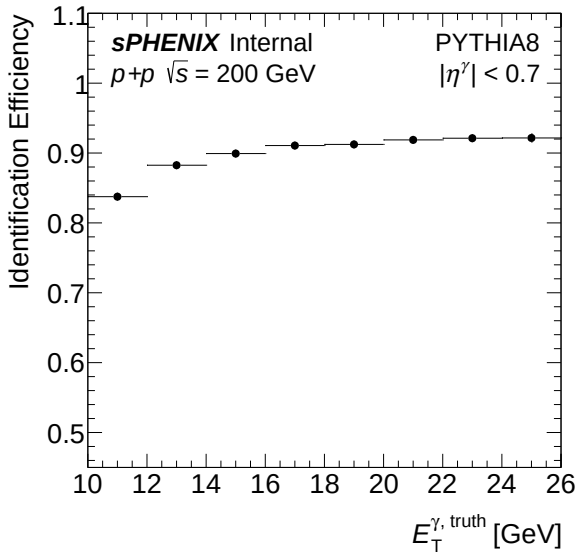
(b) Reconstruction efficiency — bdt_npb03



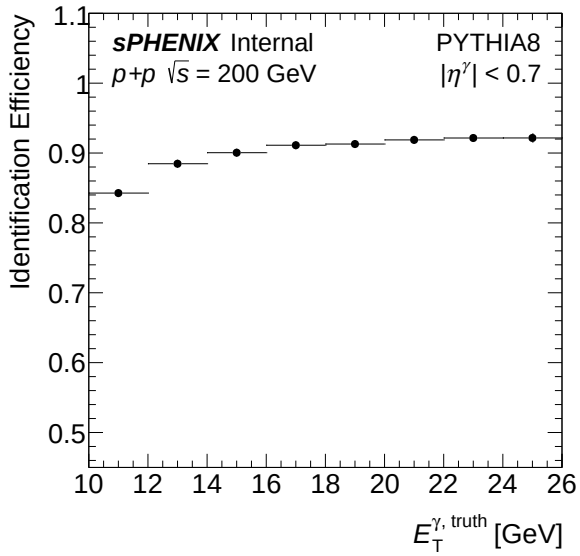
(a) Isolation efficiency — bdt_nom



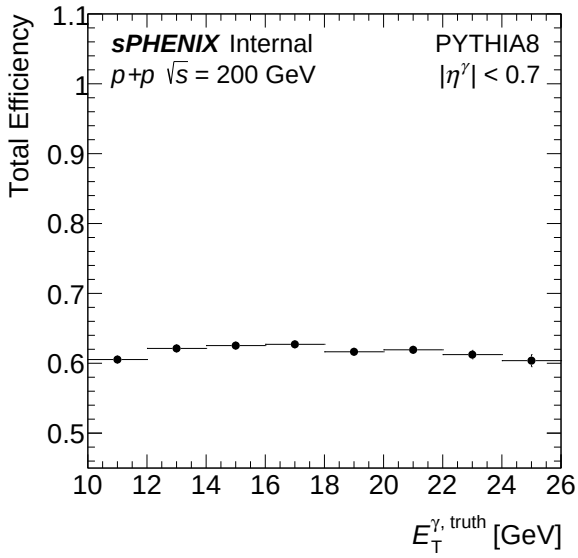
(b) Isolation efficiency — bdt_npb03



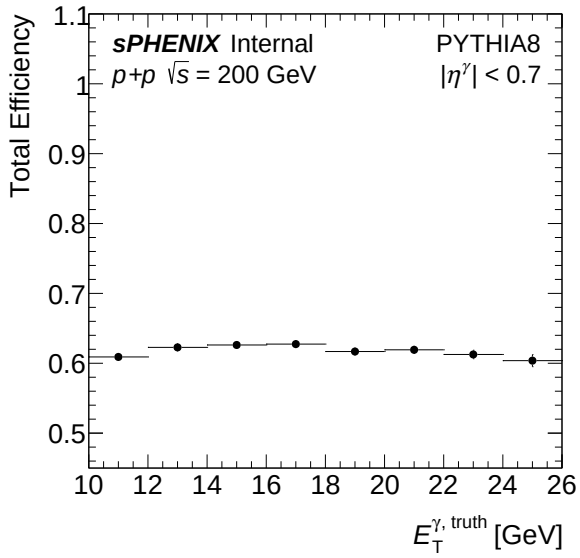
(a) Identification efficiency — bdt_nom



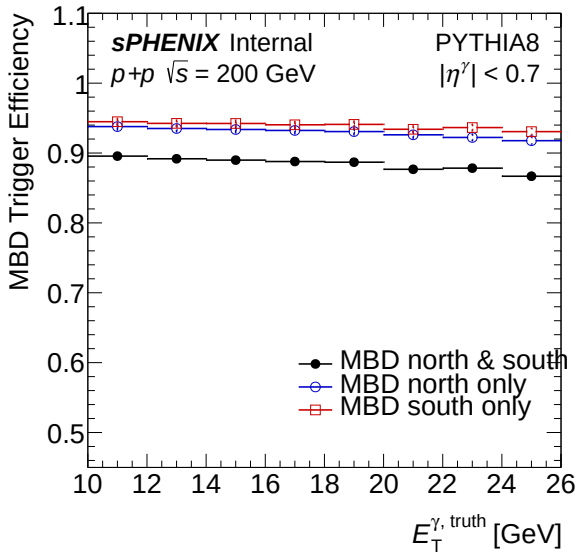
(b) Identification efficiency — bdt_npb03



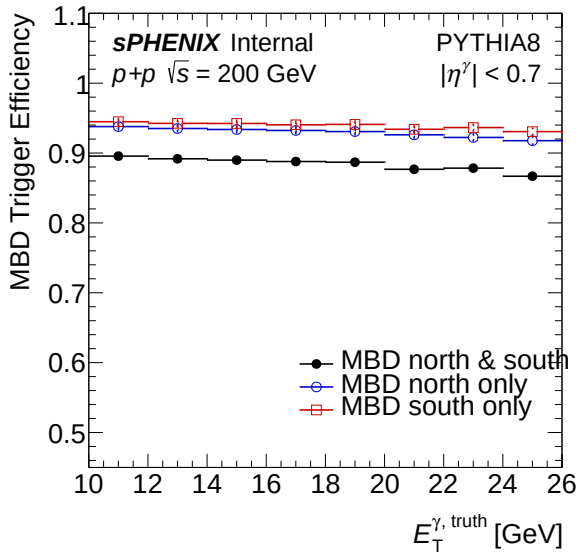
(a) Total efficiency — bdt_nom



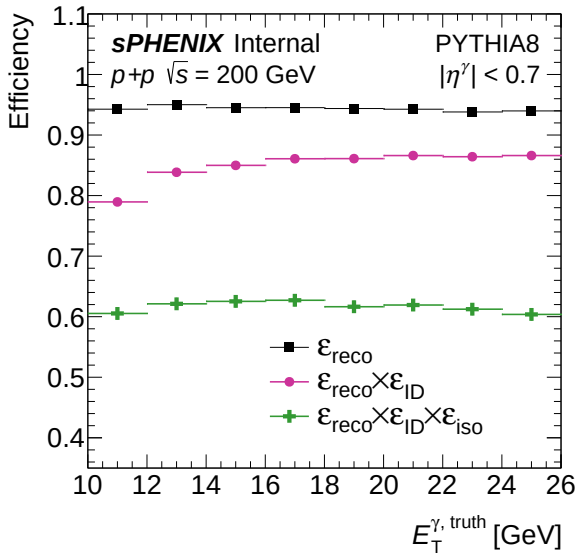
(b) Total efficiency — bdt_npb03



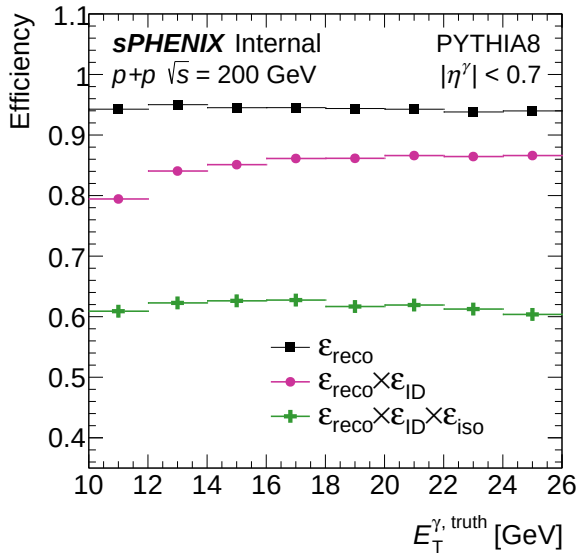
(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_npb03

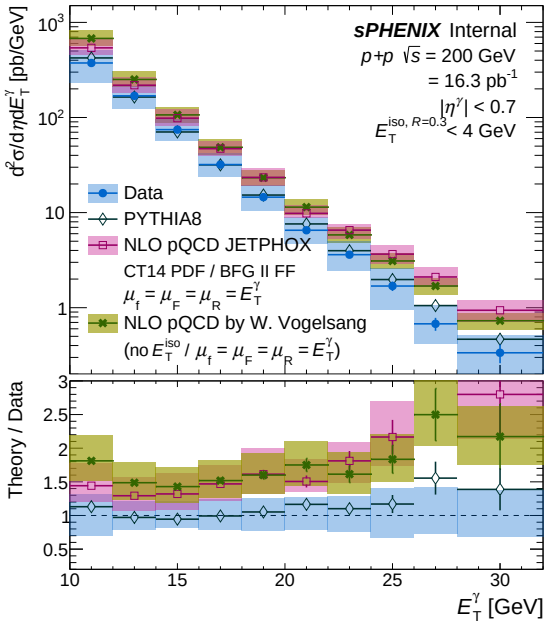


(a) Combined photon efficiency — bdt_nom

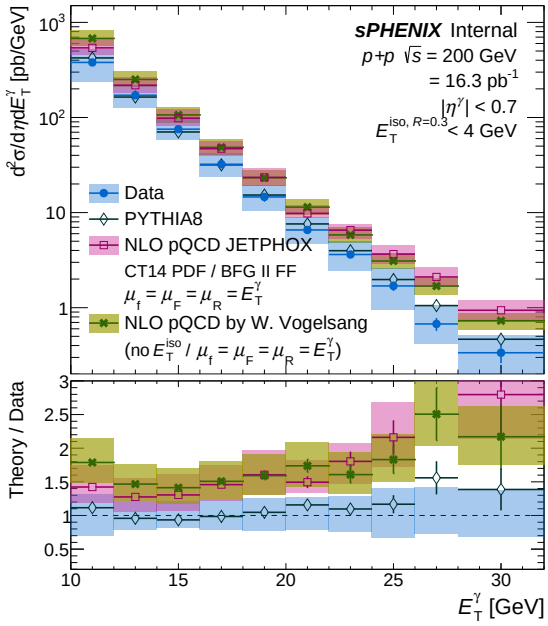


(b) Combined photon efficiency — bdt_npb03

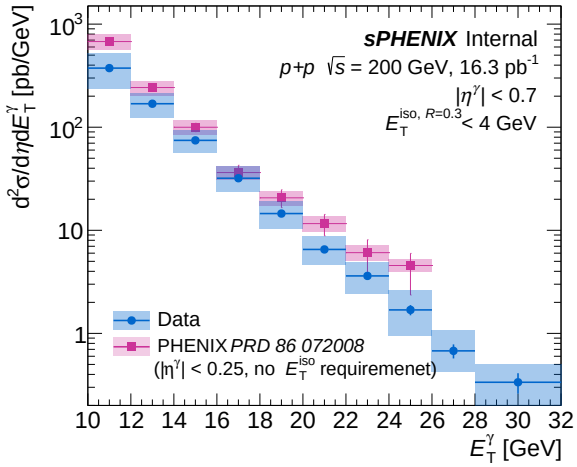
2.5 Final result



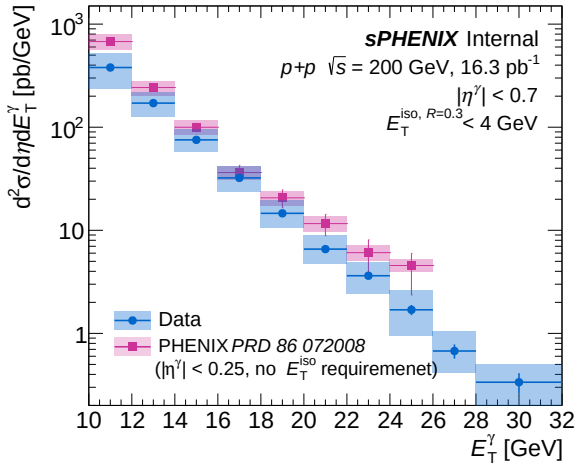
(a) Final cross section result — bdt_nom



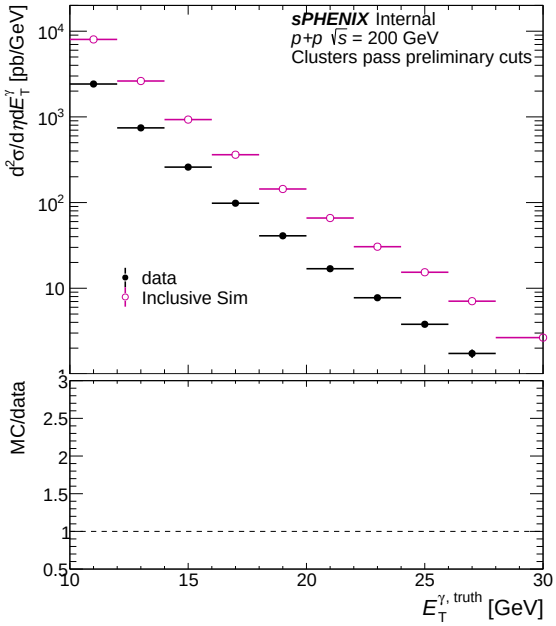
(b) Final cross section result — bdt_npb03



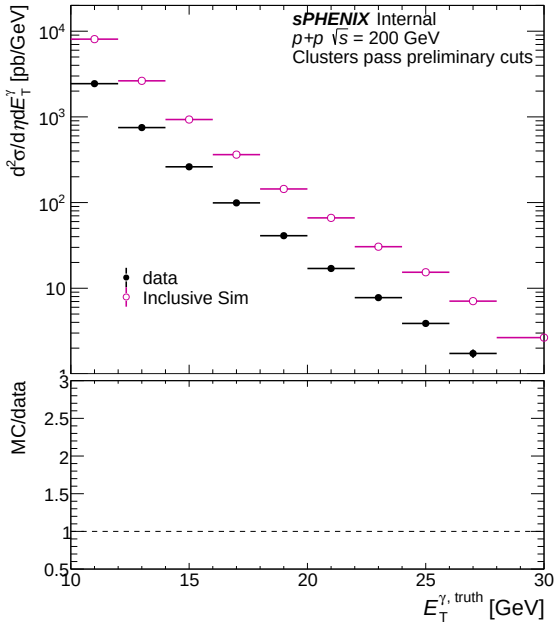
(a) Comparison with PHENIX — bdt_nom



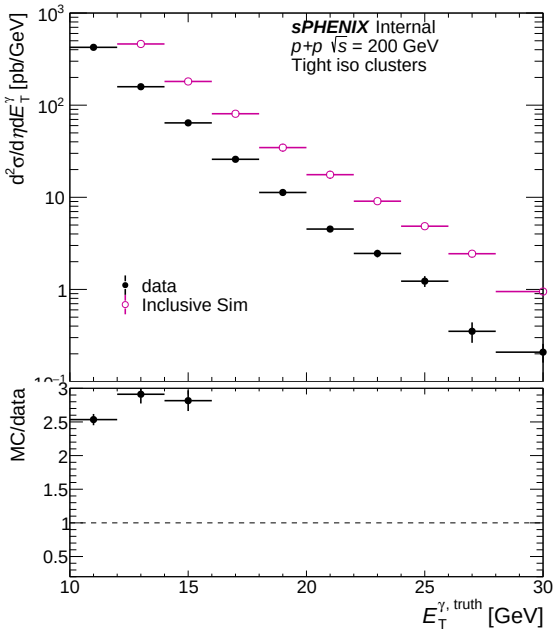
(b) Comparison with PHENIX — bdt_npb03



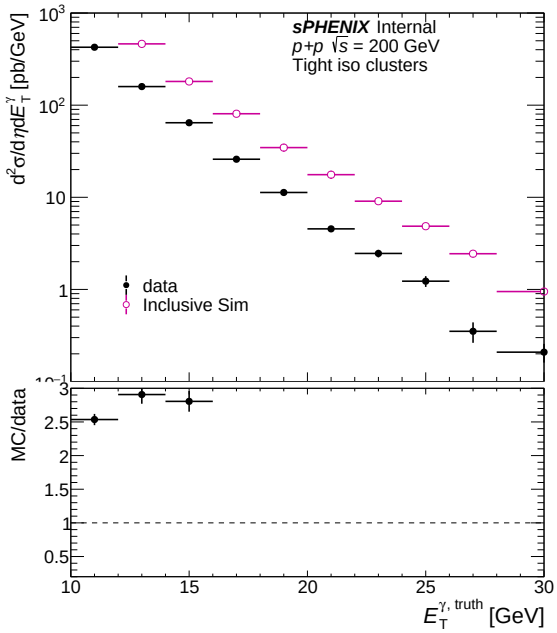
(a) Common cluster yield (data vs MC) — bdt_nom



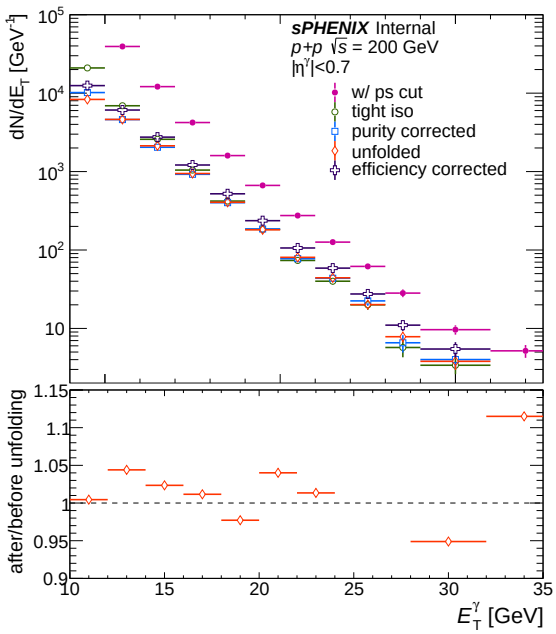
(b) Common cluster yield (data vs MC) — bdt_npb03



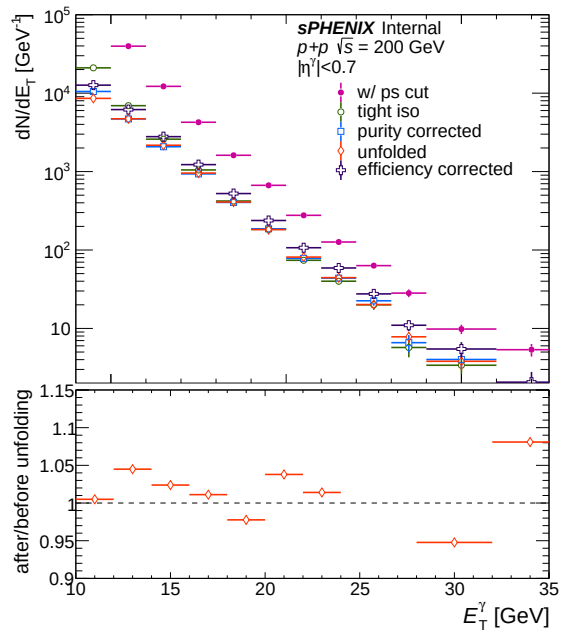
(a) Tight iso cluster yield (data vs MC) — bdt_nom



(b) Tight iso cluster yield (data vs MC) — bdt_npb03

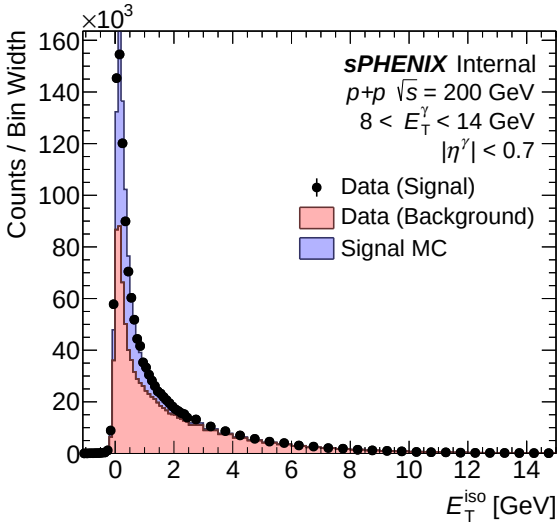


(a) All correction steps — bdt_nom

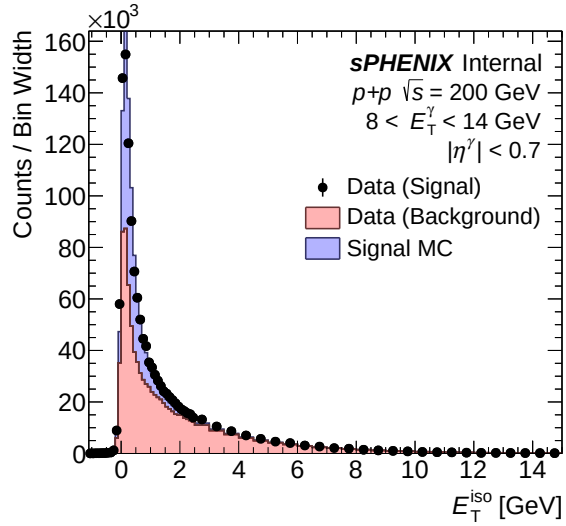


(b) All correction steps — bdt_npb03

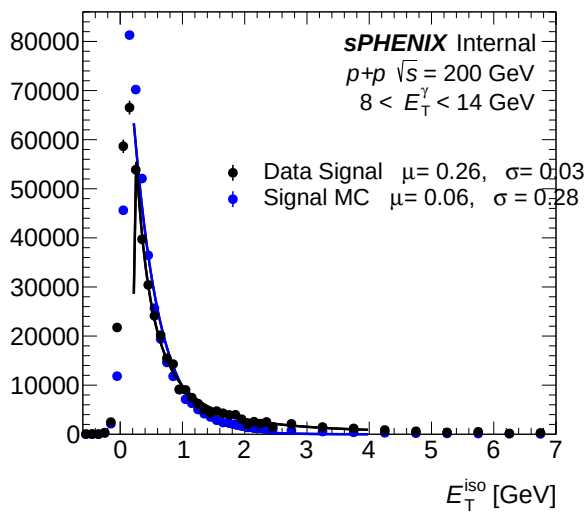
2.6 Isolation fit (CONF)



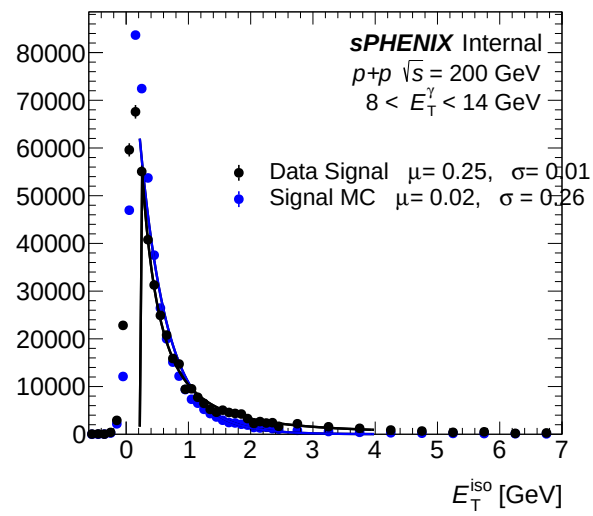
(a) IsoET template (pT bins 0–2) — bdt_nom



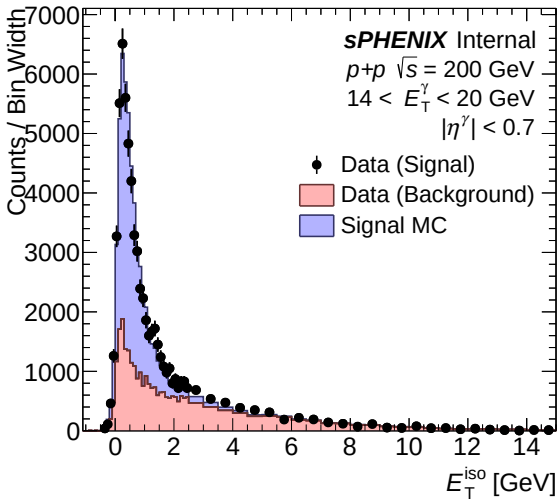
(b) IsoET template (pT bins 0–2) — bdt_npb03



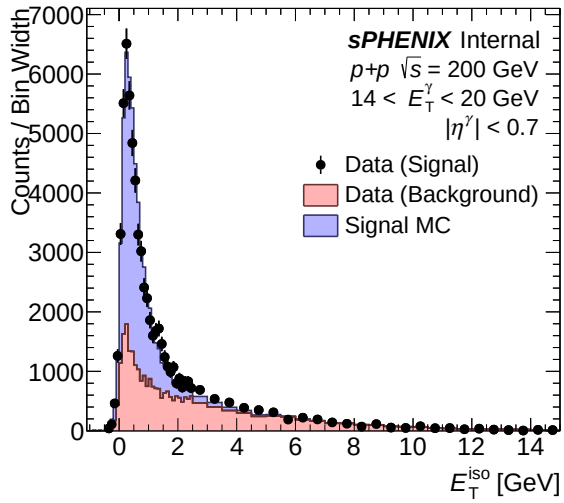
(a) IsoET fit template (pT bins 0–2) — bdt_nom



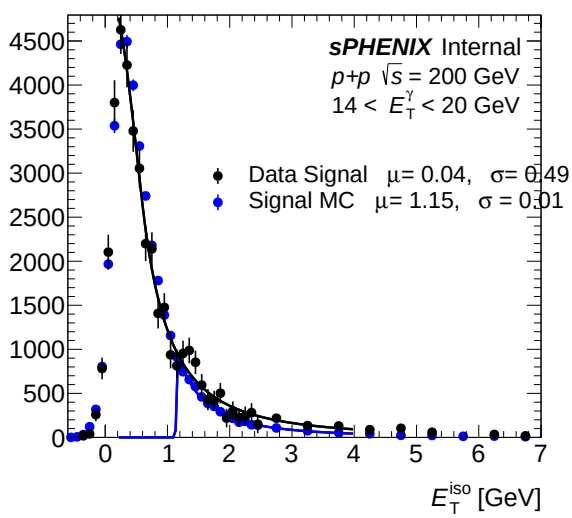
(b) IsoET fit template (pT bins 0–2) — bdt_npb03



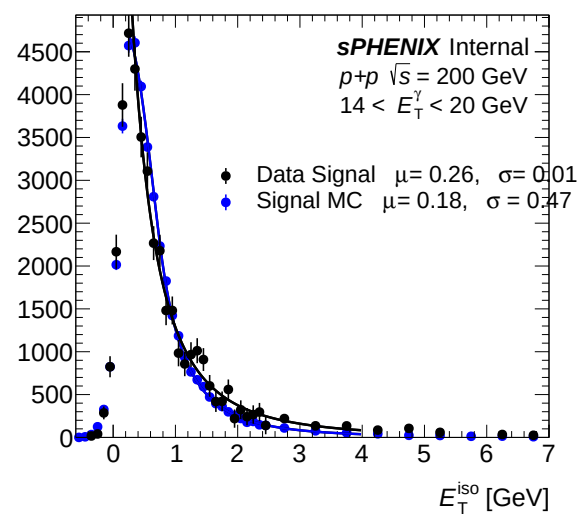
(a) IsoET template (pT bins 3–5) — bdt_nom



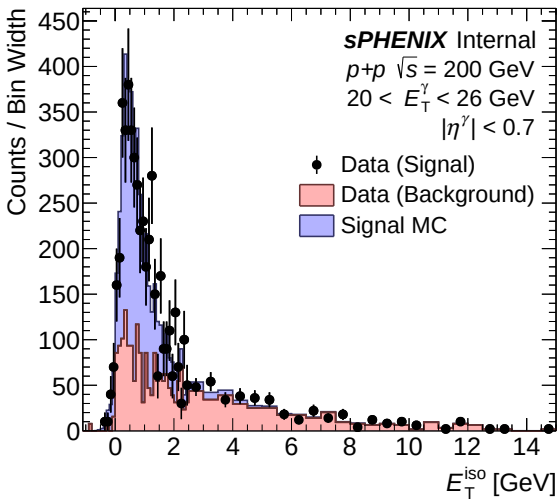
(b) IsoET template (pT bins 3–5) — bdt_npb03



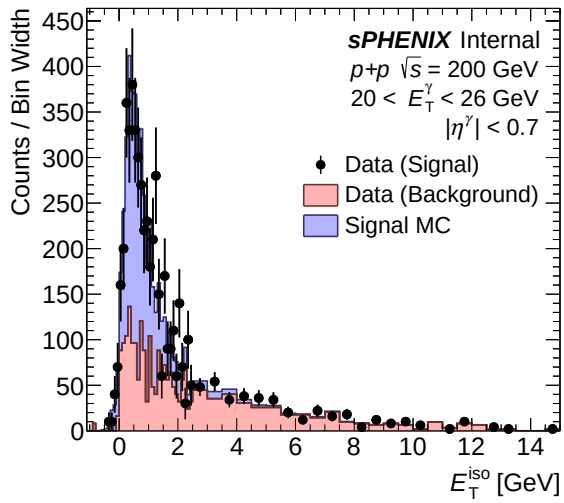
(a) IsoET fit template (pT bins 3–5) — bdt_nom



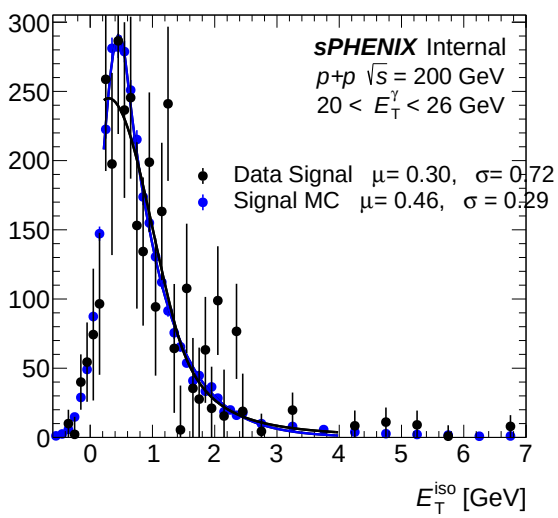
(b) IsoET fit template (pT bins 3–5) — bdt_npb03



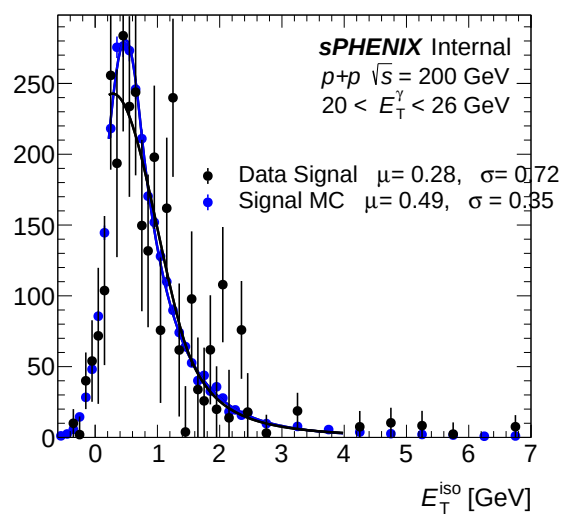
(a) IsoET template (pT bins 6–8) — bdt_nom



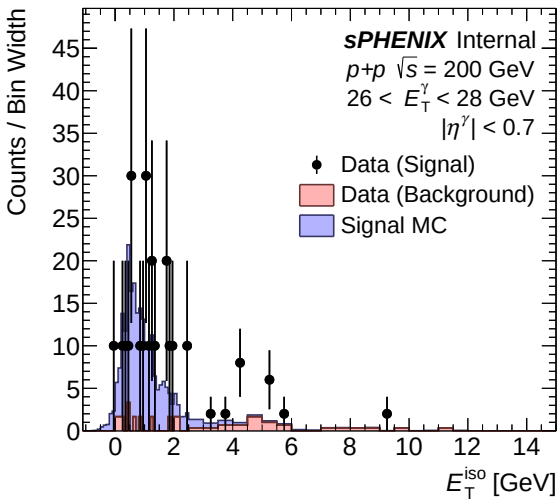
(b) IsoET template (pT bins 6–8) — bdt_npb03



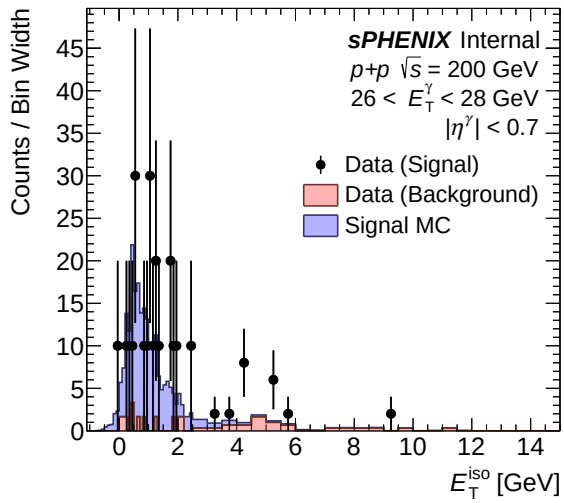
(a) IsoET fit template (pT bins 6–8) — bdt_nom



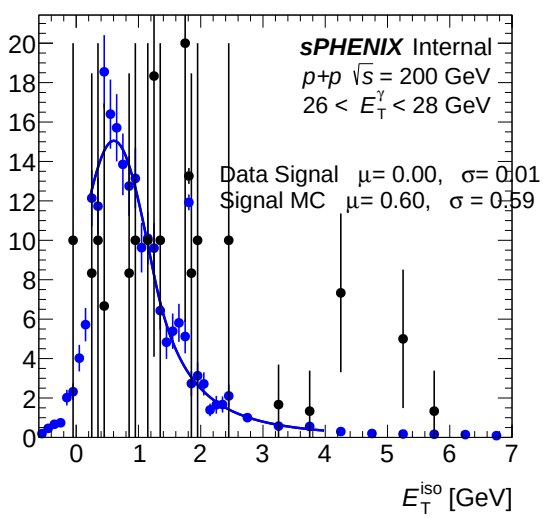
(b) IsoET fit template (pT bins 6–8) — bdt_npb03



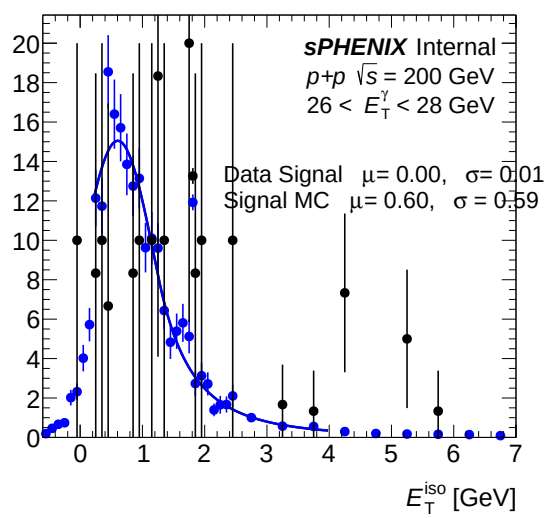
(a) IsoET template (pT bins 9–9) — bdt_nom



(b) IsoET template (pT bins 9–9) — bdt_npb03

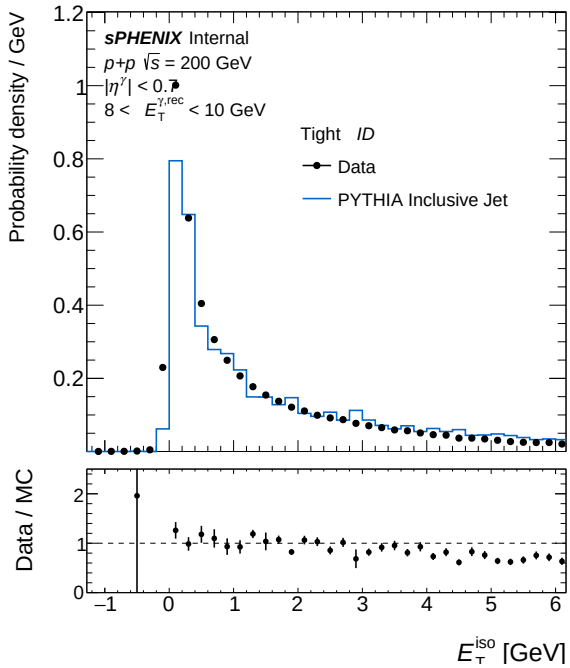


(a) IsoET fit template (pT bins 9–9) — bdt_nom

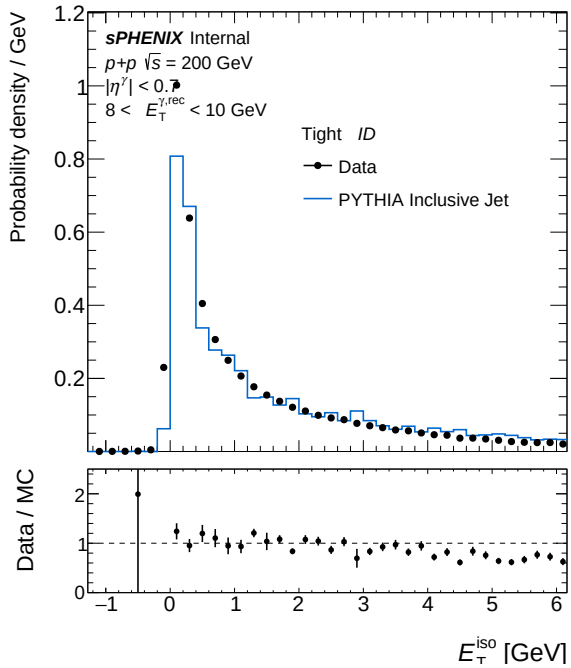


(b) IsoET fit template (pT bins 9–9) — bdt_npb03

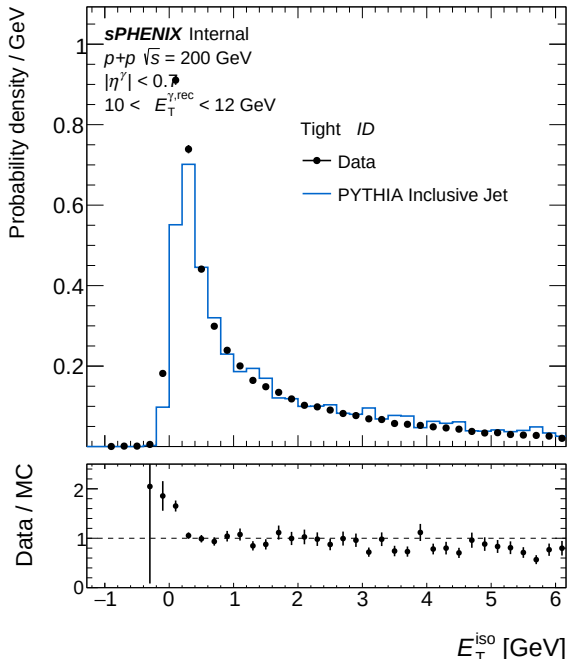
2.7 Isolation ET



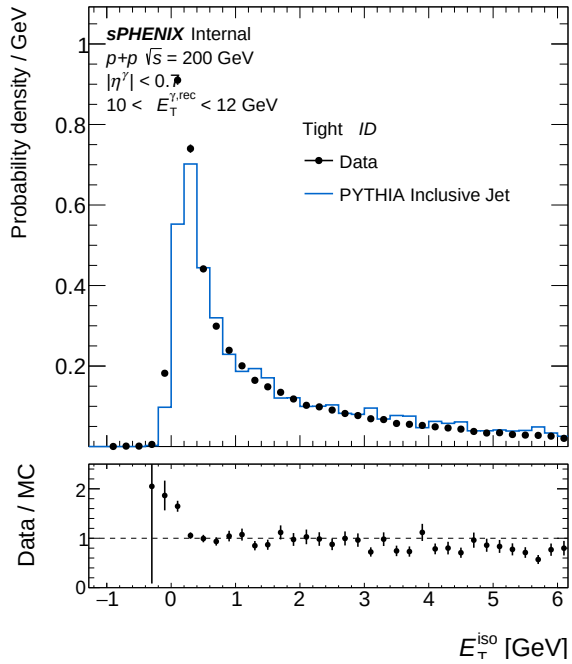
(a) IsoET Tight, pT bin 0 — bdt_nom



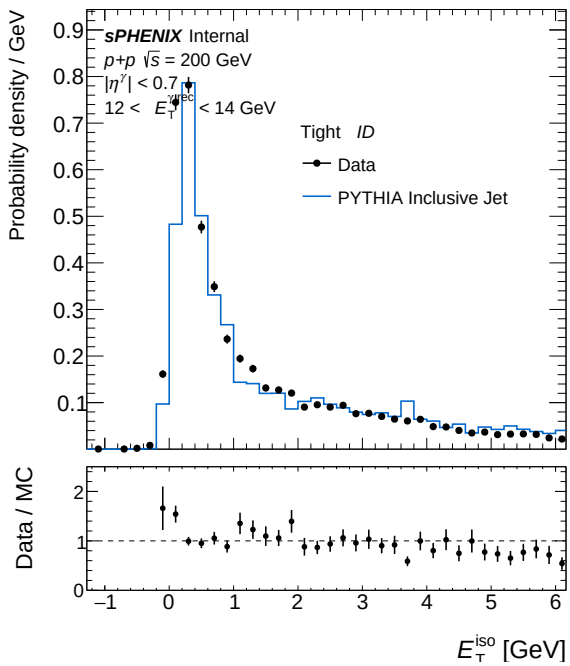
(b) IsoET Tight, pT bin 0 — bdt_npb03



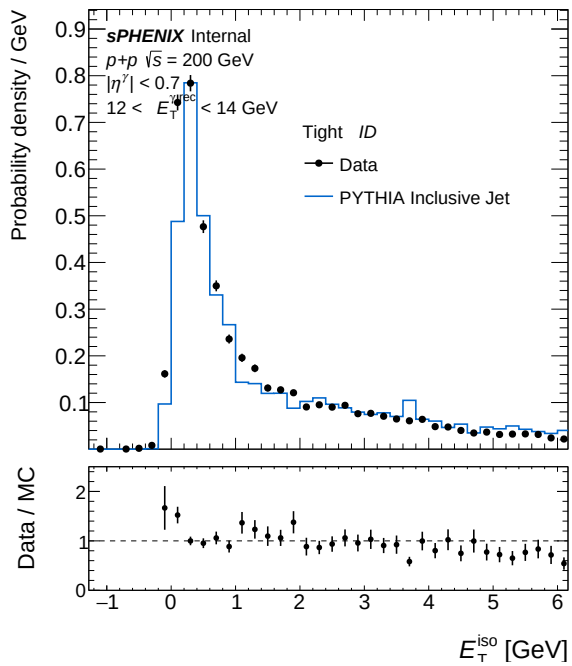
(a) IsoET Tight, pT bin 1 — bdt_nom



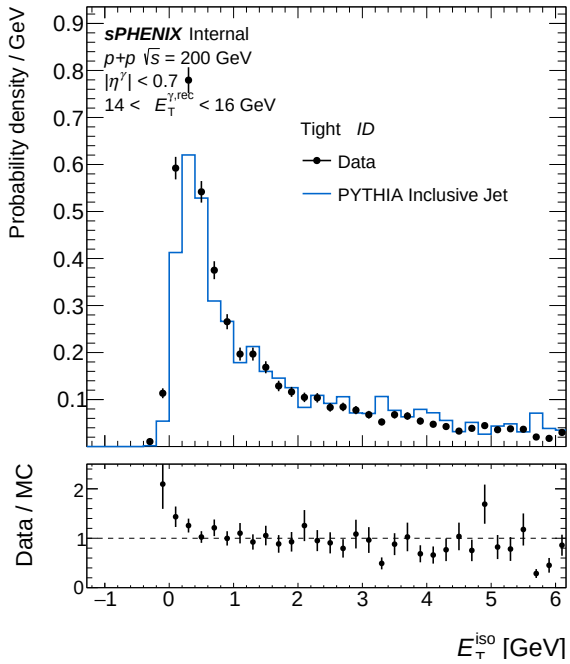
(b) IsoET Tight, pT bin 1 — bdt_npb03



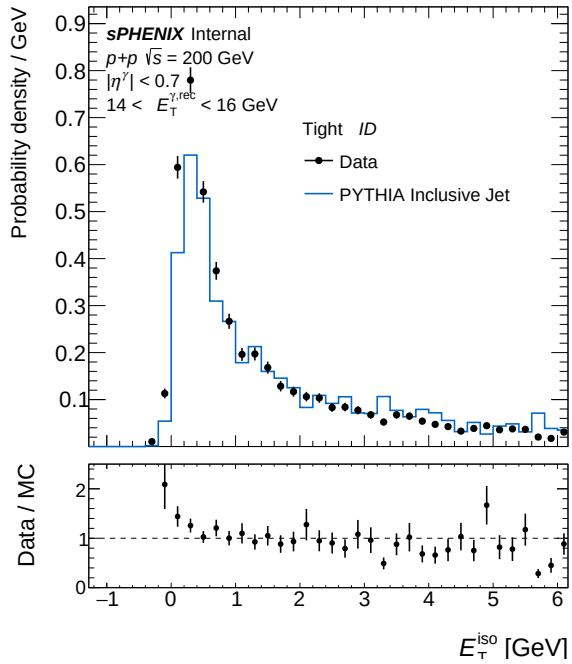
(a) IsoET Tight, pT bin 2 — bdt_nom



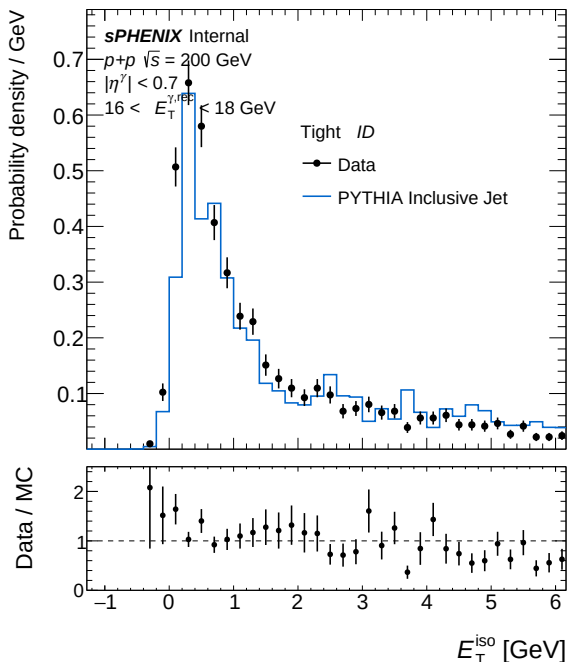
(b) IsoET Tight, pT bin 2 — bdt_npb03



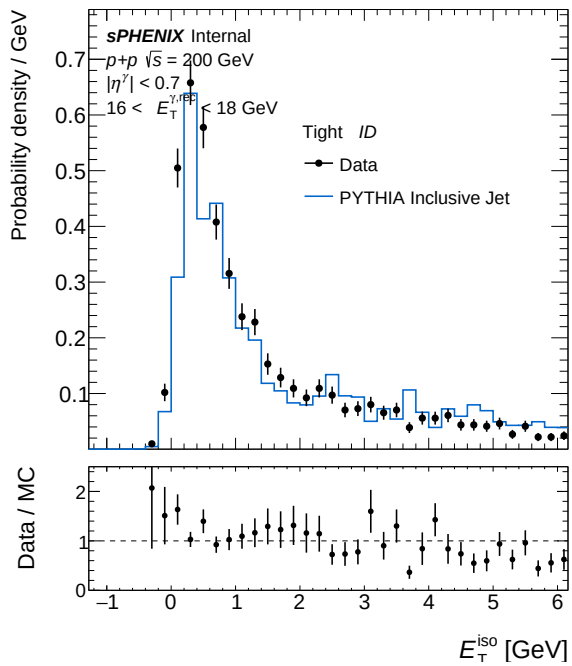
(a) IsoET Tight, pT bin 3 — bdt_nom



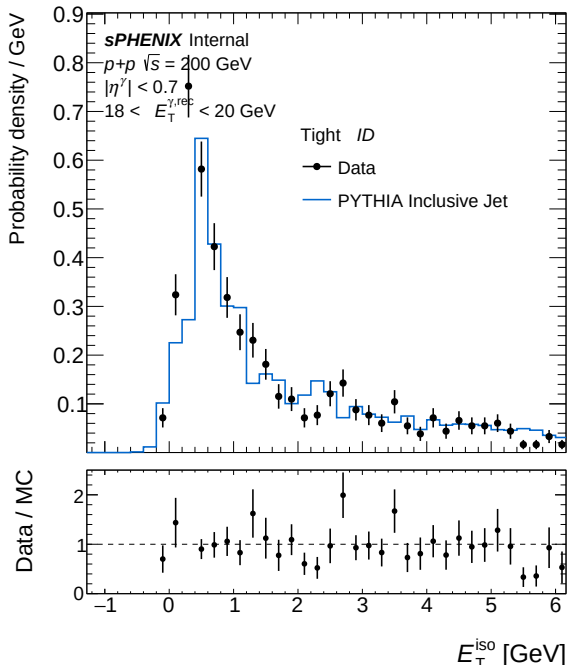
(b) IsoET Tight, pT bin 3 — bdt_npb03



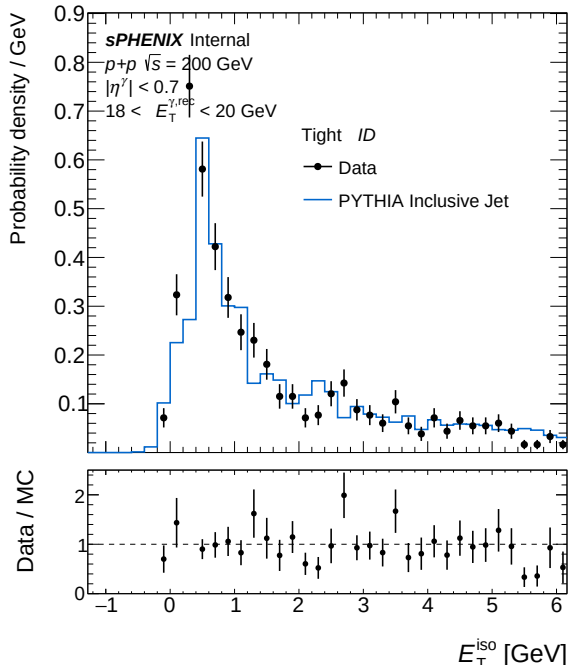
(a) IsoET Tight, pT bin 4 — bdt_nom



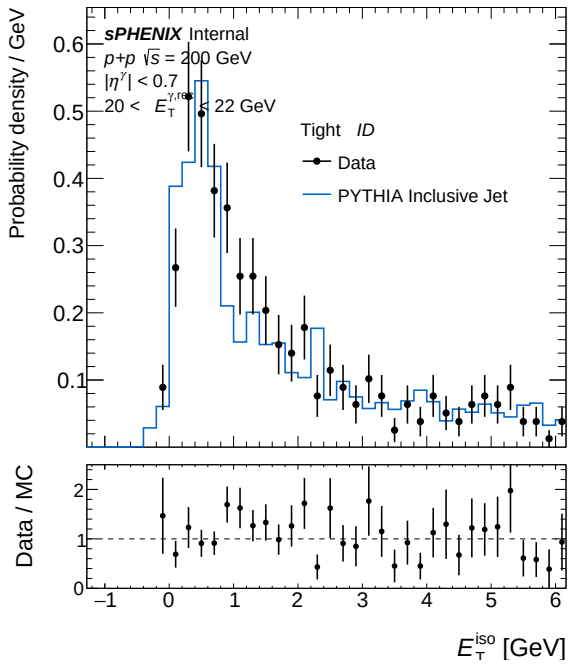
(b) IsoET Tight, pT bin 4 — bdt_npb03



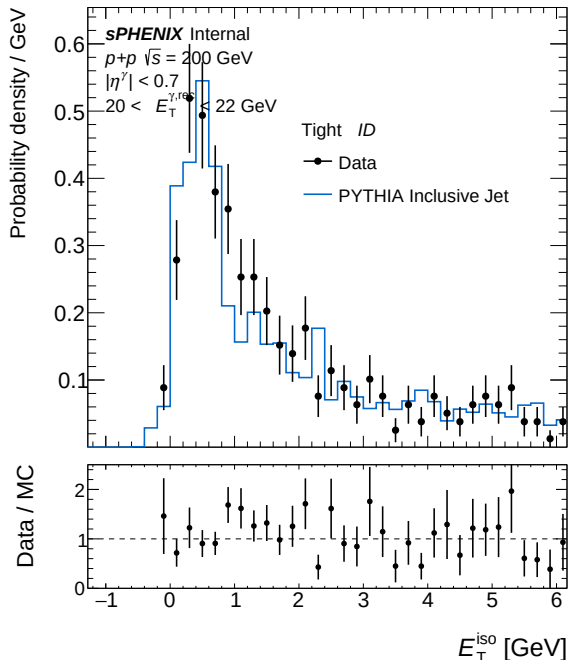
(a) IsoET Tight, pT bin 5 — bdt_nom



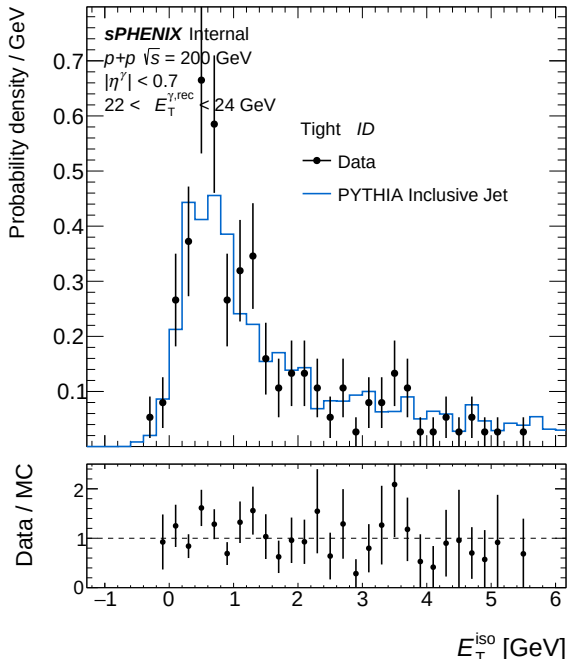
(b) IsoET Tight, pT bin 5 — bdt_npb03



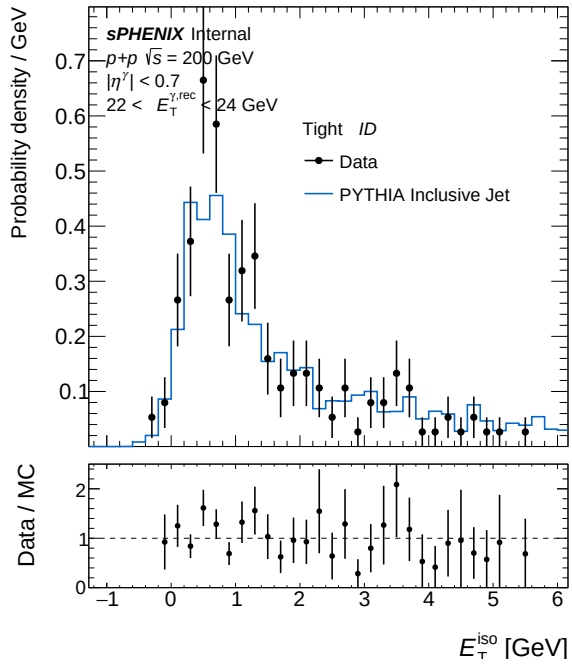
(a) IsoET Tight, pT bin 6 — bdt_nom



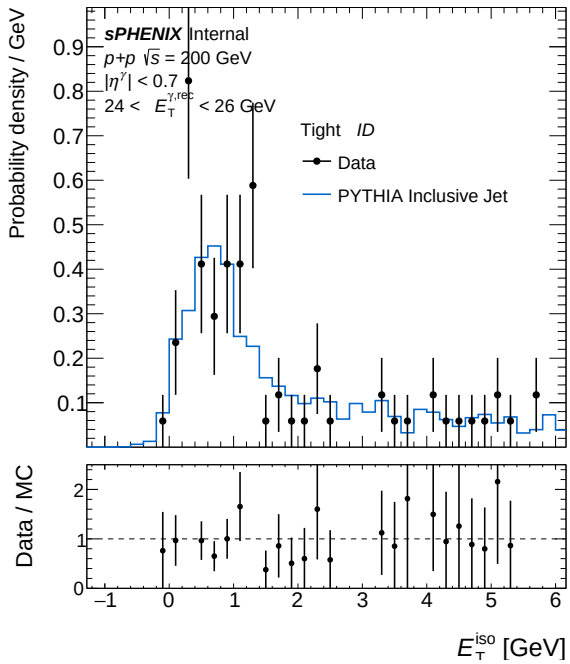
(b) IsoET Tight, pT bin 6 — bdt_npb03



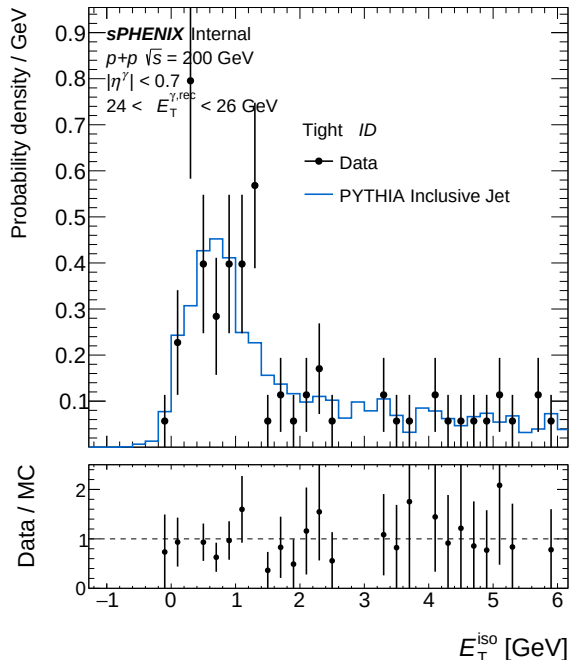
(a) IsoET Tight, pT bin 7 — bdt_nom



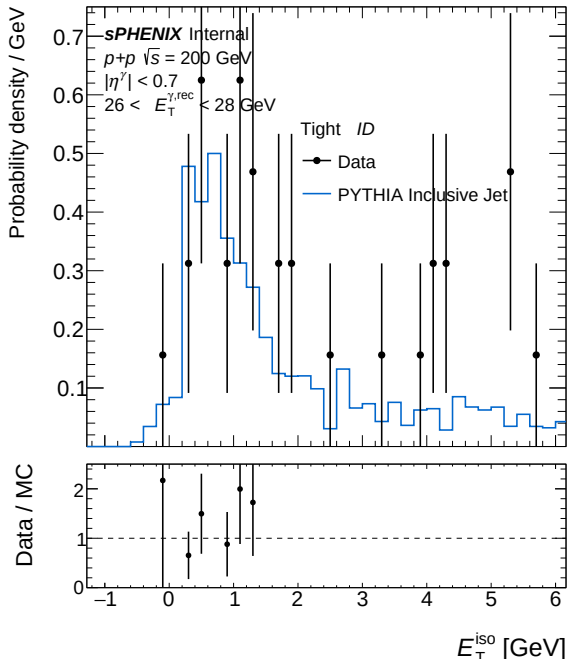
(b) IsoET Tight, pT bin 7 — bdt_npb03



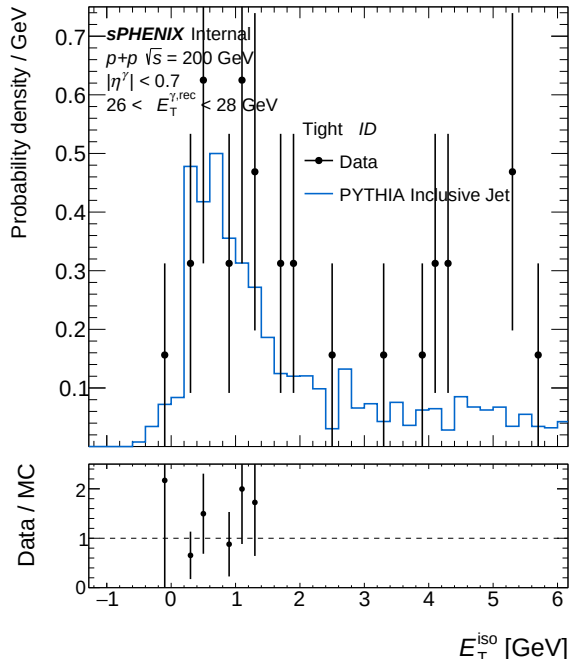
(a) IsoET Tight, pT bin 8 — bdt_nom



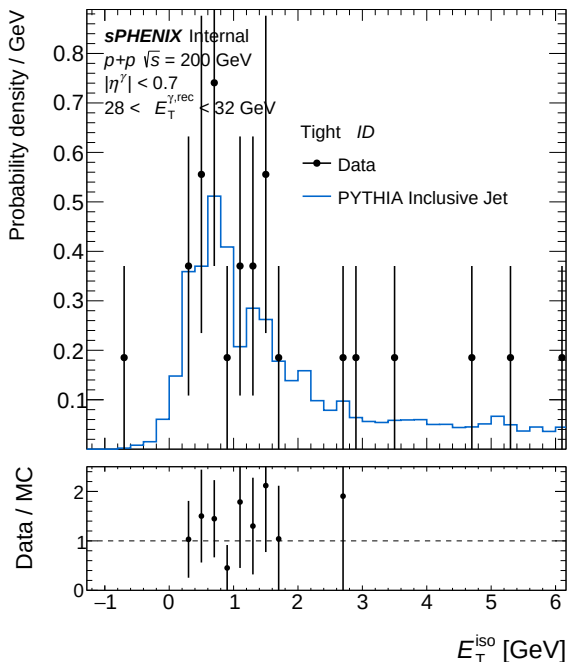
(b) IsoET Tight, pT bin 8 — bdt_npb03



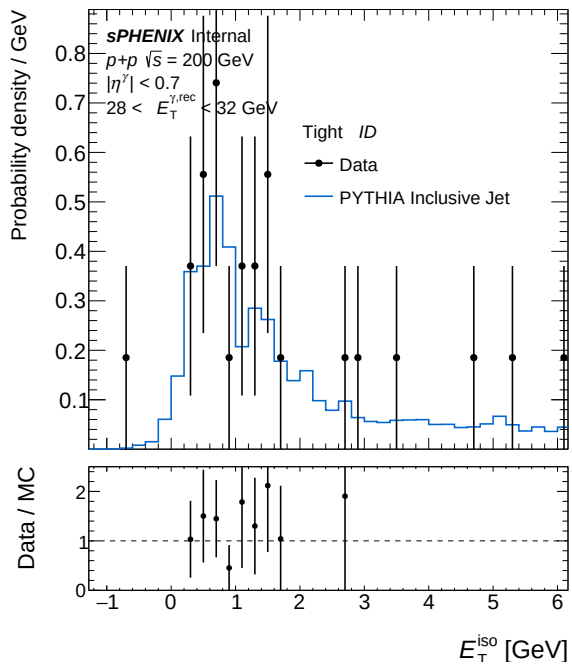
(a) IsoET Tight, pT bin 9 — bdt_nom



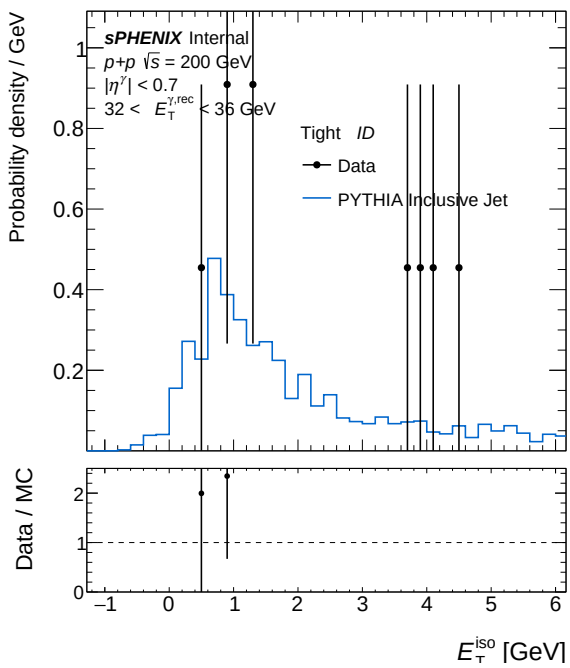
(b) IsoET Tight, pT bin 9 — bdt_npb03



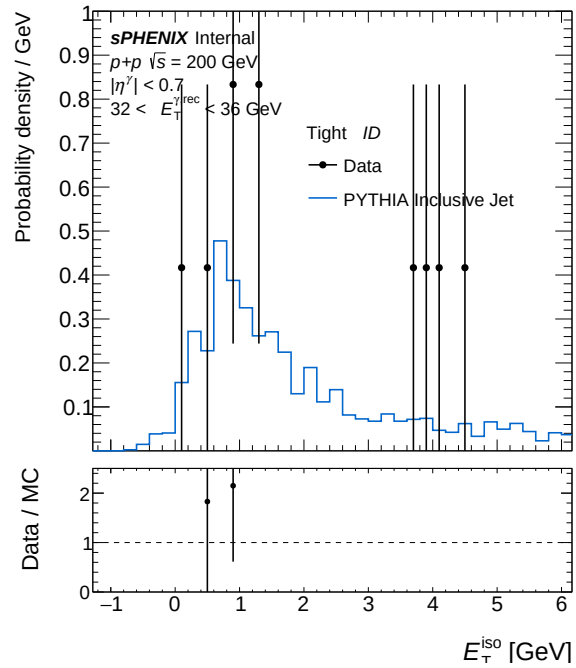
(a) IsoET Tight, pT bin 10 — bdt_nom



(b) IsoET Tight, pT bin 10 — bdt_npb03



(a) IsoET Tight, pT bin 11 — bdt_nom



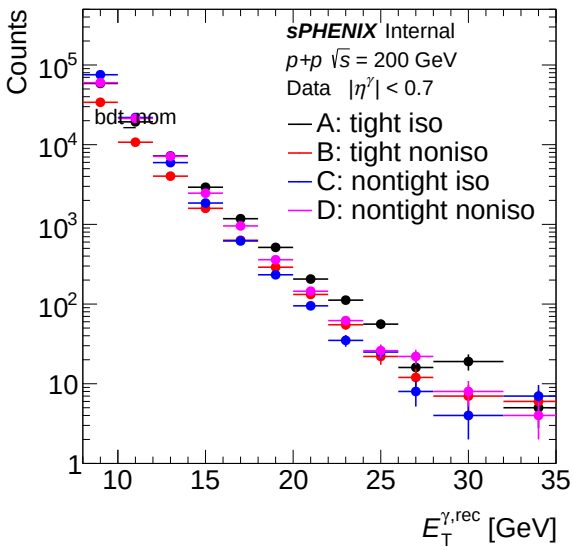
(b) IsoET Tight, pT bin 11 — bdt_npb03

3 bdt_nom vs bdt_nosplit

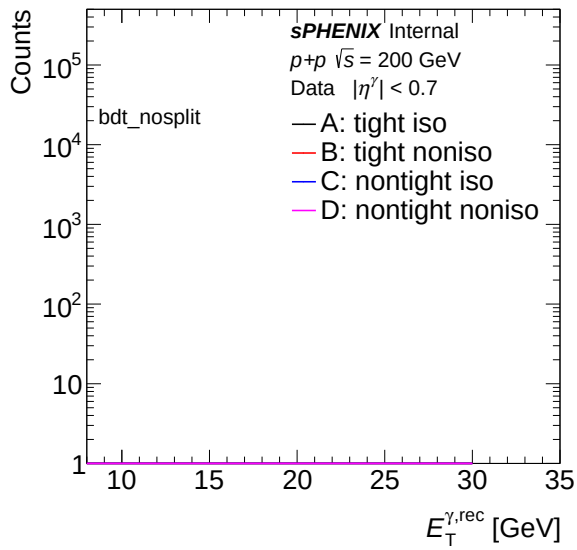
Config diff (A → B)

```diff
--- config_bdt_nom.yaml
+++ config_bdt_nosplit.yaml
@@ -7,5 +7,5 @@
   data_file: "/sphenix/user/shuhangli/ppg12/anatreemaker/macro_maketree/data/ana521/condorout/part_*_wit
   tree: "slimtree"
-  cluster_node_name: "CLUSTERINFO_CEMC"
+  cluster_node_name: "CLUSTERINFO_CEMC_NO_SPLIT"
   bdt_model_name: "base_E"
   # Optional: use different BDT models in different ET bins.
@@ -20,5 +20,5 @@
   response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
   data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
+  var_type: "bdt_nosplit"
   final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"

@@ -77,11 +77,4 @@
   # Leave empty ("") to use the pre-made vertex_reweight_file instead.
   vertex_scan_data_file: ""
-  # Data-driven iterative truth-vertex reweight (standalone tool in
-  # efficiencytool/truth_vertex_reweight/). When on, forces vertex_reweight_on
-  # off and applies w(z_h) (single MC) or w(z_h)*w(z_mb) (double MC) per event.
-  # Flip truth_vertex_reweight_on to 1 to switch the nominal 1.5 mrad
-  # analysis from the f(z_r) reco-level reweight to the truth-level one.
-  truth_vertex_reweight_on: 0
-  truth_vertex_reweight_file: "/sphenix/user/shuhangli/ppg12/efficiencytool/truth_vertex_reweight/output
   fit_option: 1
   cluster_escale: 1.0
```

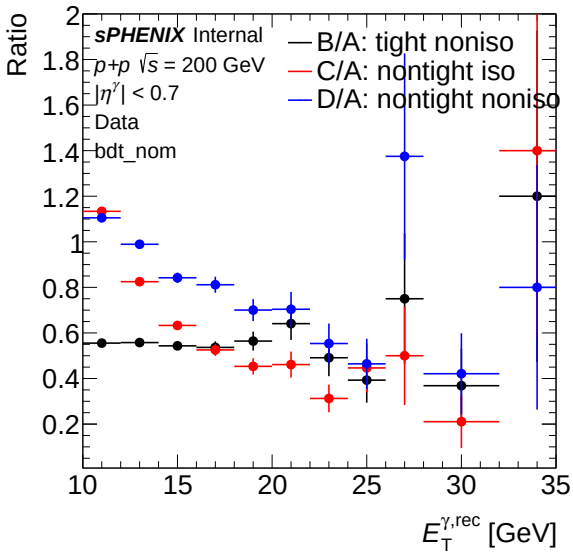
3.1 Sideband (Data)



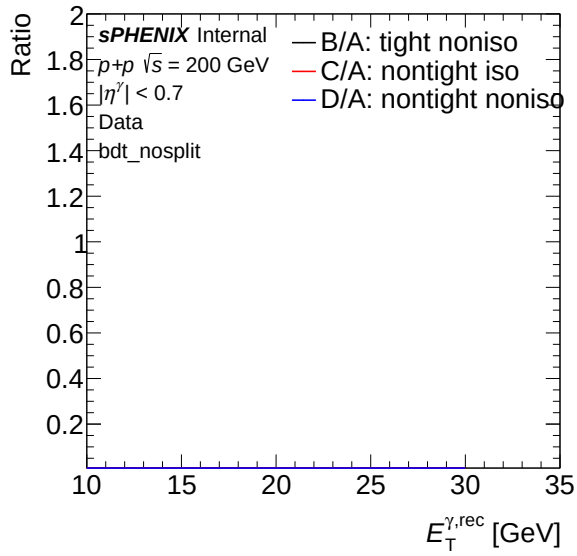
(a) Sideband ET spectrum (data) — bdt_nom



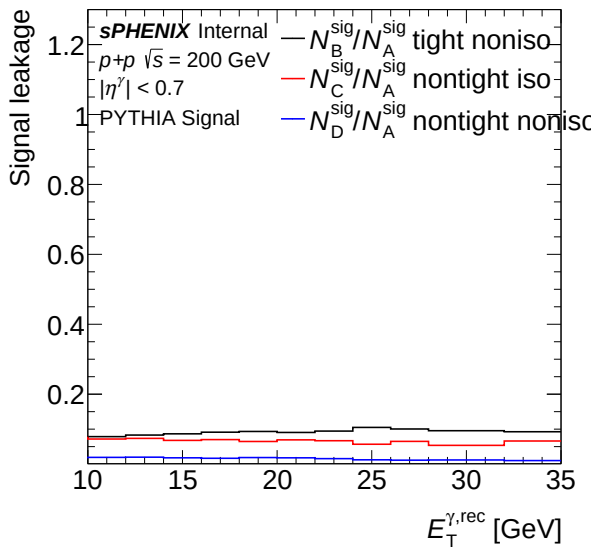
(b) Sideband ET spectrum (data) — bdt_nosplit



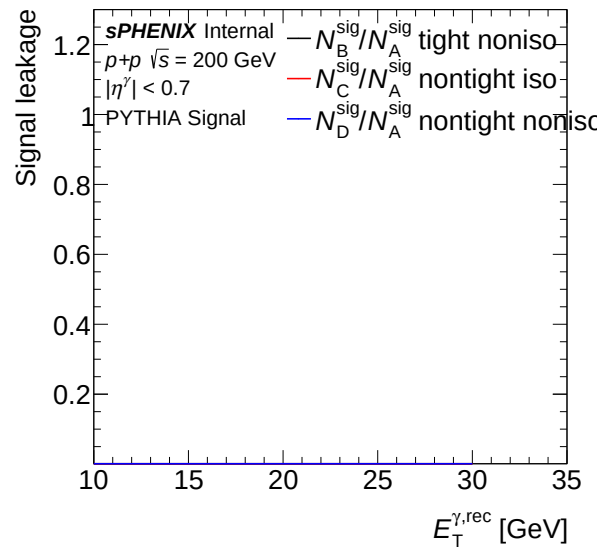
(a) Sideband ET ratio (data) — bdt_nom



(b) Sideband ET ratio (data) — bdt_nosplit

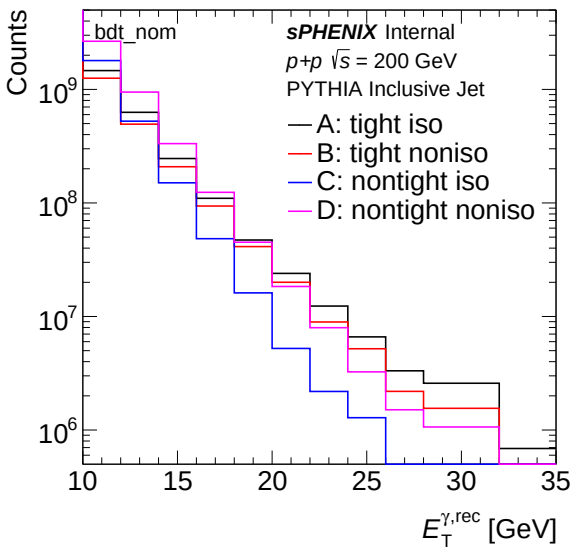


(a) Leakage fraction vs ET (data) — bdt_nom

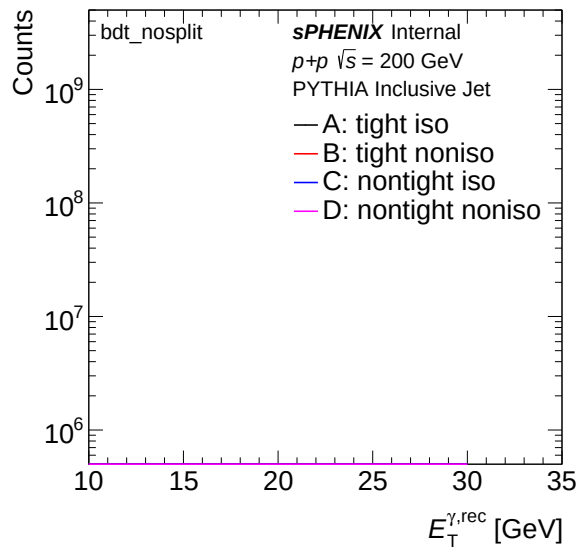


(b) Leakage fraction vs ET (data) — bdt_nosplit

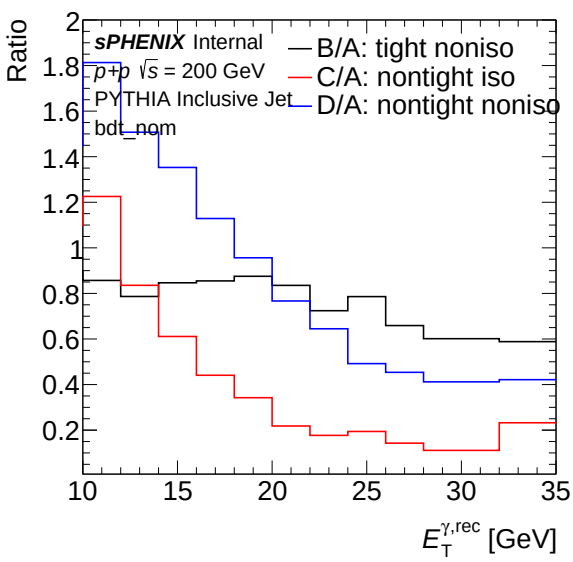
3.2 Sideband (MC)



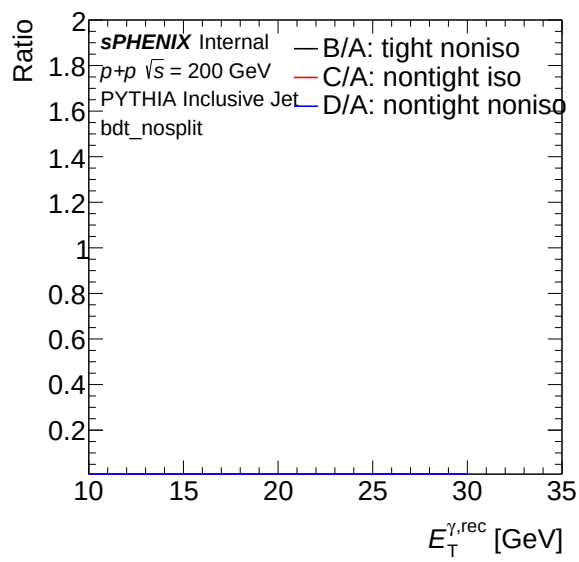
(a) Sideband ET spectrum (MC) — bdt_nom



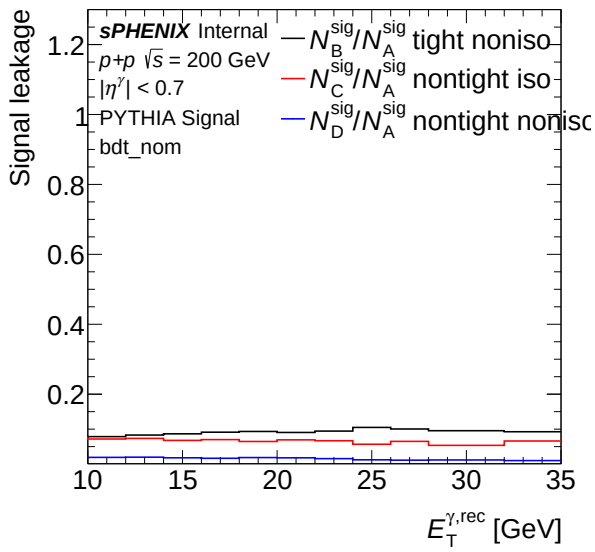
(b) Sideband ET spectrum (MC) — bdt_nosplit



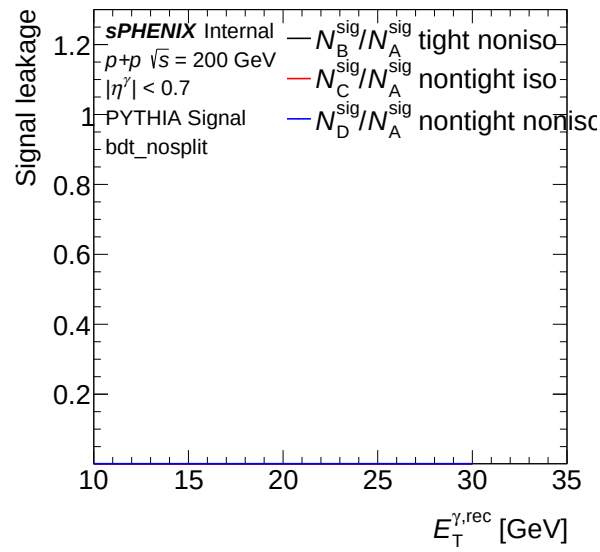
(a) Sideband ET ratio (MC) — bdt_nom



(b) Sideband ET ratio (MC) — bdt_nosplit

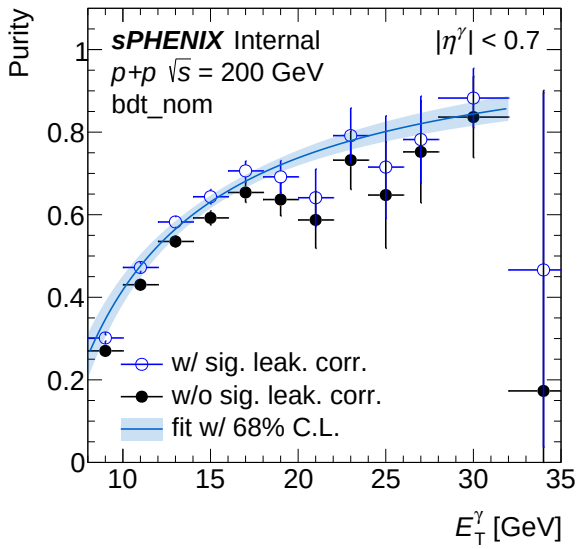


(a) Leakage fraction vs ET (MC) — bdt_nom

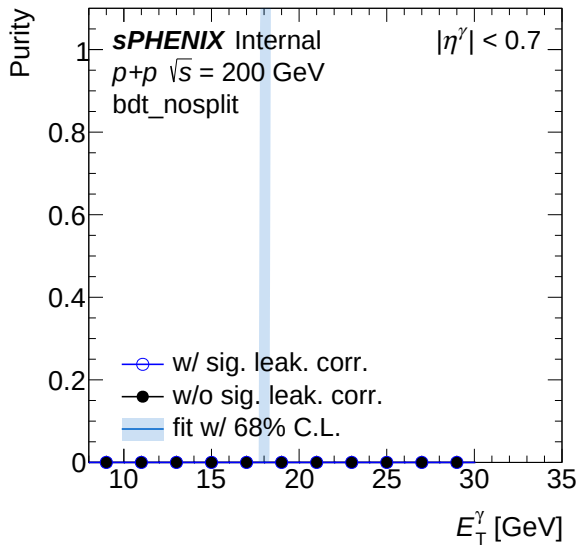


(b) Leakage fraction vs ET (MC) — bdt_nosplit

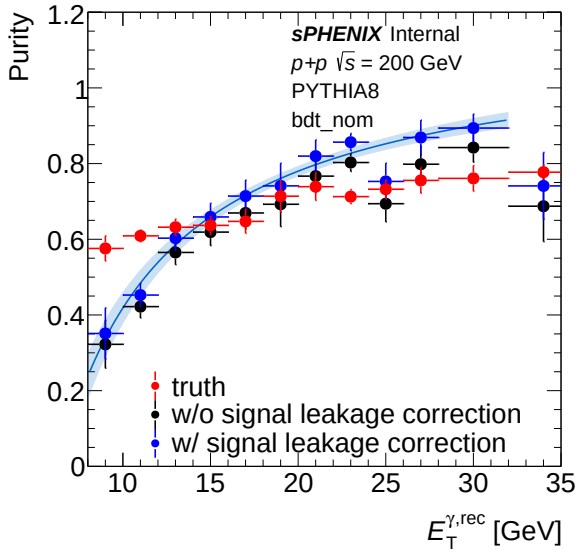
3.3 Purity



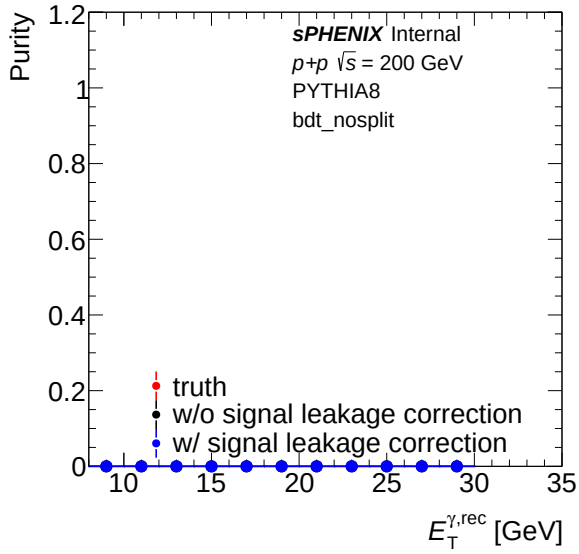
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_nosplit

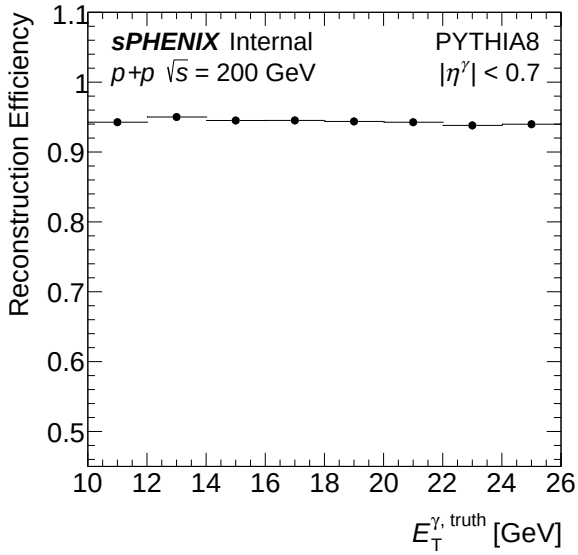


(a) Purity (MC) — bdt_nom

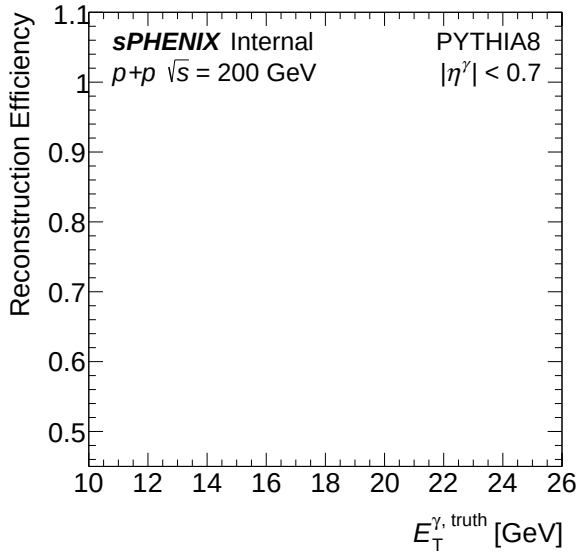


(b) Purity (MC) — bdt_nosplit

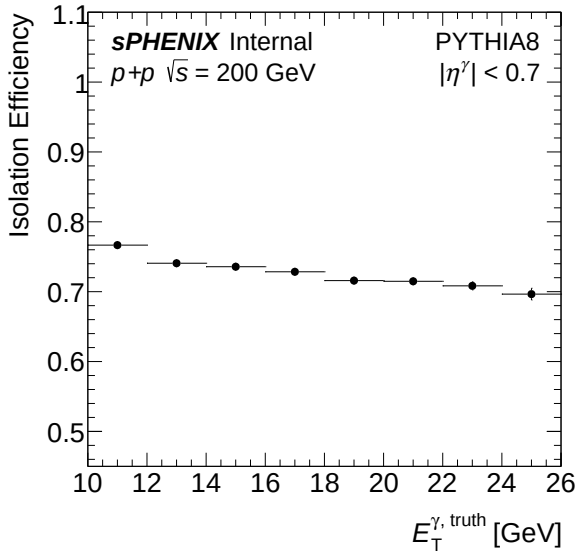
3.4 Efficiency



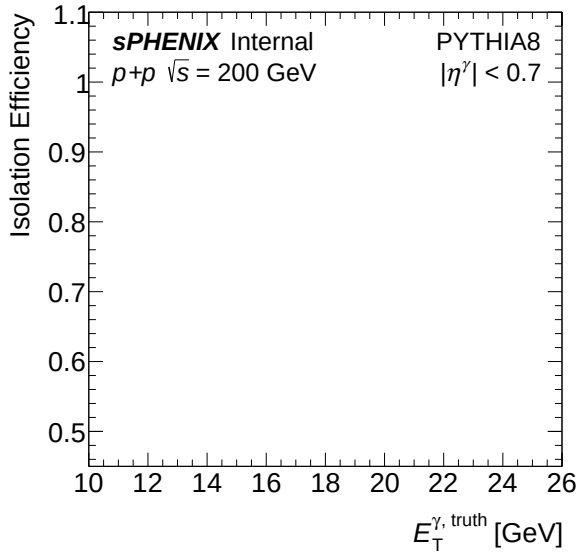
(a) Reconstruction efficiency — bdt_nom



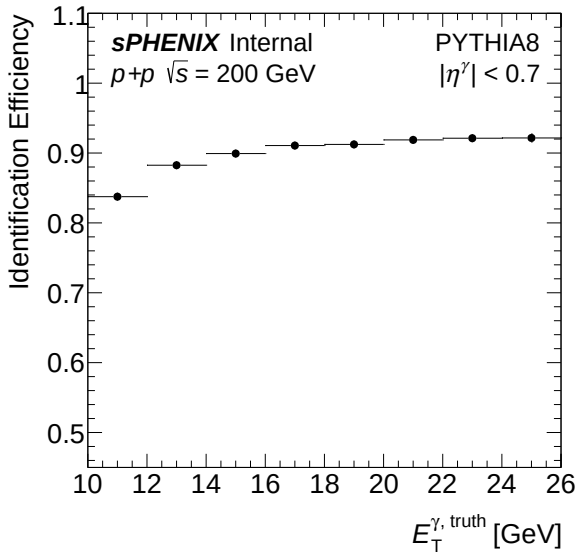
(b) Reconstruction efficiency — bdt_nosplit



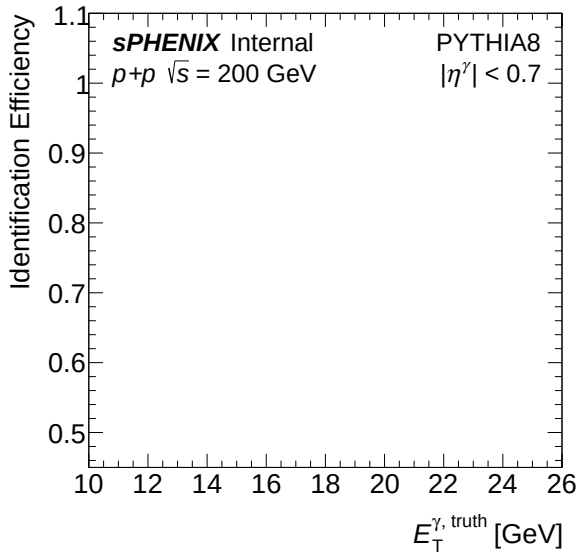
(a) Isolation efficiency — bdt_nom



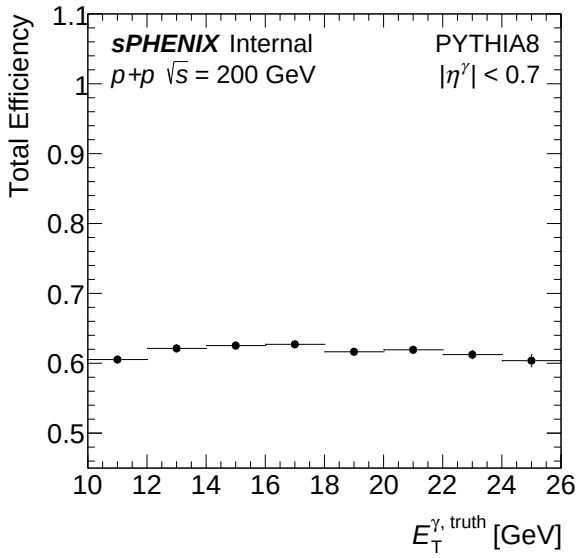
(b) Isolation efficiency — bdt_nosplit



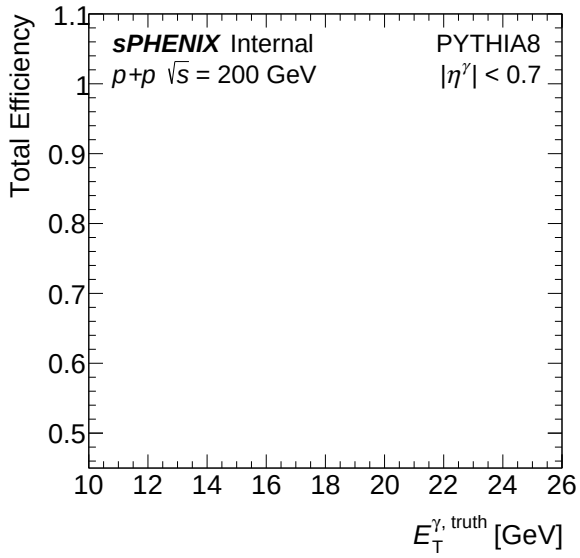
(a) Identification efficiency — bdt_nom



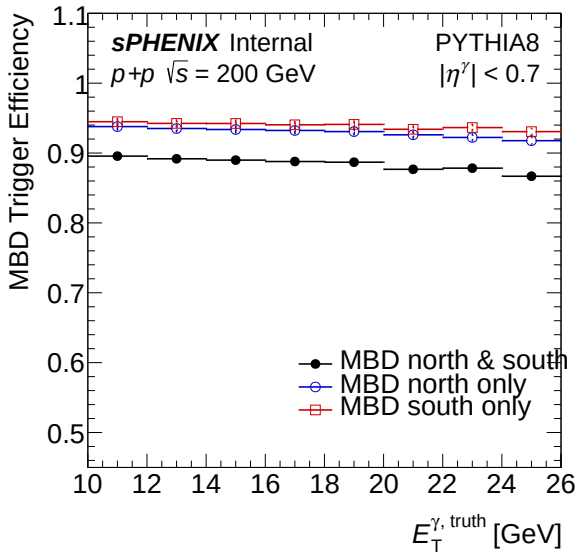
(b) Identification efficiency — bdt_nosplit



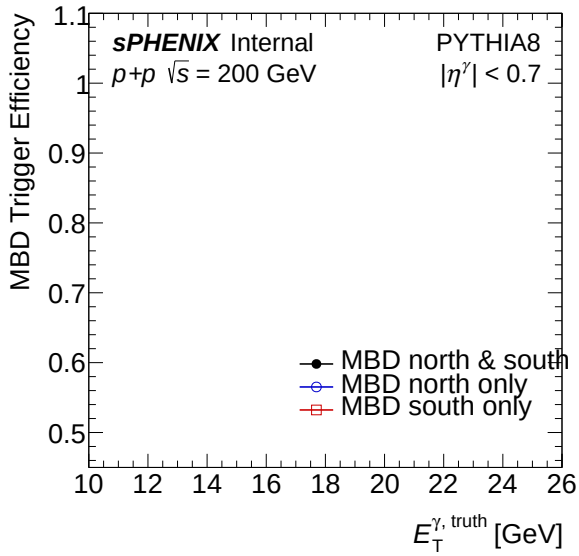
(a) Total efficiency — bdt_nom



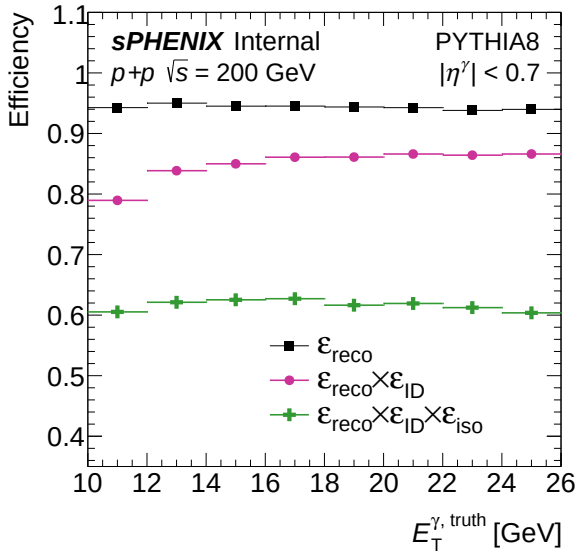
(b) Total efficiency — bdt_nosplit



(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_nosplit

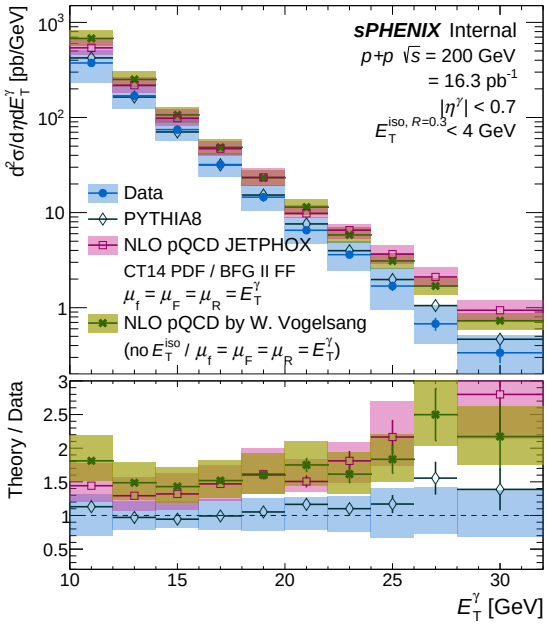


(a) Combined photon efficiency — bdt_nom

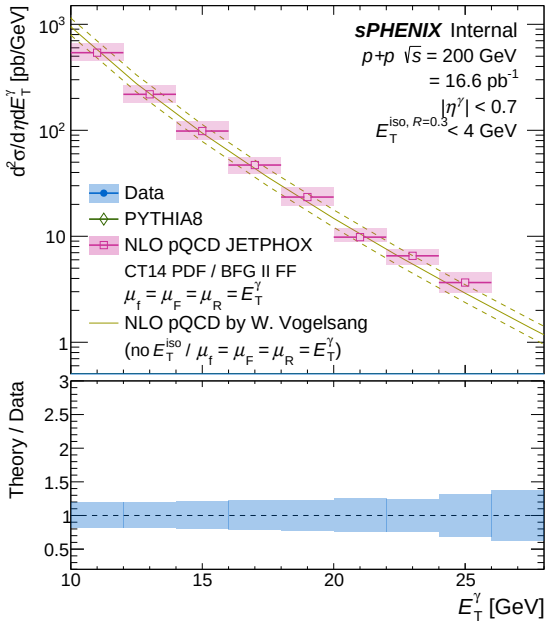
not found

(b) Combined photon efficiency — bdt_nosplit

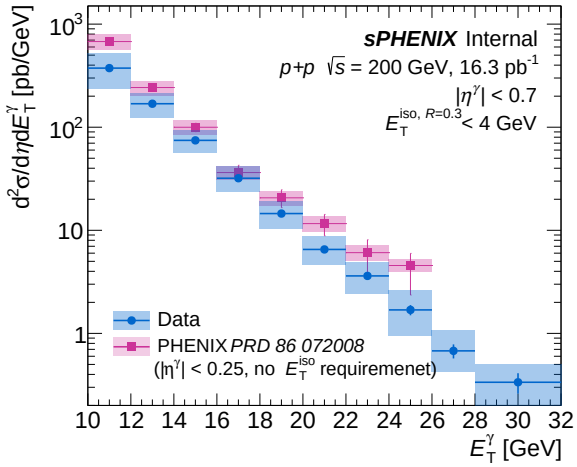
3.5 Final result



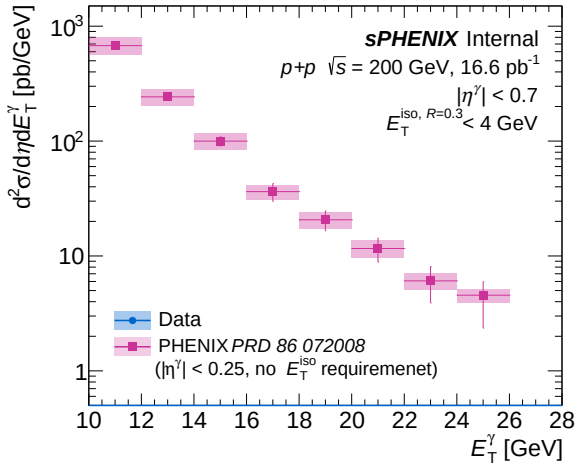
(a) Final cross section result — bdt_nom



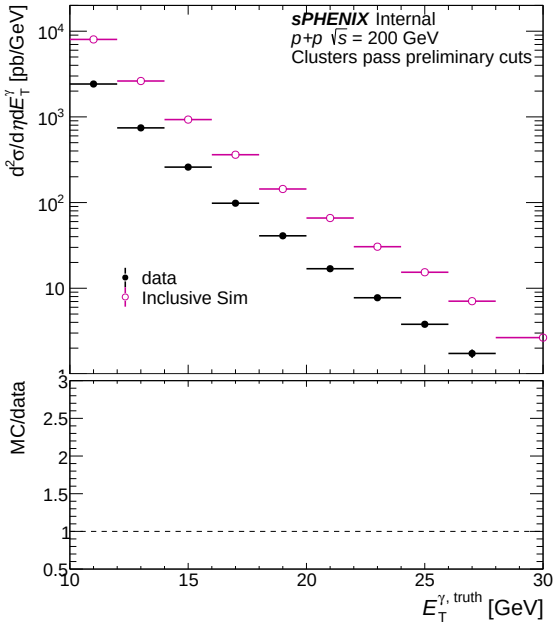
(b) Final cross section result — bdt_nosplit



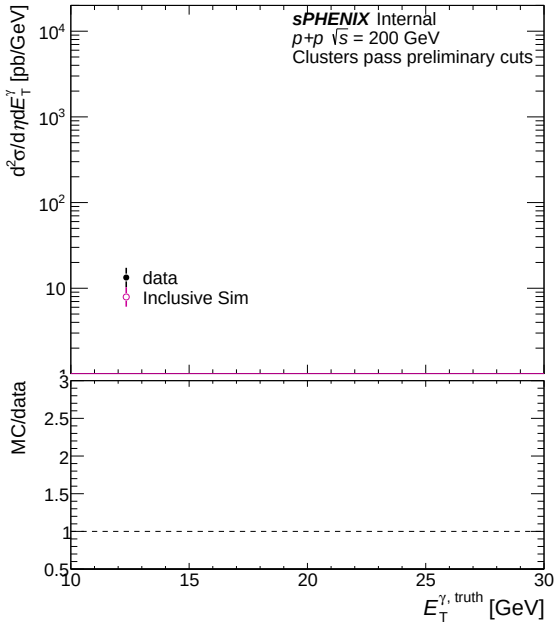
(a) Comparison with PHENIX — bdt_nom



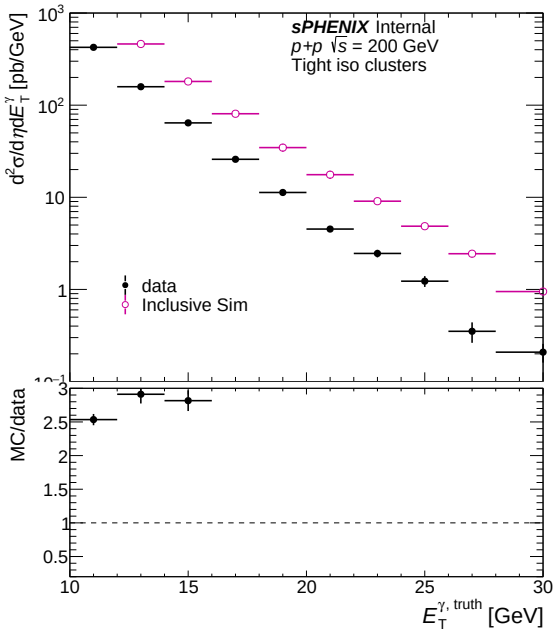
(b) Comparison with PHENIX — bdt_nosplit



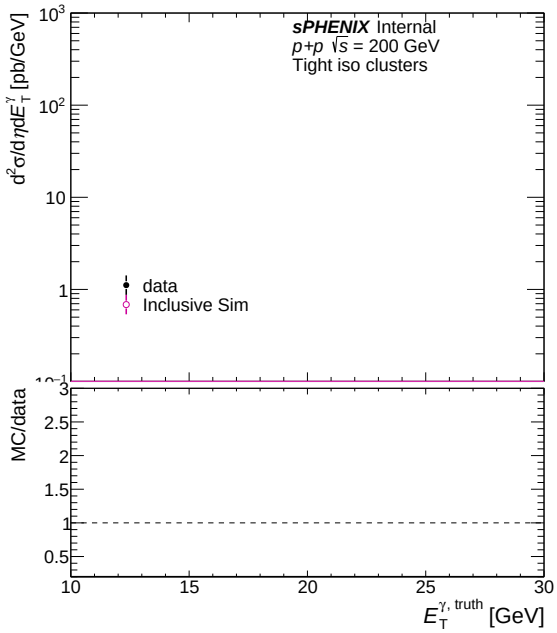
(a) Common cluster yield (data vs MC) — bdt_nom



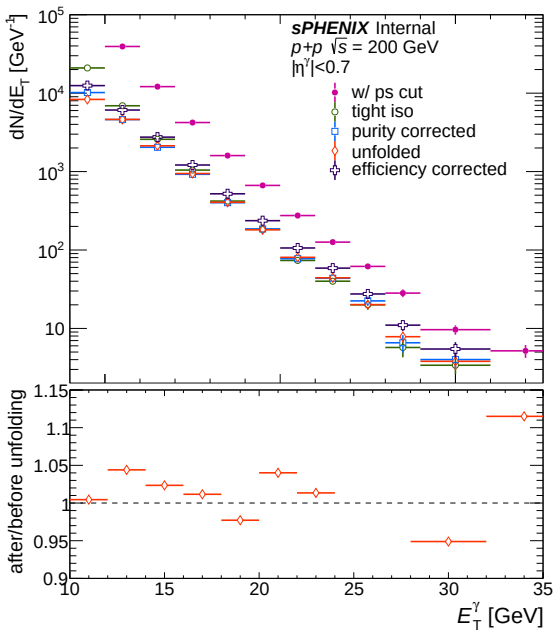
(b) Common cluster yield (data vs MC) — bdt_nosplit



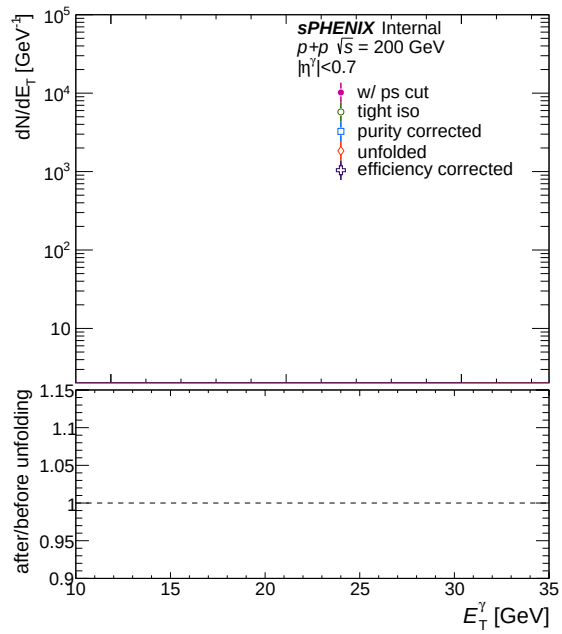
(a) Tight iso cluster yield (data vs MC) — bdt_nom



(b) Tight iso cluster yield (data vs MC) — bdt_nosplit

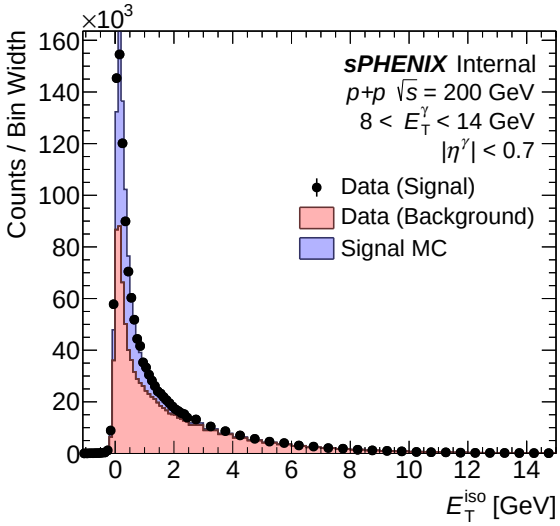


(a) All correction steps — bdt_nom

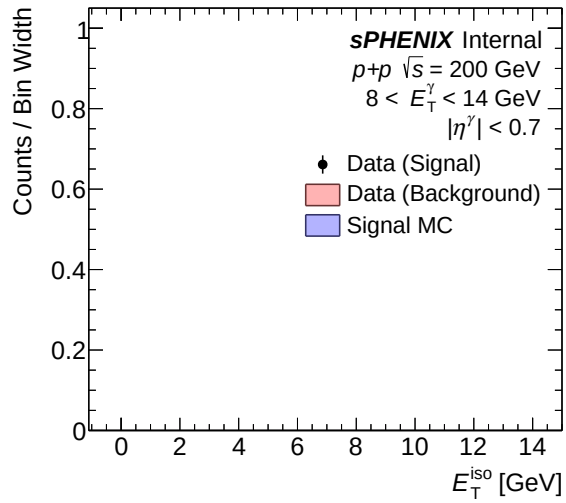


(b) All correction steps — bdt_nosplit

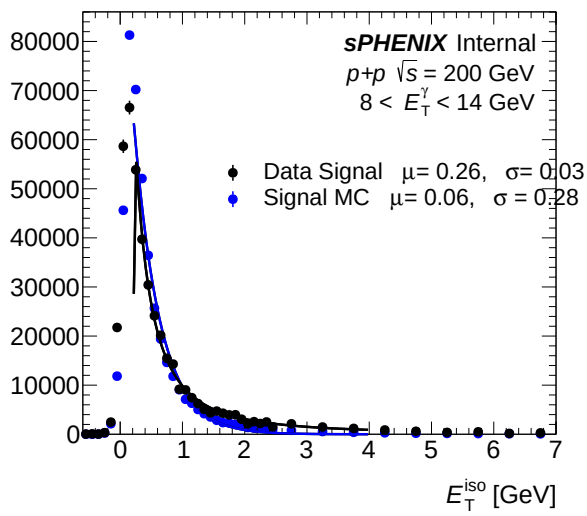
3.6 Isolation fit (CONF)



(a) IsoET template (pT bins 0–2) — bdt_nom

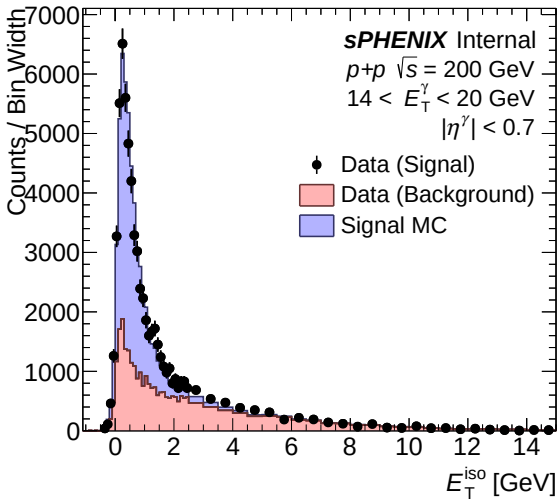


(b) IsoET template (pT bins 0–2) — bdt_nosplit

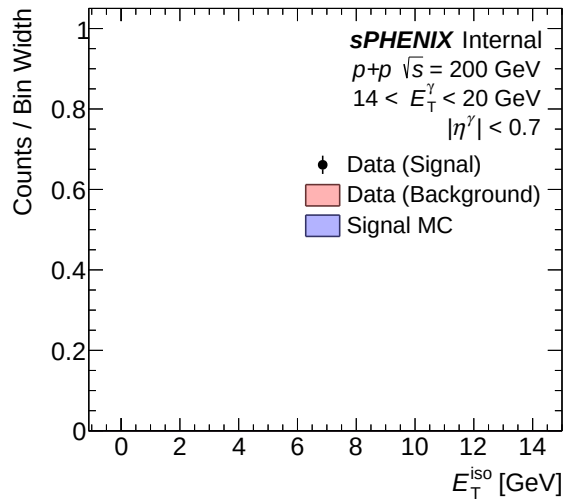


(a) IsoET fit template (pT bins 0–2) — bdt_nom

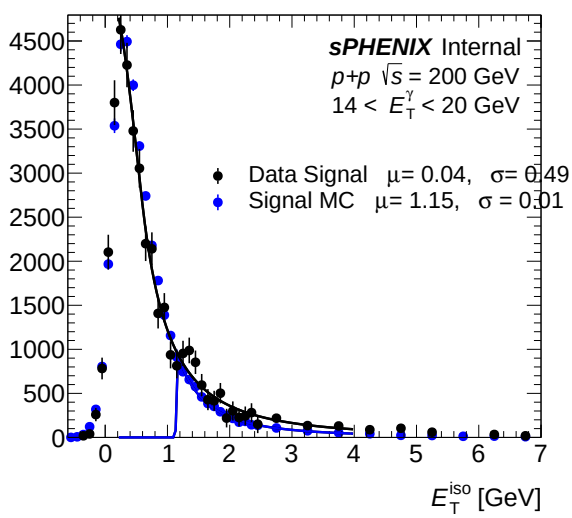
(b) IsoET fit template (pT bins 0–2) — bdt_nosplit



(a) IsoET template (pT bins 3–5) — bdt_nom

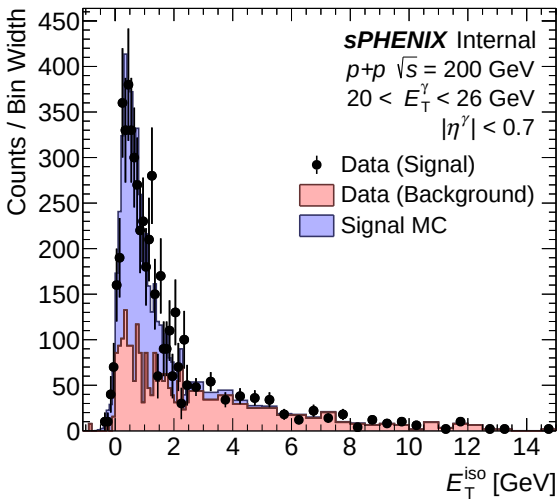


(b) IsoET template (pT bins 3–5) — bdt_nosplit

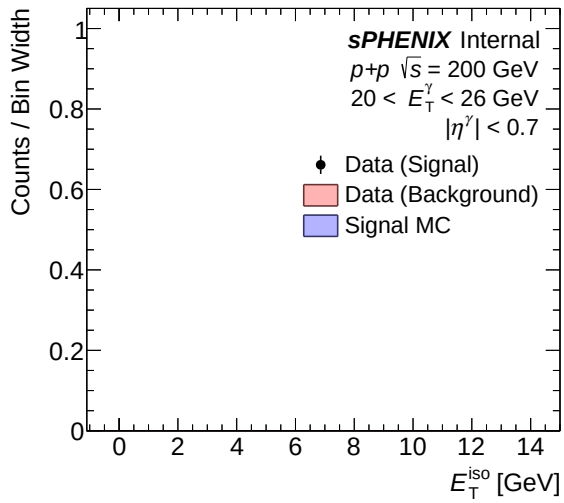


(a) IsoET fit template (pT bins 3–5) — bdt_nom

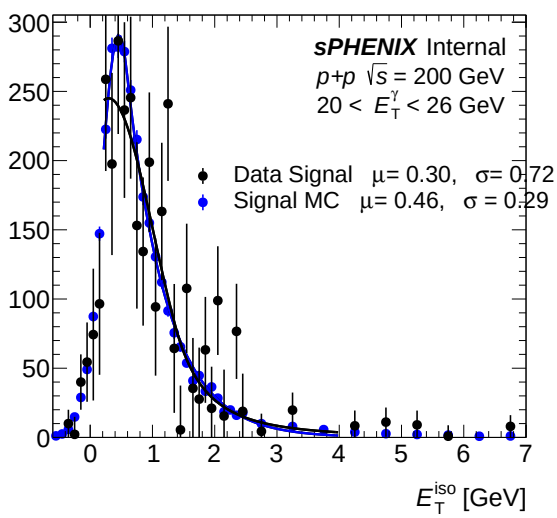
(b) IsoET fit template (pT bins 3–5) — bdt_nosplit



(a) IsoET template (pT bins 6–8) — bdt_nom

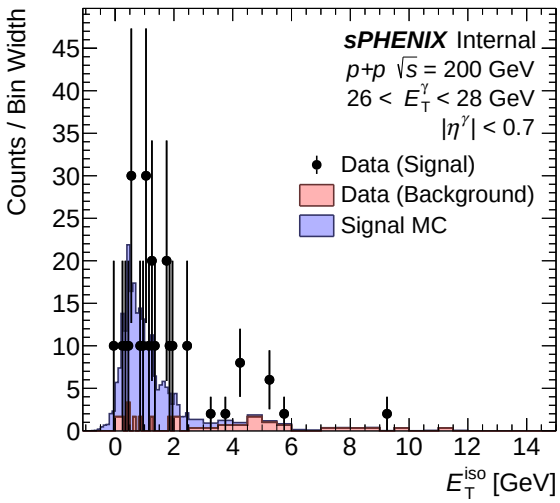


(b) IsoET template (pT bins 6–8) — bdt_nosplit

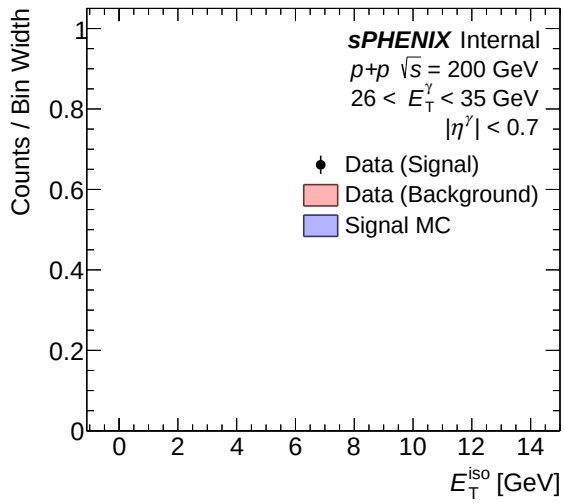


(a) IsoET fit template (pT bins 6–8) — bdt_nom

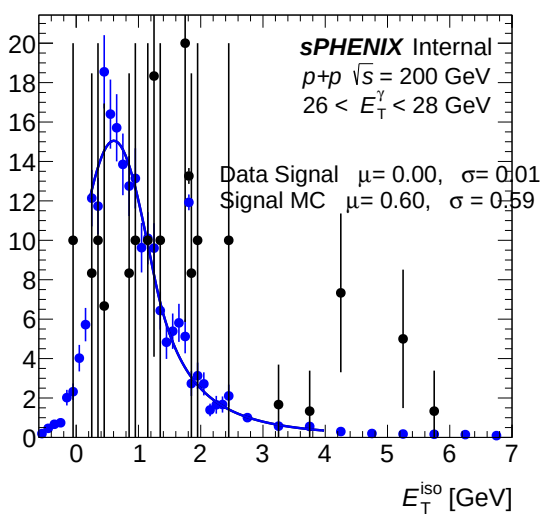
(b) IsoET fit template (pT bins 6–8) — bdt_nosplit



(a) IsoET template (pT bins 9–9) — bdt_nom



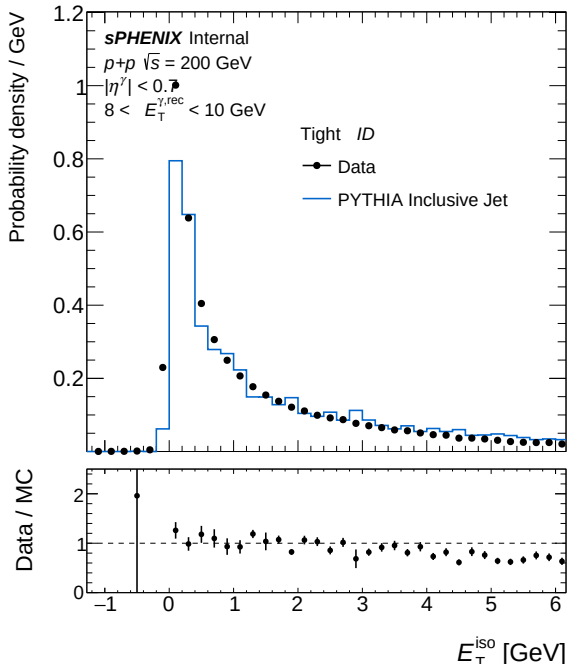
(b) IsoET template (pT bins 9–9) — bdt_nosplit



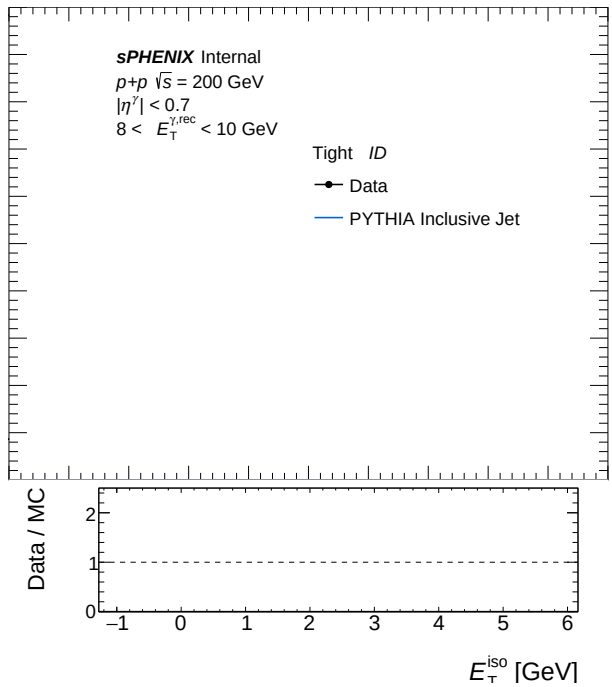
(a) IsoET fit template (pT bins 9–9) — bdt_nom

(b) IsoET fit template (pT bins 9–9) — bdt_nosplit

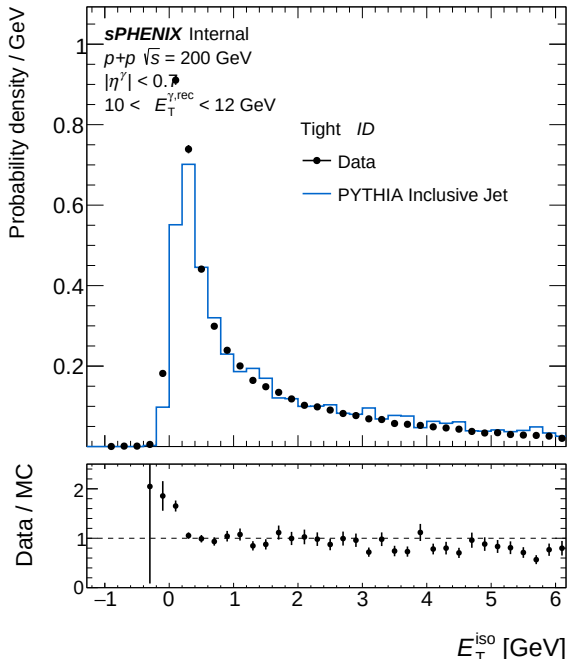
3.7 Isolation ET



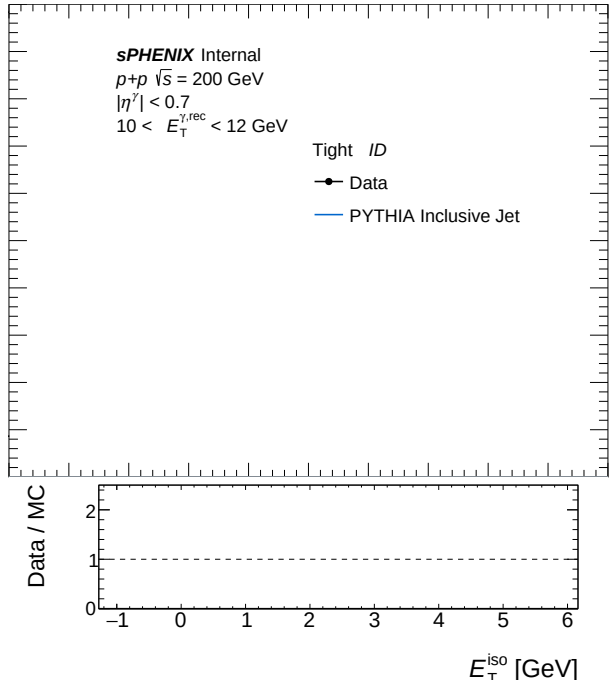
(a) IsoET Tight, pT bin 0 — bdt_nom



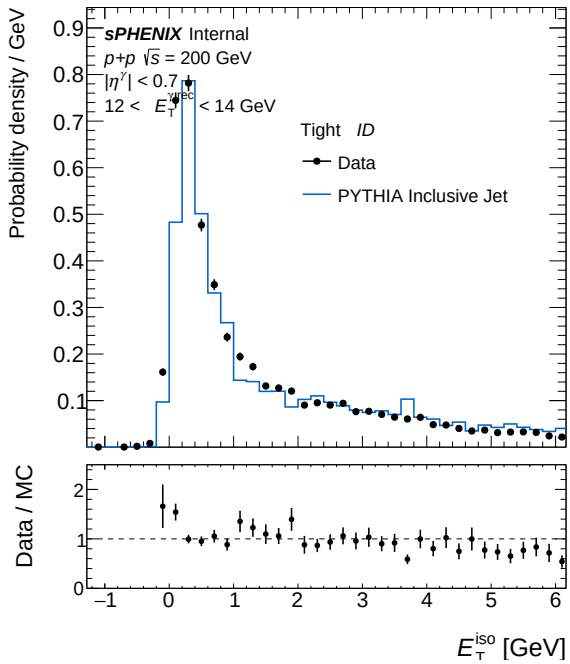
(b) IsoET Tight, pT bin 0 — bdt_nosplit



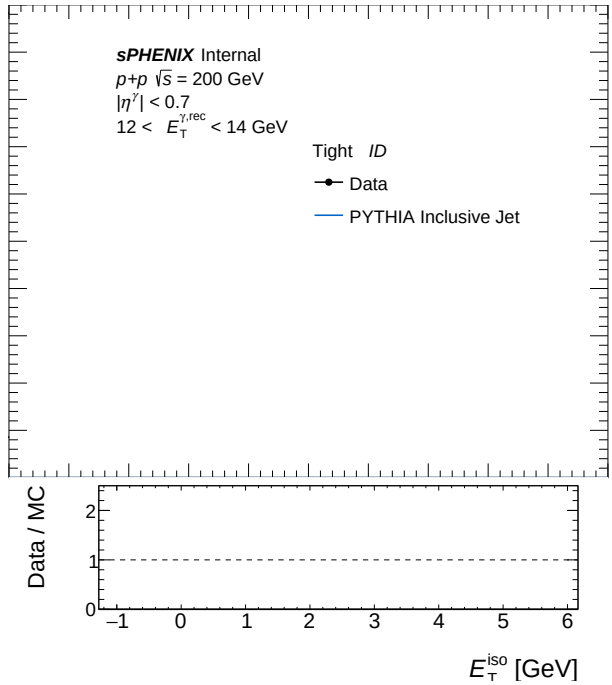
(a) IsoET Tight, pT bin 1 — bdt_nom



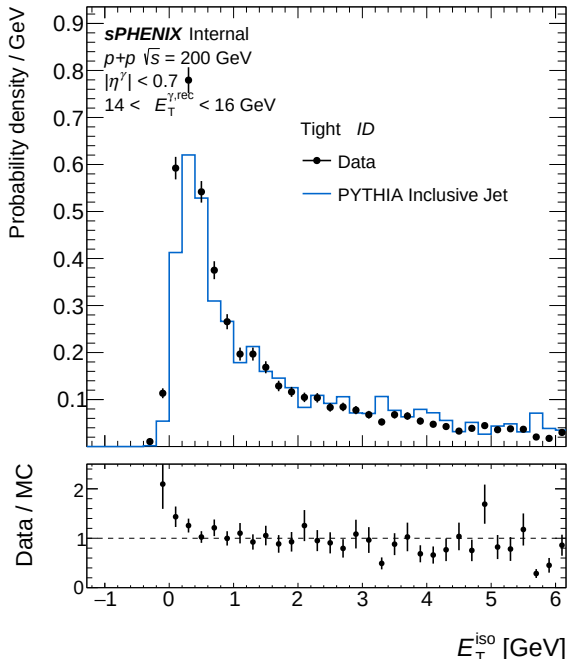
(b) IsoET Tight, pT bin 1 — bdt_nosplit



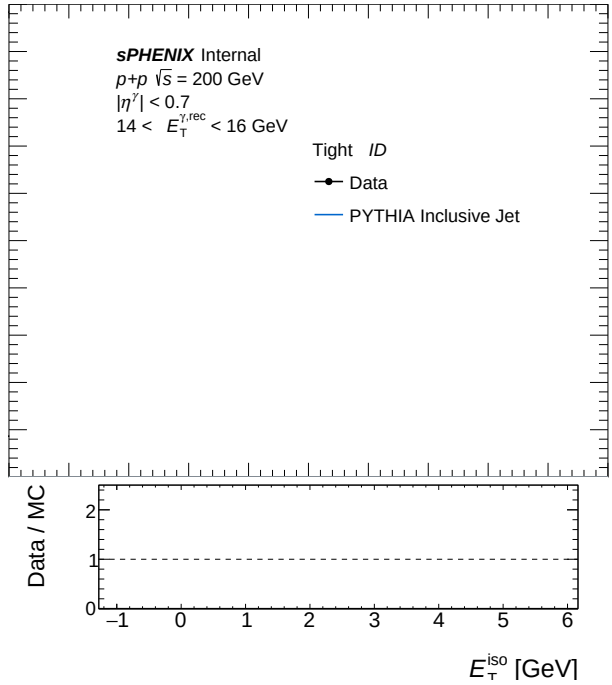
(a) IsoET Tight, pT bin 2 — bdt_nom



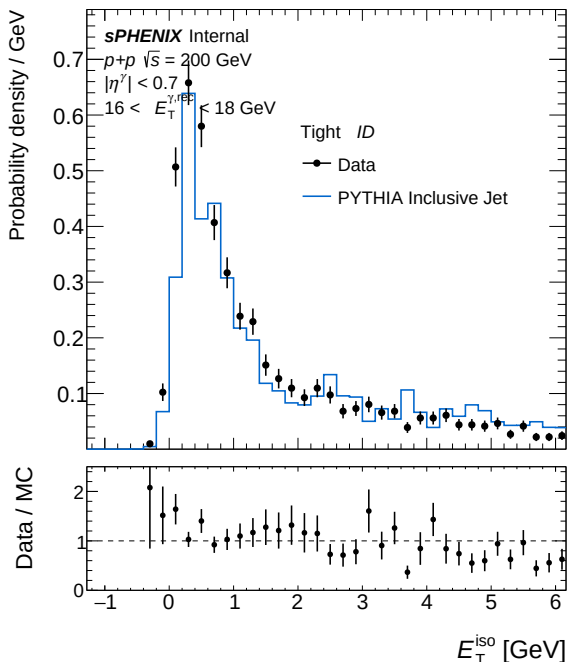
(b) IsoET Tight, pT bin 2 — bdt_nosplit



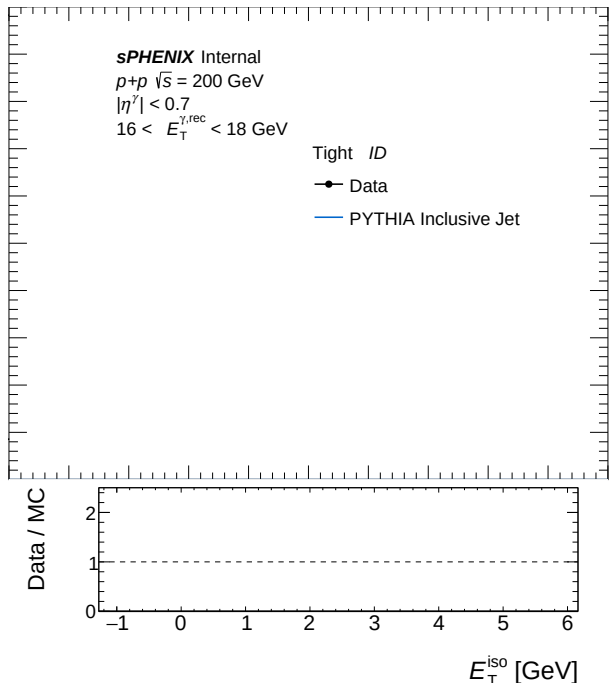
(a) IsoET Tight, pT bin 3 — bdt_nom



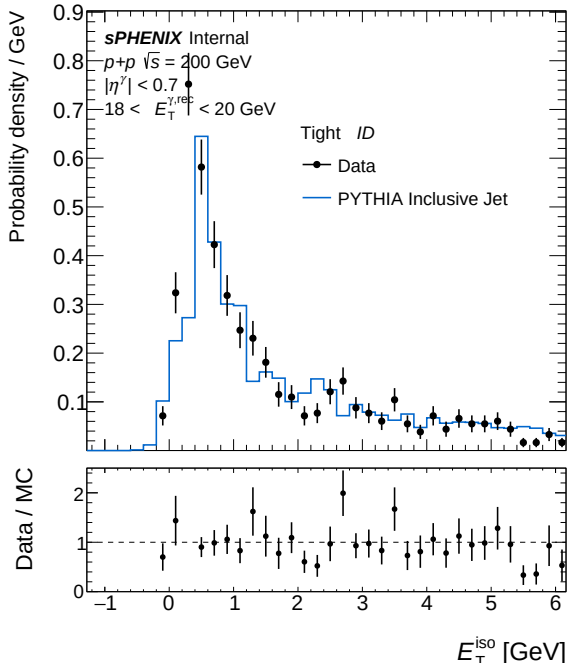
(b) IsoET Tight, pT bin 3 — bdt_nosplit



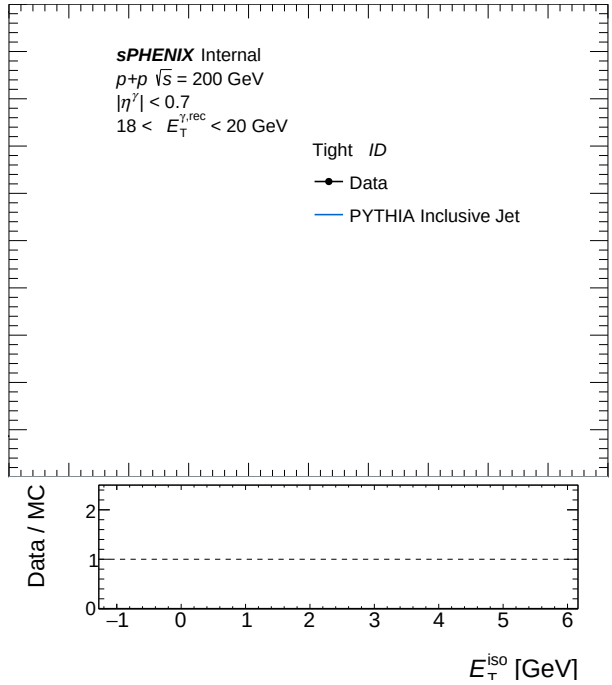
(a) IsoET Tight, pT bin 4 — bdt_nom



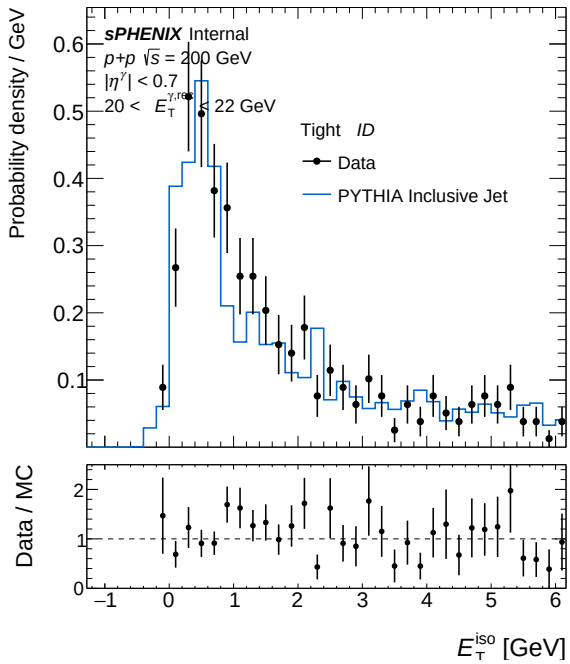
(b) IsoET Tight, pT bin 4 — bdt_nosplit



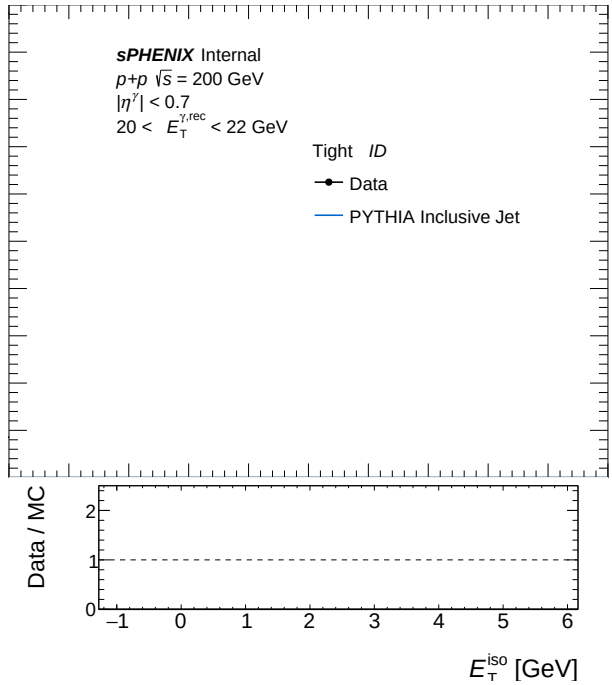
(a) IsoET Tight, pT bin 5 — bdt_nom



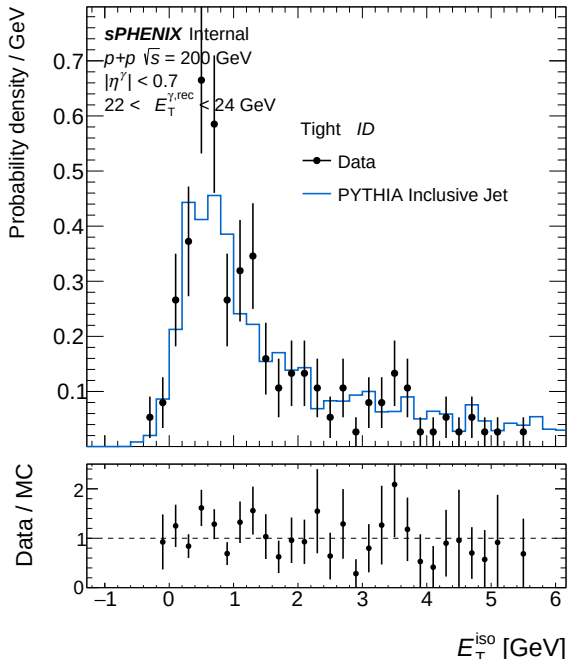
(b) IsoET Tight, pT bin 5 — bdt_nosplit



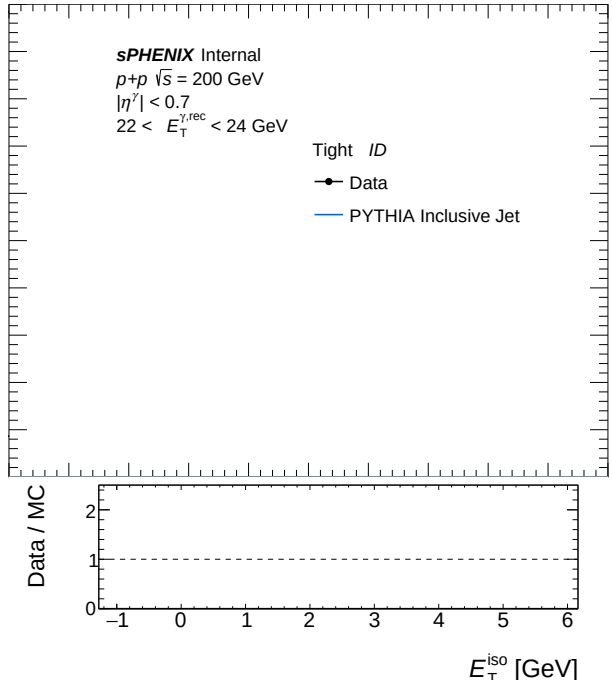
(a) IsoET Tight, pT bin 6 — bdt_nom



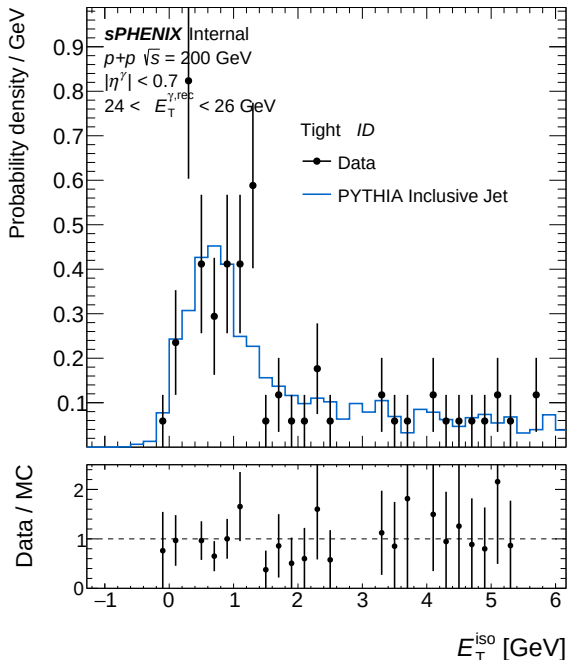
(b) IsoET Tight, pT bin 6 — bdt_nosplit



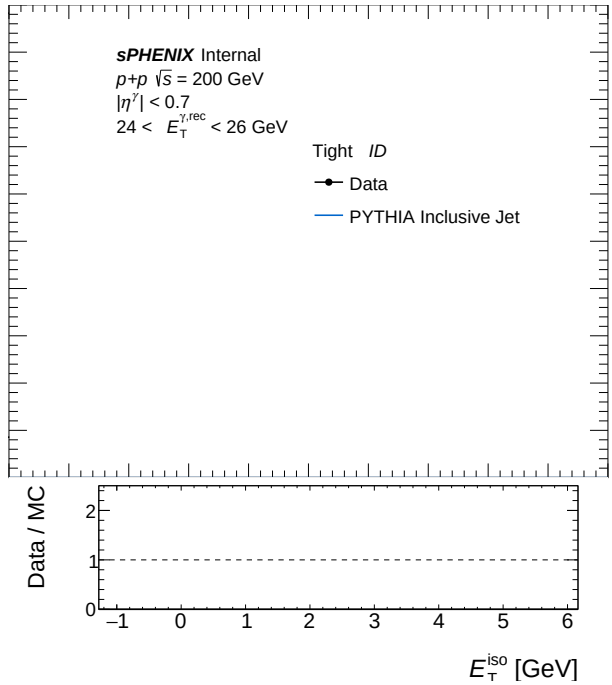
(a) IsoET Tight, pT bin 7 — bdt_nom



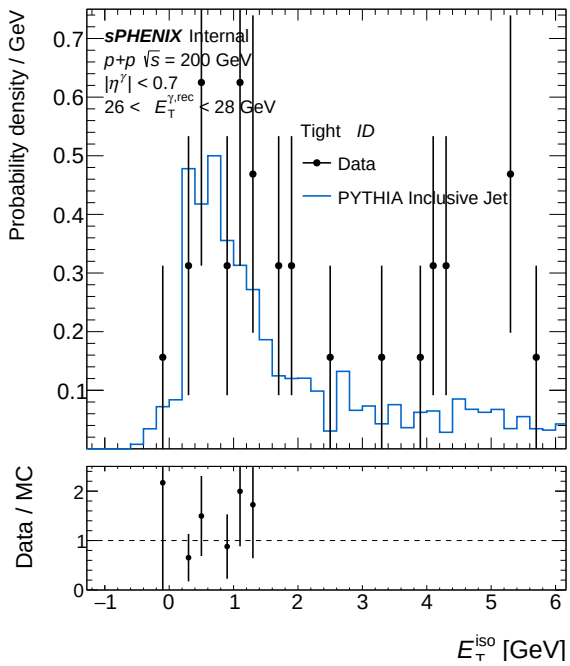
(b) IsoET Tight, pT bin 7 — bdt_nosplit



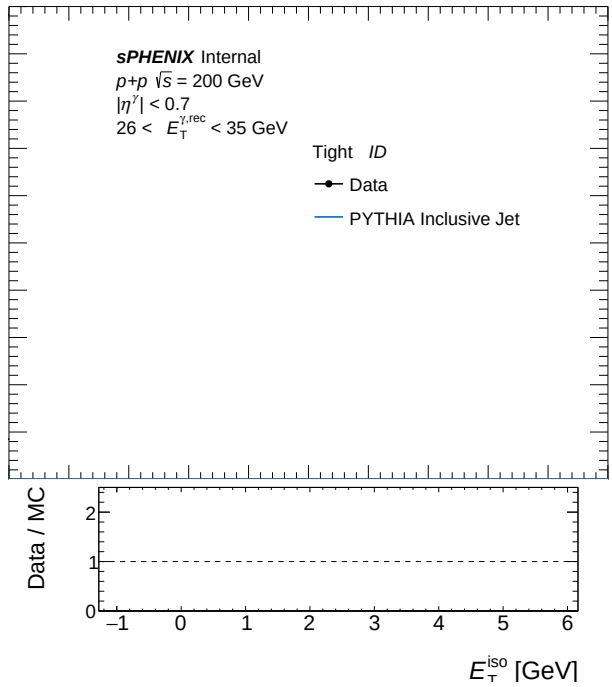
(a) IsoET Tight, pT bin 8 — bdt_nom



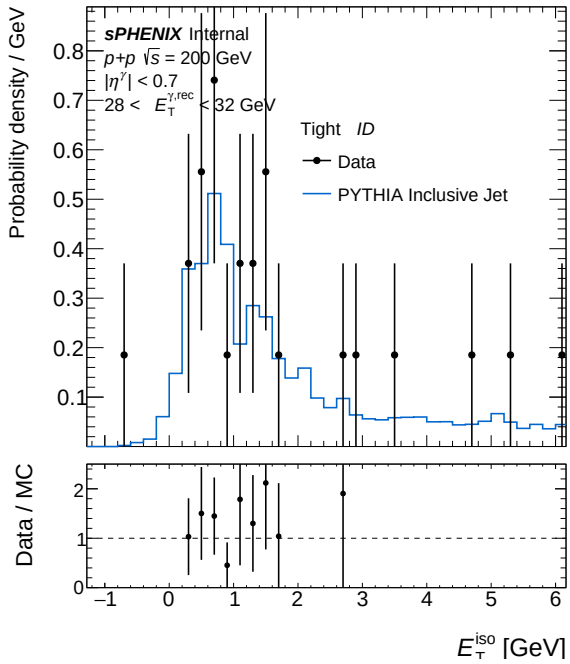
(b) IsoET Tight, pT bin 8 — bdt_nosplit



(a) IsoET Tight, pT bin 9 — `bdt_nom`



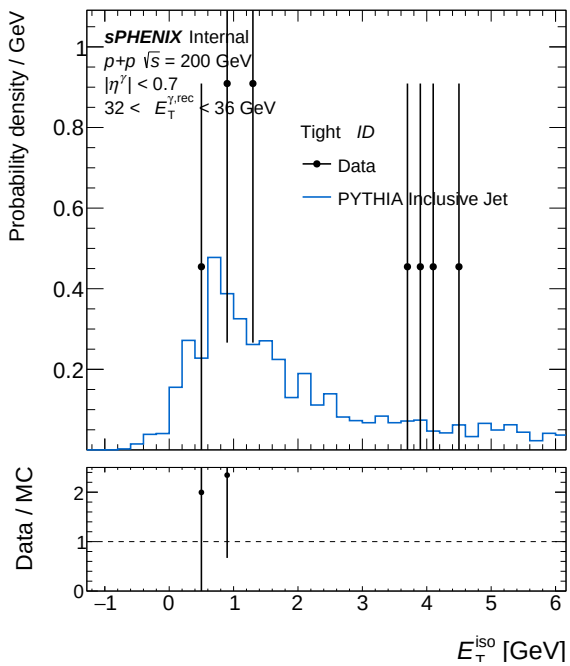
(b) IsoET Tight, pT bin 9 — `bdt_nosplit`



(a) IsoET Tight, pT bin 10 — `bdt_nom`

not found

(b) IsoET Tight, pT bin 10 — `bdt_nosplit`



(a) IsoET Tight, pT bin 11 — `bdt_nom`

not found

(b) IsoET Tight, pT bin 11 — `bdt_nosplit`